Cell 1 — setup + load the validated tract-year panel for the new cluster-time workflow

In [1]:

```
# Cell 1 — setup + load the validated tract-year panel for the new cluster-t

from pathlib import Path
import numpy as np
import pandas as pd

pd.set_option("display.max_columns", 200)
pd.set_option("display.width", 200)

# ----------------------------
# Paths / core settings
# ----------------------------
DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
OUT_DIR = DATA_DIR / "processed"

LOW_IGS_THRESHOLD = 45
ANALYSIS_YEARS = list(range(2017, 2025))   # ACS-aligned years only
PRE_COVID_YEARS = [2017, 2018, 2019]
COVID_YEARS = [2020, 2021]
RECOVERY_YEARS = [2022, 2023, 2024]

ECON_VARS = [
    "igs_economy",
    "poverty_rate",
    "unemp_rate",
    "median_household_income",
    "lfpr_16p",
]

CLUSTER_CORE_VARS = [
    "igs_total",
    "igs_economy",
    "poverty_rate",
    "unemp_rate",
    "median_household_income",
    "lfpr_16p",
    "pop_total",
]

DISPLAY_COLS = [
    "display_state",
    "display_county",
    "display_name",
]

REQUIRED_COLS = [
    "geoid",
    "year",
    "igs_total",
```

```python
    "igs_economy",
    "poverty_rate",
    "unemp_rate",
    "median_household_income",
    "lfpr_16p",
    "pop_total",
]

PANEL_CANDIDATES = [
    OUT_DIR / "eda_panel_clean.parquet",
    OUT_DIR / "igs_x_acs_full.parquet",
]

def normalize_geoid_series(series: pd.Series) -> pd.Series:
    s = series.astype("string").str.strip()
    numeric = pd.to_numeric(s, errors="coerce")
    numeric_mask = numeric.notna()
    s = s.copy()
    s.loc[numeric_mask] = numeric.loc[numeric_mask].astype("Int64").astype("
    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    return s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

def add_display_fields(df: pd.DataFrame) -> pd.DataFrame:
    out = df.copy()

    if "display_state" not in out.columns:
        if "State" in out.columns or "state" in out.columns:
            out["display_state"] = out.get("State", pd.Series(index=out.inde
                out.get("state", pd.Series(index=out.index))
            )
        else:
            out["display_state"] = pd.NA

    if "display_county" not in out.columns:
        if "County" in out.columns or "county" in out.columns:
            out["display_county"] = out.get("County", pd.Series(index=out.ir
                out.get("county", pd.Series(index=out.index))
            )
        else:
            out["display_county"] = pd.NA

    if "display_name" not in out.columns:
        if "NAME" in out.columns:
            out["display_name"] = out["NAME"].astype("string")
        elif "Census Tract FIPS code" in out.columns:
            out["display_name"] = "Census Tract " + out["Census Tract FIPS c
        else:
            out["display_name"] = out["geoid"].astype("string")

    return out

def assign_period(year: int) -> str:
    if year in PRE_COVID_YEARS:
        return "pre_covid"
```

```python
    if year in COVID_YEARS:
        return "covid"
    if year in RECOVERY_YEARS:
        return "recovery"
    return "outside_scope"

def load_analysis_panel() -> tuple[pd.DataFrame, str]:
    # Prefer in-memory validated objects if they already exist in the notebo
    if "eda_df" in globals():
        df = eda_df.copy()
        source = "existing global: eda_df"
    elif "model_df_full" in globals():
        df = model_df_full.copy()
        source = "existing global: model_df_full"
    else:
        df = None
        source = None
        for p in PANEL_CANDIDATES:
            if p.exists():
                df = pd.read_parquet(p)
                source = f"file: {p}"
                break
        if df is None:
            searched = ", ".join(str(p) for p in PANEL_CANDIDATES)
            raise FileNotFoundError(
                "Could not find a processed merged panel. "
                f"Expected one of: {searched}"
            )

    # Standardize core keys
    df = df.copy()
    df["geoid"] = normalize_geoid_series(df["geoid"])
    df["year"] = pd.to_numeric(df["year"], errors="coerce").astype("Int64")

    # Restrict to ACS-aligned period only
    df = df[df["year"].isin(ANALYSIS_YEARS)].copy()
    df["year"] = df["year"].astype(int)

    # Add consistent display fields
    df = add_display_fields(df)

    # Remove duplicate tract-year rows if any slipped in
    df = (
        df.sort_values(["geoid", "year"])
          .drop_duplicates(["geoid", "year"])
          .reset_index(drop=True)
    )

    # Validate required columns
    missing = [c for c in REQUIRED_COLS if c not in df.columns]
    if missing:
        raise ValueError(f"Missing required columns for new workflow: {missi

    # Add time-period labels for the new method
    df["analysis_period"] = df["year"].map(assign_period)
```

```python
    return df, source

panel_df, PANEL_SOURCE = load_analysis_panel()

analysis_cols = (
    ["geoid", "year", "analysis_period"] +
    DISPLAY_COLS +
    CLUSTER_CORE_VARS
)

analysis_panel = panel_df[analysis_cols].copy()

print("Loaded panel from:", PANEL_SOURCE)
print("Analysis panel shape:", analysis_panel.shape)
print("Unique tracts:", analysis_panel["geoid"].nunique())
print("Years:", sorted(analysis_panel["year"].unique().tolist()))
print("Periods:", analysis_panel["analysis_period"].value_counts(dropna=Fals
print()
print("Missingness in core vars:")
print(analysis_panel[CLUSTER_CORE_VARS].isna().mean().sort_values(ascending=
print()
print(analysis_panel.head())
```

```
Loaded panel from: file: data\processed\eda_panel_clean.parquet
Analysis panel shape: (680256, 13)
Unique tracts: 85032
Years: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]
Periods: {'pre_covid': 255096, 'recovery': 255096, 'covid': 170064}

Missingness in core vars:
median_household_income    0.116411
poverty_rate               0.111802
unemp_rate                 0.111357
lfpr_16p                   0.109857
pop_total                  0.104783
igs_total                  0.008863
igs_economy                0.001313
dtype: float64

         geoid  year analysis_period display_state  display_county
display_name  igs_total  igs_economy  poverty_rate  unemp_rate  median_house
hold_income  lfpr_16p  \
0  01001020100  2017       pre_covid       Alabama  Autauga County  Census T
ract 201, Autauga County, Alabama       47.0         34.0      0.106775
0.045504                  67826.0  0.622387
1  01001020100  2018       pre_covid       Alabama  Autauga County  Census T
ract 201, Autauga County, Alabama       52.0         48.0      0.113365
0.036232                  58625.0  0.618438
2  01001020100  2019       pre_covid       Alabama  Autauga County  Census T
ract 201, Autauga County, Alabama       46.0         40.0      0.166583
0.028571                  60208.0  0.606061
3  01001020100  2020           covid       Alabama  Autauga County  Census T
ract 201, Autauga County, Alabama       46.0         48.0      0.136528
0.021127                  60388.0  0.546855
4  01001020100  2021           covid       Alabama  Autauga County  Census T
ract 201, Autauga County, Alabama       38.0         32.0      0.153546
0.024967                  57399.0  0.534035

   pop_total
0     1845.0
1     1923.0
2     1993.0
3     1941.0
4     1791.0
```

Cell 2 — reuse or rebuild tract -> cluster mapping from the validated adjacency work

In [2]:

```
# Cell 2 — broadened rebuild of tract -> cluster mapping
# Goal:
# 1) keep the same vulnerability logic
# 2) broaden the actual candidate universe upstream
# 3) prevent the seed-county stage from collapsing into only one geography
# 4) keep adjacency clustering as the foundation

from pathlib import Path
from collections import defaultdict, deque
import zipfile
```

```python
import requests
import geopandas as gpd
import numpy as np
import pandas as pd

# ---------------------------------------------------
# Settings
# ---------------------------------------------------
MIN_CLUSTER_TRACTS = 3
MIN_CLUSTER_POP = 5000

# Larger national pool used only to identify where strong signals exist
COUNTY_SIGNAL_POOL_N = 1500

# Diversified county selection
TARGET_SEED_COUNTIES_N = 40
PER_STATE_BASE_COUNTIES = 1
PER_STATE_MAX_COUNTIES = 3
MIN_SIGNAL_TRACTS_PER_COUNTY = 2

# County expansion after seed selection
BROADER_MAX_PER_COUNTY = 35

# Add one-hop tract connectors so communities do not fragment
CONNECTOR_HOPS = 1

TRACT_BASE_URL = "https://www2.census.gov/geo/tiger/TIGER2024/TRACT/"
TRACT_ROOT_DIR = RAW_DIR / "tl_2024_state_tracts"
TRACT_ROOT_DIR.mkdir(parents=True, exist_ok=True)

# ---------------------------------------------------
# Helpers
# ---------------------------------------------------
def bad_when_high(series: pd.Series) -> pd.Series:
    s = pd.to_numeric(series, errors="coerce")
    return s.rank(pct=True, method="average")

def bad_when_low(series: pd.Series) -> pd.Series:
    s = pd.to_numeric(series, errors="coerce")
    return (-s).rank(pct=True, method="average")

def load_state_tract_geometry(state_fips_list):
    tract_gdf_list = []

    for st in state_fips_list:
        zip_name = f"tl_2024_{st}_tract.zip"
        shp_name = f"tl_2024_{st}_tract.shp"
        zip_path = TRACT_ROOT_DIR / zip_name
        extract_dir = TRACT_ROOT_DIR / f"tl_2024_{st}_tract"
        shp_path = extract_dir / shp_name

        if not shp_path.exists():
            extract_dir.mkdir(parents=True, exist_ok=True)

            if not zip_path.exists():
                url = f"{TRACT_BASE_URL}{zip_name}"
```

```python
                print(f"Downloading {url}")
                r = requests.get(url, timeout=120)
                r.raise_for_status()
                with open(zip_path, "wb") as f:
                    f.write(r.content)

            print(f"Extracting {zip_name}")
            with zipfile.ZipFile(zip_path, "r") as zf:
                zf.extractall(extract_dir)

        if not shp_path.exists():
            raise FileNotFoundError(f"Missing shapefile after extraction: {s

        gdf_state = gpd.read_file(shp_path)
        tract_gdf_list.append(gdf_state)

    tract_gdf = pd.concat(tract_gdf_list, ignore_index=True)
    tract_gdf = gpd.GeoDataFrame(
        tract_gdf,
        geometry="geometry",
        crs=tract_gdf_list[0].crs if tract_gdf_list else None
    )

    tract_gdf["geoid"] = normalize_geoid_series(tract_gdf["GEOID"])
    tract_gdf = tract_gdf.dropna(subset=["geoid", "geometry"]).copy()
    tract_gdf = tract_gdf[tract_gdf.geometry.notna()].copy()
    tract_gdf = tract_gdf[~tract_gdf.geometry.is_empty].copy()
    tract_gdf = tract_gdf.drop_duplicates(subset=["geoid"]).copy()

    return tract_gdf

def build_adjacency_edges_fast(gdf: gpd.GeoDataFrame) -> pd.DataFrame:
    """
    Build tract-touching edges within each county.
    """
    edge_frames = []

    for (state_name, county_name), grp in gdf.groupby(
        ["display_state", "display_county"],
        dropna=False
    ):
        grp = grp[["geoid", "display_state", "display_county", "geometry"]].
        grp = grp.dropna(subset=["geoid", "geometry"]).copy()

        if len(grp) < 2:
            continue

        left = grp[["geoid", "geometry"]].copy()
        right = grp[["geoid", "geometry"]].copy()

        joined = gpd.sjoin(left, right, how="inner", predicate="touches")
        if joined.empty:
            continue

        joined = joined.reset_index(drop=True)
        joined = joined.rename(columns={"geoid_left": "geoid_1", "geoid_righ
```

```
        joined = joined[joined["geoid_1"] != joined["geoid_2"]].copy()
        if joined.empty:
            continue

        joined["a"] = joined[["geoid_1", "geoid_2"]].min(axis=1)
        joined["b"] = joined[["geoid_1", "geoid_2"]].max(axis=1)
        joined = joined.drop_duplicates(subset=["a", "b"]).copy()

        out = joined[["a", "b"]].rename(columns={"a": "geoid_1", "b": "geoic
        out["display_state"] = state_name
        out["display_county"] = county_name
        edge_frames.append(out)

    if not edge_frames:
        return pd.DataFrame(columns=["display_state", "display_county", "gec

    return pd.concat(edge_frames, ignore_index=True)

def connected_components_from_edges(nodes, edges_df):
    neighbors = defaultdict(set)

    for n in nodes:
        neighbors[n] = set()

    if edges_df is not None and len(edges_df) > 0:
        for _, row in edges_df.iterrows():
            a = str(row["geoid_1"])
            b = str(row["geoid_2"])
            neighbors[a].add(b)
            neighbors[b].add(a)

    visited = set()
    node_to_cluster_num = {}
    cluster_num = 0

    for n in nodes:
        if n in visited:
            continue

        cluster_num += 1
        q = deque([n])
        visited.add(n)

        while q:
            cur = q.popleft()
            node_to_cluster_num[cur] = cluster_num

            for nxt in neighbors[cur]:
                if nxt not in visited:
                    visited.add(nxt)
                    q.append(nxt)

    return node_to_cluster_num

def diversified_seed_counties(
    county_signal_df: pd.DataFrame,
```

```
    target_n: int,
    per_state_base: int,
    per_state_max: int,
) -> pd.DataFrame:
    """
    Step 1: take a base number of strong counties per state/territory.
    Step 2: backfill nationally until target_n, while respecting state caps.
    """
    county_signal_df = county_signal_df.copy()

    county_signal_df = county_signal_df.sort_values(
        ["display_state", "county_seed_score", "n_signal_tracts", "mean_scop
        ascending=[True, False, False, False, True]
    ).reset_index(drop=True)

    county_signal_df["state_rank"] = (
        county_signal_df.groupby("display_state").cumcount() + 1
    )

    base = county_signal_df.loc[county_signal_df["state_rank"] <= per_state_

    selected = base[
        ["display_state", "display_county"]
    ].drop_duplicates()

    selected_state_counts = (
        selected.groupby("display_state").size().to_dict()
        if len(selected) > 0 else {}
    )

    remaining = county_signal_df.merge(
        selected,
        on=["display_state", "display_county"],
        how="left",
        indicator=True
    )
    remaining = remaining.loc[remaining["_merge"] == "left_only"].drop(colum
    remaining = remaining.sort_values(
        ["county_seed_score", "n_signal_tracts", "mean_scope_score", "mean_i
        ascending=[False, False, False, True]
    ).reset_index(drop=True)

    chosen_rows = [selected.copy()] if len(selected) > 0 else []

    for _, row in remaining.iterrows():
        state_name = row["display_state"]

        current_ct = selected_state_counts.get(state_name, 0)
        if current_ct >= per_state_max:
            continue

        candidate = pd.DataFrame([{
            "display_state": row["display_state"],
            "display_county": row["display_county"]
        }])
```

```
        chosen_rows.append(candidate)
        selected_state_counts[state_name] = current_ct + 1

        total_selected = sum(selected_state_counts.values())
        if total_selected >= target_n:
            break

    if len(chosen_rows) == 0:
        return pd.DataFrame(columns=["display_state", "display_county"])

    out = (
        pd.concat(chosen_rows, ignore_index=True)
        .drop_duplicates(subset=["display_state", "display_county"])
        .reset_index(drop=True)
    )

    return out

# --------------------------------------------------
# A. Build recovery-period tract scoring base
# --------------------------------------------------
tract_recovery = (
    analysis_panel.loc[analysis_panel["analysis_period"] == "recovery"]
    .groupby(
        ["geoid", "display_state", "display_county", "display_name"],
        as_index=False
    )
    .agg({
        "igs_total": "mean",
        "igs_economy": "mean",
        "poverty_rate": "mean",
        "unemp_rate": "mean",
        "median_household_income": "mean",
        "lfpr_16p": "mean",
        "pop_total": "mean",
    })
    .copy()
)

tract_recovery["score_low_igs"] = (
    bad_when_low(tract_recovery["igs_total"]) +
    bad_when_low(tract_recovery["igs_economy"])
) / 2

tract_recovery["score_econ_vulnerability"] = (
    bad_when_low(tract_recovery["igs_economy"]) +
    bad_when_high(tract_recovery["poverty_rate"]) +
    bad_when_high(tract_recovery["unemp_rate"]) +
    bad_when_low(tract_recovery["median_household_income"]) +
    bad_when_low(tract_recovery["lfpr_16p"])
) / 5

tract_recovery["scope_score"] = 100 * (
    0.50 * tract_recovery["score_low_igs"] +
    0.50 * tract_recovery["score_econ_vulnerability"]
)
```

```
tract_recovery = tract_recovery.sort_values(
    ["scope_score", "igs_total"],
    ascending=[False, True]
).reset_index(drop=True)

# --------------------------------------------------
# B. Use a much larger signal pool to identify strong counties
# --------------------------------------------------
signal_pool = tract_recovery.head(COUNTY_SIGNAL_POOL_N).copy()

county_signal = (
    signal_pool
    .groupby(["display_state", "display_county"], as_index=False)
    .agg(
        n_signal_tracts=("geoid", "nunique"),
        mean_scope_score=("scope_score", "mean"),
        median_scope_score=("scope_score", "median"),
        mean_igs_total=("igs_total", "mean"),
        min_igs_total=("igs_total", "min"),
        total_signal_pop=("pop_total", "sum"),
    )
    .copy()
)

county_signal = county_signal.loc[
    county_signal["n_signal_tracts"] >= MIN_SIGNAL_TRACTS_PER_COUNTY
].copy()

county_signal["county_seed_score"] = (
    0.40 * bad_when_high(county_signal["n_signal_tracts"]) +
    0.35 * bad_when_high(county_signal["mean_scope_score"]) +
    0.15 * bad_when_low(county_signal["mean_igs_total"]) +
    0.10 * bad_when_low(county_signal["min_igs_total"])
)

county_signal = county_signal.sort_values(
    ["county_seed_score", "n_signal_tracts", "mean_scope_score", "mean_igs_t
    ascending=[False, False, False, True]
).reset_index(drop=True)

seed_counties = diversified_seed_counties(
    county_signal_df=county_signal,
    target_n=TARGET_SEED_COUNTIES_N,
    per_state_base=PER_STATE_BASE_COUNTIES,
    per_state_max=PER_STATE_MAX_COUNTIES,
)

seed_counties = (
    seed_counties
    .merge(
        county_signal[
            ["display_state", "display_county", "n_signal_tracts", "mean_sco
        ],
        on=["display_state", "display_county"],
        how="left",
```

```
        validate="1:1"
    )
    .sort_values(["county_seed_score", "n_signal_tracts", "mean_scope_score"
    .reset_index(drop=True)
)

# ---------------------------------------------------
# C. Expand within selected counties
# ---------------------------------------------------
broader_county_pool = (
    tract_recovery
    .merge(
        seed_counties[["display_state", "display_county"]],
        on=["display_state", "display_county"],
        how="inner"
    )
    .copy()
)

broader_county_pool["county_rank_by_score"] = (
    broader_county_pool
    .groupby(["display_state", "display_county"])["scope_score"]
    .rank(method="first", ascending=False)
)

core_scope_df = (
    broader_county_pool
    .loc[broader_county_pool["county_rank_by_score"] <= BROADER_MAX_PER_COUN
    .drop(columns=["county_rank_by_score"])
    .drop_duplicates(subset=["geoid"])
    .sort_values(["display_state", "display_county", "scope_score"], ascendi
    .reset_index(drop=True)
)

# ---------------------------------------------------
# D. Load geometry only for selected states
# ---------------------------------------------------
needed_state_fips = sorted(
    core_scope_df["geoid"].astype(str).str[:2].dropna().unique().tolist()
)

tract_gdf = load_state_tract_geometry(needed_state_fips)

tract_geo_scored = tract_gdf.merge(
    tract_recovery,
    on="geoid",
    how="inner",
    validate="1:1"
)

seed_county_gdf = (
    tract_geo_scored
    .merge(
        seed_counties[["display_state", "display_county"]],
        on=["display_state", "display_county"],
        how="inner"
```

```
    )
    .copy()
)

seed_county_gdf = seed_county_gdf[seed_county_gdf.geometry.notna()].copy()
seed_county_gdf = seed_county_gdf[~seed_county_gdf.geometry.is_empty].copy()

# ----------------------------------------------------
# E. Add one-hop geographic connector tracts
# ----------------------------------------------------
full_seed_edges = build_adjacency_edges_fast(seed_county_gdf)

core_nodes = set(core_scope_df["geoid"].astype(str).tolist())
expanded_nodes = set(core_nodes)

for _ in range(CONNECTOR_HOPS):
    if full_seed_edges.empty:
        break

    touching_core = full_seed_edges[
        full_seed_edges["geoid_1"].astype(str).isin(expanded_nodes) |
        full_seed_edges["geoid_2"].astype(str).isin(expanded_nodes)
    ].copy()

    neighbor_nodes = set(touching_core["geoid_1"].astype(str).tolist()) | se
        touching_core["geoid_2"].astype(str).tolist()
    )

    expanded_nodes = expanded_nodes | neighbor_nodes

CLUSTER_SCOPE_DF = (
    seed_county_gdf.loc[seed_county_gdf["geoid"].astype(str).isin(expanded_n
    .drop_duplicates(subset=["geoid"])
    .sort_values(["display_state", "display_county", "scope_score"], ascendi
    .reset_index(drop=True)
)

adjacency_gdf = CLUSTER_SCOPE_DF.copy()

adjacency_edges = full_seed_edges.loc[
    full_seed_edges["geoid_1"].astype(str).isin(set(adjacency_gdf["geoid"].a
    full_seed_edges["geoid_2"].astype(str).isin(set(adjacency_gdf["geoid"].a
].copy()

# ----------------------------------------------------
# F. Connected components on broadened scope
# ----------------------------------------------------
all_candidate_nodes = adjacency_gdf["geoid"].astype(str).unique().tolist()
node_to_cluster_num = connected_components_from_edges(all_candidate_nodes, a

adjacency_gdf["cluster_id_num"] = adjacency_gdf["geoid"].astype(str).map(noc
adjacency_gdf["cluster_id"] = (
    adjacency_gdf["display_state"].astype(str)
    + " | "
    + adjacency_gdf["display_county"].astype(str)
    + " | cluster_"
```

```
    + adjacency_gdf["cluster_id_num"].astype(str)
)

tract_cluster_map = (
    adjacency_gdf[
        ["geoid", "cluster_id", "display_state", "display_county", "display_
    ]
    .drop_duplicates(subset=["geoid"])
    .copy()
)

tract_cluster_map["n_cluster_tracts"] = (
    tract_cluster_map.groupby("cluster_id")["geoid"].transform("nunique")
)

# ---------------------------------------------------
# G. Save outputs
# ---------------------------------------------------
OUT_DIR.mkdir(parents=True, exist_ok=True)

tract_cluster_map.to_parquet(OUT_DIR / "tract_cluster_map.parquet", index=Fa
CLUSTER_SCOPE_DF.drop(columns="geometry").to_parquet(OUT_DIR / "cluster_scop
adjacency_gdf.drop(columns="geometry").to_parquet(OUT_DIR / "adjacency_clust
seed_counties.to_parquet(OUT_DIR / "seed_counties.parquet", index=False)
county_signal.to_parquet(OUT_DIR / "county_signal.parquet", index=False)

# ---------------------------------------------------
# H. Diagnostics
# ---------------------------------------------------
print("Recovery tract universe:", tract_recovery.shape)
print("Signal pool shape:", signal_pool.shape)
print("County signal shape:", county_signal.shape)
print("Seed counties selected:", seed_counties.shape[0])
print("Needed state FIPS:", needed_state_fips)
print()

print("Seed counties by state:")
print(seed_counties["display_state"].value_counts(dropna=False).sort_values(
print()

print("County-expanded core scope:", core_scope_df.shape)
print("Seed-county tract universe:", seed_county_gdf.shape)
print("Connector hops:", CONNECTOR_HOPS)
print("Final CLUSTER_SCOPE_DF shape:", CLUSTER_SCOPE_DF.shape)
print("Connector tracts added:", len(set(CLUSTER_SCOPE_DF["geoid"]) - set(co
print()

print("Adjacency merge shape:", adjacency_gdf.shape)
print("Adjacency edges shape:", adjacency_edges.shape)
print()

print("tract_cluster_map shape:", tract_cluster_map.shape)
print("Unique tracts mapped:", tract_cluster_map["geoid"].nunique())
print("Unique clusters:", tract_cluster_map["cluster_id"].nunique())
print("Unique states in cluster universe:", tract_cluster_map["display_state
print("Unique counties in cluster universe:", tract_cluster_map["display_cou
```

```
print()

print("Top 25 seed counties:")
print(
    seed_counties[
        ["display_state", "display_county", "n_signal_tracts", "mean_scope_s
    ].head(25)
)

print()
print("Top 25 clusters by size:")
print(
    tract_cluster_map.groupby(["cluster_id", "display_state", "display_count
    .agg(n_cluster_tracts=("geoid", "nunique"))
    .sort_values(["n_cluster_tracts", "cluster_id"], ascending=[False, True]
    .head(25)
)
```

```
Recovery tract universe: (84153, 14)
Signal pool shape: (1500, 14)
County signal shape: (190, 9)
Seed counties selected: 40
Needed state FIPS: ['01', '04', '05', '06', '11', '12', '13', '17', '18', '2
0', '21', '22', '24', '25', '26', '28', '29', '32', '34', '35', '36', '37',
'39', '40', '42', '45', '47', '48', '51', '54', '55', '72']

Seed counties by state:
display_state
Puerto Rico             3
Arizona                 3
New Mexico              2
Ohio                    2
Michigan                2
New York                2
Georgia                 1
Indiana                 1
Pennsylvania            1
Illinois                1
Tennessee               1
Wisconsin               1
California              1
Maryland                1
Louisiana               1
Alabama                 1
Missouri                1
Kentucky                1
West Virginia           1
Mississippi             1
New Jersey              1
District of Columbia    1
South Carolina          1
Florida                 1
North Carolina          1
Oklahoma                1
Arkansas                1
Texas                   1
Nevada                  1
Virginia                1
Massachusetts           1
Kansas                  1
Name: count, dtype: int64

County-expanded core scope: (1227, 14)
Seed-county tract universe: (11039, 28)
Connector hops: 1
Final CLUSTER_SCOPE_DF shape: (2942, 28)
Connector tracts added: 1715

Adjacency merge shape: (2942, 30)
Adjacency edges shape: (7518, 4)

tract_cluster_map shape: (2942, 6)
Unique tracts mapped: 2942
Unique clusters: 86
```

Unique states in cluster universe: 32
Unique counties in cluster universe: 40

Top 25 seed counties:

	display_state	display_county	n_signal_tracts	mean_scope_score	mean_igs_total	county_seed_score
0	Puerto Rico	Utuado Municipio	8	98.872748	24.520833	0.892895
1	Puerto Rico	Mayagüez Municipio	22	97.791283	28.303030	0.887895
2	Puerto Rico	Río Grande Municipio	8	97.771098	19.829167	0.875000
3	Arizona	Navajo County	7	98.028190	22.428571	0.868684
4	Arizona	Apache County	9	96.936950	23.703704	0.830526
5	Alabama	Mobile County	14	96.645521	24.611905	0.822105
6	Indiana	Lake County	14	96.568595	26.057143	0.803947
7	Ohio	Mahoning County	8	96.964295	28.541667	0.762895
8	Georgia	Bibb County	7	97.479408	28.452381	0.744737
9	New Mexico	McKinley County	9	96.417365	27.129630	0.744211
10	Pennsylvania	Allegheny County	11	96.098523	27.939394	0.730263
11	Michigan	Wayne County	74	95.904149	29.763964	0.724211
12	Illinois	Cook County	58	95.719054	27.893678	0.716842
13	Michigan	Genesee County	16	96.172425	29.562500	0.716053
14	Tennessee	Shelby County	26	95.753324	27.750000	0.714079
15	Ohio	Cuyahoga County	38	95.771998	30.440351	0.679737
16	New Mexico	San Juan County	5	96.112597	23.033333	0.670789
17	Wisconsin	Milwaukee County	22	95.326373	28.757576	0.635263
18	California	Los Angeles County	7	96.143938	28.828571	0.622368
19	Maryland	Baltimore city	17	95.596453	30.107843	0.621053
20	Louisiana	Orleans Parish	14	95.693053	29.959524	0.614737
21	Missouri	St. Louis city	8	95.473045	27.629167	0.612632
22	Arizona	Coconino County	2	97.468963	14.666667	0.598947
23	New York	Kings County	21	95.887047	32.341270	0.594737
24	New York	Monroe County	16	95.684108	31.525000	0.592895

```
Top 25 clusters by size:
                                          cluster_id        display_state
display_county  n_cluster_tracts
37               Michigan | Wayne County | cluster_38            Michigan
Wayne County              152
13        California | Los Angeles County | cluster_8          California
Los Angeles County              148
32            Louisiana | Orleans Parish | cluster_33           Louisiana
Orleans Parish              107
62                Ohio | Cuyahoga County | cluster_63                Ohio
Cuyahoga County              107
33             Maryland | Baltimore city | cluster_34            Maryland
Baltimore city              105
85          Wisconsin | Milwaukee County | cluster_86           Wisconsin
Milwaukee County               98
76             Tennessee | Shelby County | cluster_77           Tennessee
Shelby County               96
42                  Nevada | Clark County | cluster_43              Nevada
Clark County               95
23                Illinois | Cook County | cluster_24            Illinois
Cook County              85
0                 Alabama | Mobile County | cluster_1             Alabama
Mobile County               81
66               Oklahoma | Tulsa County | cluster_67            Oklahoma
Tulsa County               77
60              New York | Monroe County | cluster_61            New York
Monroe County               75
75      South Carolina | Richland County | cluster_76       South Carolina
Richland County               73
77                  Texas | Harris County | cluster_78               Texas
Harris County               72
36             Michigan | Genesee County | cluster_37            Michigan
Genesee County               71
29                  Indiana | Lake County | cluster_30             Indiana
Lake County              69
83              Virginia | Richmond city | cluster_84            Virginia
Richmond city               67
15  District of Columbia | District of Columbia | ...  District of Columbia
District of Columbia                65
34        Massachusetts | Hampden County | cluster_35        Massachusetts
Hampden County               64
68       Pennsylvania | Allegheny County | cluster_69         Pennsylvania
Allegheny County               62
52               New York | Kings County | cluster_53            New York
Kings County              58
30             Kansas | Wyandotte County | cluster_31              Kansas
Wyandotte County               54
17           Florida | Palm Beach County | cluster_18             Florida
Palm Beach County               52
64                Ohio | Mahoning County | cluster_65                Ohio
Mahoning County               52
46            New Jersey | Camden County | cluster_47          New Jersey
Camden County               50
```

Cell 3 — join tract panel to clusters and build a cluster-year panel

In [3]:
```
# Cell 3 — join tract panel to clusters and build a cluster-year panel

def safe_weighted_mean(values: pd.Series, weights: pd.Series):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")

    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return values.loc[valid].mean()

    return np.average(values.loc[mask], weights=weights.loc[mask])

clustered_panel = (
    analysis_panel
    .merge(
        tract_cluster_map[
            ["geoid", "cluster_id", "display_state", "display_county", "n_cl
        ],
        on="geoid",
        how="inner",
        validate="many_to_one",
        suffixes=("", "_cluster")
    )
    .copy()
)

# Prefer cluster metadata from the tract_cluster_map
clustered_panel["cluster_display_state"] = clustered_panel["display_state_cl
clustered_panel["cluster_display_county"] = clustered_panel["display_county_

# Keep only true cluster aggregation vars here
cluster_core_vars = [
    "igs_total",
    "igs_economy",
    "poverty_rate",
    "unemp_rate",
    "median_household_income",
    "lfpr_16p",
]

cluster_year_rows = []

for (cluster_id, year, period), grp in clustered_panel.groupby(
    ["cluster_id", "year", "analysis_period"],
    dropna=False
):
    igs_vals = pd.to_numeric(grp["igs_total"], errors="coerce")
    valid_igs = igs_vals.notna()
```

```
    row = {
        "cluster_id": cluster_id,
        "year": int(year),
        "analysis_period": period,
        "display_state": grp["cluster_display_state"].dropna().iloc[0] if gr
        "display_county": grp["cluster_display_county"].dropna().iloc[0] if
        "n_cluster_tracts": grp["n_cluster_tracts"].dropna().iloc[0] if grp[
        "n_observed_tracts_year": grp["geoid"].nunique(),
        "cluster_pop_total": pd.to_numeric(grp["pop_total"], errors="coerce"
        "low_igs_tract_share": (
            igs_vals.loc[valid_igs].lt(LOW_IGS_THRESHOLD).mean()
            if valid_igs.any() else np.nan
        ),
    }

    weights = pd.to_numeric(grp["pop_total"], errors="coerce")

    for col in cluster_core_vars:
        row[col] = safe_weighted_mean(grp[col], weights)

    cluster_year_rows.append(row)

cluster_year_panel = (
    pd.DataFrame(cluster_year_rows)
    .sort_values(["cluster_id", "year"])
    .reset_index(drop=True)
)

print("clustered_panel shape:", clustered_panel.shape)
print("cluster_year_panel shape:", cluster_year_panel.shape)
print("Unique clusters in panel:", cluster_year_panel["cluster_id"].nunique(
print("Years in cluster panel:", sorted(cluster_year_panel["year"].unique().
print()
print(cluster_year_panel.head(15))
```

```
clustered_panel shape: (23536, 19)
cluster_year_panel shape: (688, 15)
Unique clusters in panel: 86
Years in cluster panel: [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

                             cluster_id  year analysis_period display_state
display_county  n_cluster_tracts  n_observed_tracts_year  cluster_pop_total
low_igs_tract_share  igs_total  igs_economy  \
0   Alabama | Mobile County | cluster_1  2017      pre_covid      Alabama
Mobile County                81                      81          174074.0
0.777778  38.670840    40.815976
1   Alabama | Mobile County | cluster_1  2018      pre_covid      Alabama
Mobile County                81                      81          171909.0
0.753086  37.426682    39.276507
2   Alabama | Mobile County | cluster_1  2019      pre_covid      Alabama
Mobile County                81                      81          171726.0
0.814815  38.016493    40.396335
3   Alabama | Mobile County | cluster_1  2020          covid      Alabama
Mobile County                81                      81          215309.0
0.800000  40.528150    43.578496
4   Alabama | Mobile County | cluster_1  2021          covid      Alabama
Mobile County                81                      81          214402.0
0.787500  40.881874    42.742312
5   Alabama | Mobile County | cluster_1  2022       recovery      Alabama
Mobile County                81                      81          215200.0
0.800000  38.472657    40.568162
6   Alabama | Mobile County | cluster_1  2023       recovery      Alabama
Mobile County                81                      81          212436.0
0.737500  40.966475    39.426842
7   Alabama | Mobile County | cluster_1  2024       recovery      Alabama
Mobile County                81                      81          210593.0
0.775000  40.376465    41.524483
8   Alabama | Mobile County | cluster_2  2017      pre_covid      Alabama
Mobile County                 6                       6           21773.0
0.833333  44.995361    44.875378
9   Alabama | Mobile County | cluster_2  2018      pre_covid      Alabama
Mobile County                 6                       6           22780.0
0.833333  44.957028    47.489776
10  Alabama | Mobile County | cluster_2  2019      pre_covid      Alabama
Mobile County                 6                       6           21366.0
0.666667  45.581859    48.723205
11  Alabama | Mobile County | cluster_2  2020          covid      Alabama
Mobile County                 6                       6           21653.0
0.500000  45.797626    49.527225
12  Alabama | Mobile County | cluster_2  2021          covid      Alabama
Mobile County                 6                       6           21498.0
0.500000  48.269281    50.113555
13  Alabama | Mobile County | cluster_2  2022       recovery      Alabama
Mobile County                 6                       6           21350.0
0.666667  42.940422    46.413653
14  Alabama | Mobile County | cluster_2  2023       recovery      Alabama
Mobile County                 6                       6           20875.0
0.333333  46.593054    49.577102

    poverty_rate  unemp_rate  median_household_income  lfpr_16p
0       0.271643    0.102214             37303.201328  0.522255
```

1	0.266367	0.088608	38388.685427	0.513352
2	0.247494	0.080236	39726.689657	0.527097
3	0.221165	0.074494	45369.758710	0.534292
4	0.215452	0.072882	47611.121599	0.531995
5	0.220681	0.075142	51866.762718	0.531486
6	0.202819	0.065167	55668.732036	0.536394
7	0.202395	0.071801	56418.745776	0.540548
8	0.191061	0.063831	43784.371331	0.621807
9	0.175418	0.055939	44464.372739	0.593021
10	0.182298	0.056576	45716.814518	0.580650
11	0.160392	0.051975	47756.066411	0.564805
12	0.154294	0.050258	48741.349707	0.567284
13	0.178421	0.051287	53507.745340	0.577691
14	0.180455	0.052773	52358.897820	0.595316

In [ ]:

In [ ]:

In [ ]:

Cell 4 — build period-level summaries and transparent trajectory features

In [4]:

```
# Cell 4 — build period-level summaries and transparent trajectory features

PERIOD_ORDER = ["pre_covid", "covid", "recovery"]

# ----------------------------
# Period-level aggregation
# ----------------------------
period_metric_cols = [
    "cluster_pop_total",
    "low_igs_tract_share",
    "igs_total",
    "igs_economy",
    "poverty_rate",
    "unemp_rate",
    "median_household_income",
    "lfpr_16p",
]

agg_dict = {
    "display_state": "first",
    "display_county": "first",
    "n_cluster_tracts": "max",
    "cluster_pop_total": "mean",
    "low_igs_tract_share": "mean",
    "igs_total": "mean",
    "igs_economy": "mean",
    "poverty_rate": "mean",
    "unemp_rate": "mean",
    "median_household_income": "mean",
    "lfpr_16p": "mean",
```

```
}

# optional diagnostic if Cell 3 includes this column
if "n_observed_tracts_year" in cluster_year_panel.columns:
    agg_dict["n_observed_tracts_year"] = "min"

cluster_period_panel = (
    cluster_year_panel
    .groupby(["cluster_id", "analysis_period"], as_index=False)
    .agg(agg_dict)
    .copy()
)

cluster_period_panel["analysis_period"] = pd.Categorical(
    cluster_period_panel["analysis_period"],
    categories=PERIOD_ORDER,
    ordered=True
)

cluster_period_panel = cluster_period_panel.sort_values(
    ["cluster_id", "analysis_period"]
).reset_index(drop=True)

# ----------------------------
# Stable cluster metadata
# ----------------------------
latest_cluster_meta = (
    cluster_year_panel
    .sort_values(["cluster_id", "year"])
    .groupby("cluster_id", as_index=False)
    .tail(1)[["cluster_id", "display_state", "display_county", "n_cluster_tr
    .rename(columns={"cluster_pop_total": "latest_cluster_pop_total"})
    .copy()
)

# ----------------------------
# Wide period summary
# Only pivot true period-varying metrics
# ----------------------------
cluster_period_wide = (
    cluster_period_panel
    .pivot(index="cluster_id", columns="analysis_period", values=period_metr
)

expected_cols = pd.MultiIndex.from_product(
    [period_metric_cols, PERIOD_ORDER],
    names=["metric", "analysis_period"]
)
cluster_period_wide = cluster_period_wide.reindex(columns=expected_cols)

cluster_period_wide.columns = [
    f"{metric}_{period}" for metric, period in cluster_period_wide.columns
]
cluster_period_wide = cluster_period_wide.reset_index()

cluster_period_summary = (
```

```python
    latest_cluster_meta
    .merge(cluster_period_wide, on="cluster_id", how="left", validate="1:1")
    .copy()
)

# ------------------------------
# Period completeness diagnostics
# ------------------------------
cluster_period_summary["has_pre_covid"] = cluster_period_summary["igs_total_
cluster_period_summary["has_covid"] = cluster_period_summary["igs_total_covi
cluster_period_summary["has_recovery"] = cluster_period_summary["igs_total_r

cluster_period_summary["complete_periods_flag"] = (
    cluster_period_summary["has_pre_covid"] &
    cluster_period_summary["has_covid"] &
    cluster_period_summary["has_recovery"]
)

# ------------------------------
# Trajectory deltas
# Positive shock metrics = worse COVID hit
# Positive rebound metrics = stronger recovery
# ------------------------------

# Lower is worse: igs_total, igs_economy
cluster_period_summary["igs_total_drop_pre_to_covid"] = (
    cluster_period_summary["igs_total_pre_covid"] - cluster_period_summary["
)
cluster_period_summary["igs_economy_drop_pre_to_covid"] = (
    cluster_period_summary["igs_economy_pre_covid"] - cluster_period_summary
)
cluster_period_summary["igs_total_rebound_covid_to_recovery"] = (
    cluster_period_summary["igs_total_recovery"] - cluster_period_summary["i
)
cluster_period_summary["igs_economy_rebound_covid_to_recovery"] = (
    cluster_period_summary["igs_economy_recovery"] - cluster_period_summary[
)

# Higher is worse: poverty_rate, unemp_rate
cluster_period_summary["poverty_increase_pre_to_covid"] = (
    cluster_period_summary["poverty_rate_covid"] - cluster_period_summary["p
)
cluster_period_summary["unemp_increase_pre_to_covid"] = (
    cluster_period_summary["unemp_rate_covid"] - cluster_period_summary["une
)
cluster_period_summary["poverty_improve_covid_to_recovery"] = (
    cluster_period_summary["poverty_rate_covid"] - cluster_period_summary["p
)
cluster_period_summary["unemp_improve_covid_to_recovery"] = (
    cluster_period_summary["unemp_rate_covid"] - cluster_period_summary["une
)

# Higher is better: income, labor-force participation
cluster_period_summary["income_drop_pre_to_covid"] = (
    cluster_period_summary["median_household_income_pre_covid"] - cluster_pe
)
```

```
cluster_period_summary["lfpr_drop_pre_to_covid"] = (
    cluster_period_summary["lfpr_16p_pre_covid"] - cluster_period_summary["l
)
cluster_period_summary["income_rebound_covid_to_recovery"] = (
    cluster_period_summary["median_household_income_recovery"] - cluster_per
)
cluster_period_summary["lfpr_rebound_covid_to_recovery"] = (
    cluster_period_summary["lfpr_16p_recovery"] - cluster_period_summary["lf
)

# ----------------------------
# Structural filters
# ----------------------------
cluster_period_summary["serious_cluster_flag"] = (
    cluster_period_summary["complete_periods_flag"] &
    (cluster_period_summary["n_cluster_tracts"] >= MIN_CLUSTER_TRACTS) &
    (cluster_period_summary["latest_cluster_pop_total"] >= MIN_CLUSTER_POP)
)

cluster_period_summary["low_igs_recovery_flag"] = (
    cluster_period_summary["igs_total_recovery"] < LOW_IGS_THRESHOLD
)

cluster_period_summary["low_igs_economy_recovery_flag"] = (
    cluster_period_summary["igs_economy_recovery"] < LOW_IGS_THRESHOLD
)

# ----------------------------
# Diagnostics
# ----------------------------
print("cluster_period_panel shape:", cluster_period_panel.shape)
print("cluster_period_summary shape:", cluster_period_summary.shape)
print()
print("Complete-period cluster count:", int(cluster_period_summary["complete
print("Serious cluster count:", int(cluster_period_summary["serious_cluster_
print("Recovery low-IGS cluster count:", int(cluster_period_summary["low_igs
print()
print(
    cluster_period_summary[
        [
            "cluster_id",
            "display_state",
            "display_county",
            "n_cluster_tracts",
            "latest_cluster_pop_total",
            "igs_total_pre_covid",
            "igs_total_covid",
            "igs_total_recovery",
            "poverty_rate_pre_covid",
            "poverty_rate_covid",
            "poverty_rate_recovery",
            "complete_periods_flag",
            "serious_cluster_flag",
            "low_igs_recovery_flag",
        ]
```

```
    ].head(15)
)
```

```
cluster_period_panel shape: (258, 14)
cluster_period_summary shape: (86, 48)

Complete-period cluster count: 86
Serious cluster count: 86
Recovery low-IGS cluster count: 67

                                      cluster_id display_state      display_
county  n_cluster_tracts  latest_cluster_pop_total  igs_total_pre_covid  igs
_total_covid  igs_total_recovery  \
0            Alabama | Mobile County | cluster_1       Alabama       Mobile
County              81                  210593.0           38.038005
40.705012           39.938532
1            Alabama | Mobile County | cluster_2       Alabama       Mobile
County               6                   21037.0           45.178083
47.033454           44.644035
2            Arizona | Apache County | cluster_3       Arizona       Apache
County              18                   65341.0           31.620153
34.649302           34.156029
3          Arizona | Coconino County | cluster_4       Arizona     Coconino
County              39                  144508.0           45.213663
50.750484           48.586877
4            Arizona | Navajo County | cluster_5       Arizona       Navajo
County              37                  108415.0           36.375067
42.065258           42.164538
5           Arkansas | Chicot County | cluster_6      Arkansas       Chicot
County               4                    9765.0           40.367679
36.681542           36.814811
6   California | Los Angeles County | cluster_10    California  Los Angeles
County               7                   29599.0           40.173054
42.128379           42.515741
7   California | Los Angeles County | cluster_11    California  Los Angeles
County               5                   19914.0           35.988973
36.954608           35.686555
8   California | Los Angeles County | cluster_12    California  Los Angeles
County               8                   30308.0           33.866214
38.305995           39.672431
9   California | Los Angeles County | cluster_13    California  Los Angeles
County              20                   77992.0           36.795463
37.874687           41.582088
10  California | Los Angeles County | cluster_14    California  Los Angeles
County              10                   30902.0           40.336142
38.719994           39.750194
11  California | Los Angeles County | cluster_15    California  Los Angeles
County               6                   25714.0           43.767486
42.820748           41.891170
12   California | Los Angeles County | cluster_7    California  Los Angeles
County               5                   26371.0           32.608527
34.514928           38.426331
13   California | Los Angeles County | cluster_8    California  Los Angeles
County             148                  569438.0           34.284506
35.928396           36.441923
14   California | Los Angeles County | cluster_9    California  Los Angeles
County               6                   25642.0           38.802970
41.999744           41.251049
```

```
    poverty_rate_pre_covid  poverty_rate_covid  poverty_rate_recovery  compl
ete_periods_flag  serious_cluster_flag  low_igs_recovery_flag
0                 0.261835            0.218308               0.208632
True                  True                   True
1                 0.182926            0.157343               0.176671
True                  True                   True
2                 0.367412            0.341306               0.313574
True                  True                   True
3                 0.280143            0.214152               0.206512
True                  True                  False
4                 0.320524            0.252731               0.247988
True                  True                   True
5                 0.286492            0.287834               0.246815
True                  True                   True
6                 0.301040            0.254905               0.225949
True                  True                   True
7                 0.261123            0.273549               0.257480
True                  True                   True
8                 0.318769            0.260076               0.236495
True                  True                   True
9                 0.280843            0.213290               0.187341
True                  True                   True
10                0.261532            0.217998               0.209009
True                  True                   True
11                0.141211            0.160974               0.183929
True                  True                   True
12                0.721659            0.644663               0.613950
True                  True                   True
13                0.308654            0.267020               0.251133
True                  True                   True
14                0.203908            0.147500               0.150510
True                  True                   True
```

Cell 5 — practical shortlist logic for the new method

In [5]:

```python
# Cell 5 — practical shortlist logic for the new method

def bad_when_high(series: pd.Series) -> pd.Series:
    s = pd.to_numeric(series, errors="coerce")
    return s.rank(pct=True, method="average")

def bad_when_low(series: pd.Series) -> pd.Series:
    s = pd.to_numeric(series, errors="coerce")
    return (-s).rank(pct=True, method="average")

cluster_scored = cluster_period_summary.copy()

# -----------------------------------
# Scoring universe
# Score only clusters that are structurally usable
# Then apply final low-IGS shortlist filter afterward
# -----------------------------------
cluster_scored["eligible_for_scoring"] = (
    cluster_scored["complete_periods_flag"].fillna(False) &
    cluster_scored["serious_cluster_flag"].fillna(False)
```

```python
)

score_mask = cluster_scored["eligible_for_scoring"]
score_df = cluster_scored.loc[score_mask].copy()

# ------------------------------------
# Shock variables:
# Positive values = adverse COVID shock
# Negative values are clipped to 0 so they do not behave like "anti-shock"
# ------------------------------------
shock_cols = [
    "igs_total_drop_pre_to_covid",
    "igs_economy_drop_pre_to_covid",
    "poverty_increase_pre_to_covid",
    "unemp_increase_pre_to_covid",
    "income_drop_pre_to_covid",
    "lfpr_drop_pre_to_covid",
]

for col in shock_cols:
    score_df[f"{col}_shock_only"] = pd.to_numeric(score_df[col], errors="coe

# ------------------------------------
# Weak-recovery shortfall variables:
# Higher = weaker recovery
# ------------------------------------
score_df["weakrec_igs_total"] = -pd.to_numeric(score_df["igs_total_rebound_c
score_df["weakrec_igs_economy"] = -pd.to_numeric(score_df["igs_economy_rebou
score_df["weakrec_poverty"] = -pd.to_numeric(score_df["poverty_improve_covic
score_df["weakrec_unemp"] = -pd.to_numeric(score_df["unemp_improve_covid_to_
score_df["weakrec_income"] = -pd.to_numeric(score_df["income_rebound_covid_t
score_df["weakrec_lfpr"] = -pd.to_numeric(score_df["lfpr_rebound_covid_to_re

# ------------------------------------
# 1) Still low on IGS in recovery
# Use overall IGS + share of tracts below the IGS threshold
# This avoids double-counting igs_economy here
# ------------------------------------
score_df["score_low_igs_recovery"] = (
    bad_when_low(score_df["igs_total_recovery"]) +
    bad_when_high(score_df["low_igs_tract_share_recovery"])
) / 2

# ------------------------------------
# 2) Economically vulnerable in recovery
# Keep igs_economy only here, as the economy anchor + ACS support
# ------------------------------------
score_df["score_econ_vulnerability_recovery"] = (
    bad_when_low(score_df["igs_economy_recovery"]) +
    bad_when_high(score_df["poverty_rate_recovery"]) +
    bad_when_high(score_df["unemp_rate_recovery"]) +
    bad_when_low(score_df["median_household_income_recovery"]) +
    bad_when_low(score_df["lfpr_16p_recovery"])
) / 5

# ------------------------------------
```

```python
# 3) Harder COVID shock relative to pre-COVID baseline
# ------------------------------------
score_df["score_covid_shock"] = (
    bad_when_high(score_df["igs_total_drop_pre_to_covid_shock_only"]) +
    bad_when_high(score_df["igs_economy_drop_pre_to_covid_shock_only"]) +
    bad_when_high(score_df["poverty_increase_pre_to_covid_shock_only"]) +
    bad_when_high(score_df["unemp_increase_pre_to_covid_shock_only"]) +
    bad_when_high(score_df["income_drop_pre_to_covid_shock_only"]) +
    bad_when_high(score_df["lfpr_drop_pre_to_covid_shock_only"])
) / 6

# ------------------------------------
# 4) Weak recovery since COVID
# ------------------------------------
score_df["score_weak_recovery"] = (
    bad_when_high(score_df["weakrec_igs_total"]) +
    bad_when_high(score_df["weakrec_igs_economy"]) +
    bad_when_high(score_df["weakrec_poverty"]) +
    bad_when_high(score_df["weakrec_unemp"]) +
    bad_when_high(score_df["weakrec_income"]) +
    bad_when_high(score_df["weakrec_lfpr"])
) / 6

# ------------------------------------
# Transparent final score
# Recovery status + recovery vulnerability remain the foundation
# ------------------------------------
score_df["cluster_shortlist_score"] = (
    0.35 * score_df["score_low_igs_recovery"] +
    0.35 * score_df["score_econ_vulnerability_recovery"] +
    0.15 * score_df["score_covid_shock"] +
    0.15 * score_df["score_weak_recovery"]
)

# Write scores back into the full cluster frame
score_cols = [
    "score_low_igs_recovery",
    "score_econ_vulnerability_recovery",
    "score_covid_shock",
    "score_weak_recovery",
    "cluster_shortlist_score",
]
cluster_scored = cluster_scored.merge(
    score_df[["cluster_id"] + score_cols],
    on="cluster_id",
    how="left",
    validate="1:1",
)

# ------------------------------------
# Final practical screen for enrichment candidates
# ------------------------------------
cluster_shortlist = (
    cluster_scored
    .loc[
        cluster_scored["eligible_for_scoring"] &
```

```
        cluster_scored["low_igs_recovery_flag"]
    ]
    .sort_values(
        [
            "cluster_shortlist_score",
            "igs_total_recovery",
            "igs_economy_recovery",
            "latest_cluster_pop_total",
        ],
        ascending=[False, True, True, False]
    )
    .reset_index(drop=True)
)

cluster_shortlist["rank"] = np.arange(1, len(cluster_shortlist) + 1)

shortlist_view_cols = [
    "rank",
    "cluster_id",
    "display_state",
    "display_county",
    "n_cluster_tracts",
    "latest_cluster_pop_total",
    "igs_total_pre_covid",
    "igs_total_covid",
    "igs_total_recovery",
    "low_igs_tract_share_recovery",
    "igs_economy_recovery",
    "poverty_rate_recovery",
    "unemp_rate_recovery",
    "median_household_income_recovery",
    "lfpr_16p_recovery",
    "score_low_igs_recovery",
    "score_econ_vulnerability_recovery",
    "score_covid_shock",
    "score_weak_recovery",
    "cluster_shortlist_score",
]

print("Scoring universe shape:", score_df.shape)
print("Final shortlist shape:", cluster_shortlist.shape)
print()
print(cluster_shortlist[shortlist_view_cols].head(20))

# Optional: persist key outputs for later healthcare + business enrichment
OUT_DIR.mkdir(parents=True, exist_ok=True)
cluster_year_panel.to_parquet(OUT_DIR / "cluster_year_panel.parquet", index=
cluster_period_summary.to_parquet(OUT_DIR / "cluster_period_summary.parquet"
cluster_shortlist.to_parquet(OUT_DIR / "cluster_shortlist.parquet", index=Fa

print()
print("Saved:")
print("-", OUT_DIR / "cluster_year_panel.parquet")
print("-", OUT_DIR / "cluster_period_summary.parquet")
print("-", OUT_DIR / "cluster_shortlist.parquet")
```

```
Scoring universe shape: (86, 66)
Final shortlist shape: (67, 55)

    rank                                       cluster_id   display_state
display_county  n_cluster_tracts  latest_cluster_pop_total  igs_total_pre_co
vid  igs_total_covid  igs_total_recovery  \
0      1      Puerto Rico | Utuado Municipio | cluster_75     Puerto Rico
Utuado Municipio                8                   27621.0           27.5
37654       28.321111          25.035923
1      2    Puerto Rico | Mayagüez Municipio | cluster_73     Puerto Rico
Mayagüez Municipio               25                   70828.0           3
2.572527       33.746456          31.136263
2      3  Puerto Rico | Río Grande Municipio | cluster_74     Puerto Rico  R
ío Grande Municipio               12                   46051.0           2
7.542052       25.692985          25.284481
3      4              Ohio | Cuyahoga County | cluster_64            Ohio
Cuyahoga County               13                   31886.0           33.47
4823       36.792429          34.225025
4      5             Michigan | Wayne County | cluster_38        Michigan
Wayne County              152                  319984.0           29.91185
6       33.667928          34.793412
5      6              Illinois | Cook County | cluster_29        Illinois
Cook County                9                   27412.0           32.908060
32.684606          30.615724
6      7              Arizona | Apache County | cluster_3         Arizona
Apache County               18                   65341.0           31.6201
53       34.649302          34.156029
7      8              Illinois | Cook County | cluster_25        Illinois
Cook County               33                   81348.0           29.989644
33.926272          34.120468
8      9             Michigan | Wayne County | cluster_39        Michigan
Wayne County                9                   19947.0           27.93307
8       33.327025          29.825030
9     10              Illinois | Cook County | cluster_24        Illinois
Cook County               85                  187702.0           30.537359
34.481588          34.342235
10    11           Tennessee | Shelby County | cluster_77       Tennessee
Shelby County               96                  268298.0           35.9003
91       36.717131          35.813056
11    12        New Mexico | McKinley County | cluster_49      New Mexico
McKinley County               33                   70431.0           41.88
5820       39.464497          36.970752
12    13     California | Los Angeles County | cluster_11      California
Los Angeles County                5                   19914.0           3
5.988973       36.954608          35.686555
13    14              Ohio | Cuyahoga County | cluster_63            Ohio
Cuyahoga County              107                  210162.0           35.09
8325       36.916820          38.303060
14    15        Wisconsin | Milwaukee County | cluster_86       Wisconsin
Milwaukee County               98                  225339.0           34.1
06953       35.186728          35.715526
15    16          Louisiana | Orleans Parish | cluster_33       Louisiana
Orleans Parish              107                  218876.0           39.817
413       38.444589          37.108224
16    17             New York | Kings County | cluster_51        New York
Kings County               15                   55602.0           39.08708
```

```
1        39.677277           39.203114
17    18             New York | Kings County | cluster_53        New York
Kings County               58                  213577.0           34.98004
1        36.149592           37.669346
18    19     North Carolina | Robeson County | cluster_62  North Carolina
Robeson County               35                  116902.0           36.956
487        38.253102           35.994049
19    20      California | Los Angeles County | cluster_8      California
Los Angeles County              148                  569438.0           3
4.284506        35.928396           36.441923

    low_igs_tract_share_recovery  igs_economy_recovery  poverty_rate_recover
y  unemp_rate_recovery  median_household_income_recovery  lfpr_16p_recovery
score_low_igs_recovery  \
0                       1.000000             15.289723              0.52533
2             0.264035                      18417.685903           0.408071
0.985465
1                       1.000000             19.475426              0.53221
0             0.229062                      19229.673889           0.415725
0.962209
2                       1.000000             17.443243              0.35665
4             0.165771                      28512.143909           0.525166
0.979651
3                       1.000000             31.398833              0.37025
5             0.136437                      36960.197388           0.590425
0.944767
4                       0.951435             29.645510              0.34875
3             0.161412                      37886.810012           0.516598
0.906977
5                       1.000000             34.163963              0.23487
4             0.166658                      51537.524910           0.543750
0.968023
6                       0.851852             28.709013              0.31357
4             0.099555                      41738.341598           0.407386
0.831395
7                       0.959596             31.288560              0.30416
7             0.149948                      43383.006594           0.586529
0.936047
8                       1.000000             27.539859              0.29727
0             0.092034                      41703.874646           0.550155
0.973837
9                       0.900794             29.945366              0.30052
4             0.165977                      46121.632254           0.577286
0.883721
10                      0.896057             35.120633              0.30034
8             0.118143                      43461.126364           0.582458
0.848837
11                      0.757576             37.724026              0.34116
0             0.098756                      45567.496445           0.484736
0.691860
12                      0.933333             35.223650              0.25748
0             0.098351                      56430.097238           0.617167
0.895349
13                      0.852201             31.909575              0.32958
2             0.141196                      38280.580628           0.551952
0.726744
```

```
14                      0.880952           35.118704               0.32114
5             0.091170                    40889.939837           0.584413
0.837209
15                      0.841270           35.605207               0.27208
0             0.104481                    49657.707522           0.582639
0.750000
16                      0.755556           32.280068               0.29726
4             0.144929                    49551.621552           0.480014
0.604651
17                      0.870283           32.004432               0.28324
5             0.124499                    52456.969477           0.559224
0.761628
18                      0.904762           43.200898               0.28025
8             0.062075                    40796.621709           0.509912
0.863372
19                      0.918919           35.197972               0.25113
3             0.099923                    54797.541906           0.611680
0.866279

    score_econ_vulnerability_recovery  score_covid_shock  score_weak_recover
y  cluster_shortlist_score
0                            0.990698           0.645349               0.54263
6                 0.869855
1                            0.981395           0.515504               0.60658
9                 0.848576
2                            0.939535           0.466085               0.53294
6                 0.821570
3                            0.837209           0.438953               0.81007
8                 0.811047
4                            0.923256           0.444767               0.43604
7                 0.772703
5                            0.734884           0.543605               0.49806
2                 0.752267
6                            0.879070           0.448643               0.50775
2                 0.742122
7                            0.809302           0.450581               0.37403
1                 0.734564
8                            0.795349           0.373062               0.38759
7                 0.733314
9                            0.825581           0.373062               0.38759
7                 0.712355
10                           0.765116           0.440891               0.54069
8                 0.712122
11                           0.779070           0.595930               0.67829
5                 0.705959
12                           0.558140           0.577519               0.73449
6                 0.705523
13                           0.867442           0.452519               0.48255
8                 0.698227
14                           0.730233           0.479651               0.43023
3                 0.685087
15                           0.681395           0.679264               0.53100
8                 0.682529
16                           0.827907           0.471899               0.69961
2                 0.677122
17                           0.755814           0.508721               0.35271
```

```
3                  0.660320
18                             0.602326          0.373062            0.58720
9                  0.657035
19                             0.583721          0.442829            0.45155
0                  0.641657

Saved:
- data\processed\cluster_year_panel.parquet
- data\processed\cluster_period_summary.parquet
- data\processed\cluster_shortlist.parquet
```

Cell 6 — prepare enrichment-ready outputs for healthcare + business data

In [6]:

```
# Cell 6 — prepare enrichment-ready outputs from the main shortlist
# IMPORTANT:
# This cell stays source-agnostic.
# It packages the shortlist for later healthcare + business enrichment only.

ENRICHMENT_TOP_K = min(15, len(cluster_shortlist))

# -------------------------------
# A. pick the top clusters to enrich next
# -------------------------------
enrichment_clusters = (
    cluster_shortlist
    .head(ENRICHMENT_TOP_K)
    .copy()
    .reset_index(drop=True)
)

enrichment_clusters["enrichment_rank"] = np.arange(1, len(enrichment_cluster
enrichment_cluster_ids = enrichment_clusters["cluster_id"].astype(str).tolis

print(f"Selected top {len(enrichment_clusters)} clusters for next-step enric
print(
    enrichment_clusters[
        [
            "enrichment_rank",
            "rank",
            "cluster_id",
            "display_state",
            "display_county",
            "n_cluster_tracts",
            "latest_cluster_pop_total",
            "igs_total_recovery",
            "igs_economy_recovery",
            "poverty_rate_recovery",
            "unemp_rate_recovery",
            "median_household_income_recovery",
            "lfpr_16p_recovery",
            "cluster_shortlist_score",
        ]
    ].head(15)
)
```

```
# useful lookup for downstream merges
enrichment_rank_lookup = enrichment_clusters[
    ["cluster_id", "enrichment_rank", "rank", "cluster_shortlist_score"]
].drop_duplicates().copy()

# ----------------------------
# B. tract-level lookup table for external joins
# ----------------------------
enrichment_cluster_tracts = (
    tract_cluster_map
    .loc[tract_cluster_map["cluster_id"].astype(str).isin(enrichment_cluster
    .copy()
)

enrichment_cluster_tracts["geoid"] = normalize_geoid_series(enrichment_clust
enrichment_cluster_tracts = enrichment_cluster_tracts.dropna(subset=["geoid"

enrichment_cluster_tracts["state_fips"] = enrichment_cluster_tracts["geoid"]
enrichment_cluster_tracts["county_fips"] = enrichment_cluster_tracts["geoid"
enrichment_cluster_tracts["tract_code"] = enrichment_cluster_tracts["geoid"]

enrichment_cluster_tracts = (
    enrichment_cluster_tracts
    .merge(enrichment_rank_lookup, on="cluster_id", how="left", validate="ma
    .sort_values(["enrichment_rank", "cluster_id", "geoid"])
    .drop_duplicates(subset=["cluster_id", "geoid"])
    .reset_index(drop=True)
)

# ----------------------------
# C. county-level lookup table for external county joins
# ----------------------------
enrichment_cluster_counties = (
    enrichment_cluster_tracts
    .groupby(
        ["cluster_id", "enrichment_rank", "display_state", "display_county",
        as_index=False
    )
    .agg(
        n_cluster_tracts_in_county=("geoid", "nunique"),
        tract_list=("geoid", lambda s: sorted(set(s.astype(str))))
    )
    .sort_values(["enrichment_rank", "cluster_id", "county_fips"])
    .reset_index(drop=True)
)

cluster_county_counts = (
    enrichment_cluster_counties
    .groupby("cluster_id", as_index=False)
    .agg(n_cluster_counties=("county_fips", "nunique"))
)

# ----------------------------
# D. keep the full year-by-year history for shortlisted clusters
# ----------------------------
```

```python
enrichment_cluster_year_panel = (
    cluster_year_panel
    .loc[cluster_year_panel["cluster_id"].astype(str).isin(enrichment_cluste
    .copy()
    .merge(enrichment_rank_lookup, on="cluster_id", how="left", validate="ma
    .sort_values(["enrichment_rank", "cluster_id", "year"])
    .reset_index(drop=True)
)

# ----------------------------
# E. tract-year panel for later tract/county joins
# ----------------------------
enrichment_tract_year_panel = (
    clustered_panel
    .loc[clustered_panel["cluster_id"].astype(str).isin(enrichment_cluster_i
    .copy()
)

enrichment_tract_year_panel["geoid"] = normalize_geoid_series(enrichment_tra
enrichment_tract_year_panel = enrichment_tract_year_panel.dropna(subset=["ge

enrichment_tract_year_panel["state_fips"] = enrichment_tract_year_panel["geo
enrichment_tract_year_panel["county_fips"] = enrichment_tract_year_panel["ge
enrichment_tract_year_panel["tract_code"] = enrichment_tract_year_panel["geo

enrichment_tract_year_panel = (
    enrichment_tract_year_panel
    .merge(enrichment_rank_lookup, on="cluster_id", how="left", validate="ma
    .sort_values(["enrichment_rank", "cluster_id", "geoid", "year"])
    .drop_duplicates(subset=["cluster_id", "geoid", "year"])
    .reset_index(drop=True)
)

# ----------------------------
# F. profile table
# ----------------------------
enrichment_cluster_profiles = (
    enrichment_clusters[
        [
            "enrichment_rank",
            "rank",
            "cluster_id",
            "display_state",
            "display_county",
            "n_cluster_tracts",
            "latest_cluster_pop_total",
            "igs_total_pre_covid",
            "igs_total_covid",
            "igs_total_recovery",
            "low_igs_tract_share_recovery",
            "igs_economy_recovery",
            "poverty_rate_recovery",
            "unemp_rate_recovery",
            "median_household_income_recovery",
            "lfpr_16p_recovery",
            "score_low_igs_recovery",
```

```
            "score_econ_vulnerability_recovery",
            "score_covid_shock",
            "score_weak_recovery",
            "cluster_shortlist_score",
        ]
    ]
    .merge(cluster_county_counts, on="cluster_id", how="left", validate="1:1
    .copy()
)

# ------------------------------
# G. validation checks
# ------------------------------
expected_cluster_ids = set(enrichment_cluster_ids)
tract_cluster_ids = set(enrichment_cluster_tracts["cluster_id"].astype(str).
county_cluster_ids = set(enrichment_cluster_counties["cluster_id"].astype(st
year_cluster_ids = set(enrichment_cluster_year_panel["cluster_id"].astype(st
tract_year_cluster_ids = set(enrichment_tract_year_panel["cluster_id"].astyp
profile_cluster_ids = set(enrichment_cluster_profiles["cluster_id"].astype(s

assert tract_cluster_ids == expected_cluster_ids, "Mismatch between enrichme
assert county_cluster_ids == expected_cluster_ids, "Mismatch between enrichm
assert year_cluster_ids == expected_cluster_ids, "Mismatch between enrichmer
assert tract_year_cluster_ids == expected_cluster_ids, "Mismatch between enr
assert profile_cluster_ids == expected_cluster_ids, "Mismatch between enrich

expected_cluster_year_rows = len(expected_cluster_ids) * len(sorted(cluster_
if len(enrichment_cluster_year_panel) != expected_cluster_year_rows:
    print(
        f"Warning: enrichment_cluster_year_panel has {len(enrichment_cluster
        f"expected about {expected_cluster_year_rows} if all selected cluste
    )

# optional tract-year completeness check
tract_counts = enrichment_cluster_tracts.groupby("cluster_id")["geoid"].nuni
tract_year_counts = enrichment_tract_year_panel.groupby("cluster_id").size()
expected_years = len(sorted(enrichment_tract_year_panel["year"].dropna().uni
tract_year_check = pd.DataFrame({
    "n_cluster_tracts_lookup": tract_counts,
    "tract_year_rows": tract_year_counts
}).reset_index()
tract_year_check["expected_rows_if_full"] = tract_year_check["n_cluster_trac
tract_year_check["full_8yr_panel_flag"] = (
    tract_year_check["tract_year_rows"] == tract_year_check["expected_rows_i
)

# ------------------------------
# H. save outputs
# ------------------------------
OUT_DIR.mkdir(parents=True, exist_ok=True)

enrichment_clusters.to_parquet(OUT_DIR / "enrichment_clusters.parquet", inde
enrichment_cluster_profiles.to_parquet(OUT_DIR / "enrichment_cluster_profile
enrichment_cluster_tracts.to_parquet(OUT_DIR / "enrichment_cluster_tracts.pa
enrichment_cluster_counties.to_parquet(OUT_DIR / "enrichment_cluster_countie
enrichment_cluster_year_panel.to_parquet(OUT_DIR / "enrichment_cluster_year_
```

```
enrichment_tract_year_panel.to_parquet(OUT_DIR / "enrichment_tract_year_pane
tract_year_check.to_parquet(OUT_DIR / "enrichment_tract_year_completeness.pa

enrichment_cluster_profiles.to_csv(OUT_DIR / "enrichment_cluster_profiles.cs
enrichment_cluster_tracts.to_csv(OUT_DIR / "enrichment_cluster_tracts.csv",
enrichment_cluster_counties.to_csv(OUT_DIR / "enrichment_cluster_counties.cs
tract_year_check.to_csv(OUT_DIR / "enrichment_tract_year_completeness.csv",

print("\nSaved enrichment package.")
print("enrichment_clusters shape:", enrichment_clusters.shape)
print("enrichment_cluster_profiles shape:", enrichment_cluster_profiles.shap
print("enrichment_cluster_tracts shape:", enrichment_cluster_tracts.shape)
print("enrichment_cluster_counties shape:", enrichment_cluster_counties.shap
print("enrichment_cluster_year_panel shape:", enrichment_cluster_year_panel.
print("enrichment_tract_year_panel shape:", enrichment_tract_year_panel.shap
print("tract_year_check shape:", tract_year_check.shape)
print()
print("Tract-year completeness check:")
print(
    tract_year_check.sort_values(["full_8yr_panel_flag", "cluster_id"], asce
)
```

```
Selected top 15 clusters for next-step enrichment.
    enrichment_rank  rank                                       cluster_id d
isplay_state        display_county  n_cluster_tracts  latest_cluster_pop_tot
al  igs_total_recovery  igs_economy_recovery  \
0                 1     1      Puerto Rico | Utuado Municipio | cluster_75
Puerto Rico      Utuado Municipio                 8                  27621.
0           25.035923             15.289723
1                 2     2    Puerto Rico | Mayagüez Municipio | cluster_73
Puerto Rico    Mayagüez Municipio                25                  70828.
0           31.136263             19.475426
2                 3     3  Puerto Rico | Río Grande Municipio | cluster_74
Puerto Rico  Río Grande Municipio                12                  46051.
0           25.284481             17.443243
3                 4     4              Ohio | Cuyahoga County | cluster_64
Ohio       Cuyahoga County                13                  31886.0
34.225025             31.398833
4                 5     5             Michigan | Wayne County | cluster_38
Michigan          Wayne County               152                  319984.0
34.793412             29.645510
5                 6     6              Illinois | Cook County | cluster_29
Illinois           Cook County                 9                   27412.0
30.615724             34.163963
6                 7     7              Arizona | Apache County | cluster_3
Arizona         Apache County               18                   65341.0
34.156029             28.709013
7                 8     8              Illinois | Cook County | cluster_25
Illinois           Cook County                33                   81348.0
34.120468             31.288560
8                 9     9             Michigan | Wayne County | cluster_39
Michigan          Wayne County                 9                   19947.0
29.825030             27.539859
9                10    10              Illinois | Cook County | cluster_24
Illinois           Cook County                85                  187702.0
34.342235             29.945366
10               11    11           Tennessee | Shelby County | cluster_77
Tennessee         Shelby County                96                  268298.0
35.813056             35.120633
11               12    12        New Mexico | McKinley County | cluster_49
New Mexico       McKinley County                33                   70431.0
36.970752             37.724026
12               13    13     California | Los Angeles County | cluster_11
California    Los Angeles County                 5                   19914.0
35.686555             35.223650
13               14    14              Ohio | Cuyahoga County | cluster_63
Ohio       Cuyahoga County               107                 210162.0
38.303060             31.909575
14               15    15        Wisconsin | Milwaukee County | cluster_86
Wisconsin      Milwaukee County                98                  225339.0
35.715526             35.118704

    poverty_rate_recovery  unemp_rate_recovery  median_household_income_reco
very  lfpr_16p_recovery  cluster_shortlist_score
0                0.525332             0.264035                      18417.68
5903           0.408071                 0.869855
1                0.532210             0.229062                      19229.67
3889           0.415725                 0.848576
```

```
2               0.356654            0.165771                             28512.14
3909          0.525166                0.821570
3               0.370255            0.136437                             36960.19
7388          0.590425                0.811047
4               0.348753            0.161412                             37886.81
0012          0.516598                0.772703
5               0.234874            0.166658                             51537.52
4910          0.543750                0.752267
6               0.313574            0.099555                             41738.34
1598          0.407386                0.742122
7               0.304167            0.149948                             43383.00
6594          0.586529                0.734564
8               0.297270            0.092034                             41703.87
4646          0.550155                0.733314
9               0.300524            0.165977                             46121.63
2254          0.577286                0.712355
10              0.300348            0.118143                             43461.12
6364          0.582458                0.712122
11              0.341160            0.098756                             45567.49
6445          0.484736                0.705959
12              0.257480            0.098351                             56430.09
7238          0.617167                0.705523
13              0.329582            0.141196                             38280.58
0628          0.551952                0.698227
14              0.321145            0.091170                             40889.93
9837          0.584413                0.685087

Saved enrichment package.
enrichment_clusters shape: (15, 56)
enrichment_cluster_profiles shape: (15, 22)
enrichment_cluster_tracts shape: (703, 12)
enrichment_cluster_counties shape: (15, 7)
enrichment_cluster_year_panel shape: (120, 18)
enrichment_tract_year_panel shape: (5624, 25)
tract_year_check shape: (15, 5)

Tract-year completeness check:
                                         cluster_id  n_cluster_tracts_lookup
tract_year_rows  expected_rows_if_full  full_8yr_panel_flag
0               Arizona | Apache County | cluster_3                       18
144                    144                 True
1      California | Los Angeles County | cluster_11                        5
40                     40                 True
2              Illinois | Cook County | cluster_24                        85
680                    680                 True
3              Illinois | Cook County | cluster_25                        33
264                    264                 True
4              Illinois | Cook County | cluster_29                         9
72                     72                 True
5             Michigan | Wayne County | cluster_38                       152
1216                   1216                 True
6             Michigan | Wayne County | cluster_39                         9
72                     72                 True
7         New Mexico | McKinley County | cluster_49                        33
264                    264                 True
8              Ohio | Cuyahoga County | cluster_63                       107
```

```
856                      856                  True
9              Ohio | Cuyahoga County | cluster_64                          13
104                      104                  True
10    Puerto Rico | Mayagüez Municipio | cluster_73                         25
200                      200                  True
11  Puerto Rico | Río Grande Municipio | cluster_74                         12
96                       96                  True
12      Puerto Rico | Utuado Municipio | cluster_75                          8
64                       64                  True
13           Tennessee | Shelby County | cluster_77                         96
768                      768                  True
14         Wisconsin | Milwaukee County | cluster_86                        98
784                      784                  True
```

Cell 7A — fetch CDC PLACES tract-level data for Shelby County

In [7]:

```python
# ============================================================
# Cell 7A — fetch CDC PLACES tract-level data for Shelby County
# NOTE:
# - We use PLACES as a tract-level local snapshot, not a time-trend source.
# - We keep all returned years because some concepts may be carried from
#   an earlier release year.
# ============================================================

import requests
import pandas as pd
import numpy as np
import re

PLACES_TRACT_RESOURCE = "cwsq-ngmh"   # CDC PLACES census tract dataset
PLACES_TRACT_URL = f"https://data.cdc.gov/resource/{PLACES_TRACT_RESOURCE}.j

TARGET_STATE_ABBR = "TN"
TARGET_COUNTY_FIPS = "47157"   # Shelby County, TN
PLACES_LIMIT = 50000

# Fallback helper in case your notebook session lost it
if "normalize_geoid_series" not in globals():
    def normalize_geoid_series(s):
        s = pd.Series(s, copy=False)
        out = (
            s.astype("string")
             .str.extract(r"(\d{11})", expand=False)
        )
        return out

def fetch_places_tract_rows(state_abbr: str, county_fips: str, limit: int =
    session = requests.Session()
    rows = []
    offset = 0

    while True:
        params = {
            "$where": f"stateabbr='{state_abbr}' AND countyfips='{county_fip
```

```
            "$limit": limit,
            "$offset": offset,
        }
        r = session.get(PLACES_TRACT_URL, params=params, timeout=120)
        r.raise_for_status()
        chunk = r.json()

        if not chunk:
            break

        rows.extend(chunk)

        if len(chunk) < limit:
            break

        offset += limit

    df = pd.DataFrame(rows)
    if df.empty:
        raise ValueError(
            f"No CDC PLACES tract rows returned for stateabbr={state_abbr},
        )

    df.columns = [str(c).strip().lower() for c in df.columns]

    # Ensure expected columns exist
    expected_cols = [
        "year", "stateabbr", "statedesc", "countyname", "countyfips",
        "locationname", "measure", "measureid", "short_question_text",
        "category", "data_value_type", "datavaluetypeid", "data_value",
        "data_value_unit", "low_confidence_limit", "high_confidence_limit",
        "totalpopulation"
    ]
    for c in expected_cols:
        if c not in df.columns:
            df[c] = pd.NA

    # Robust tract GEOID extraction
    geo_candidates = []
    for c in ["tractfips", "locationname", "geoid", "tract", "tract_fips"]:
        if c in df.columns:
            norm = normalize_geoid_series(df[c])
            geo_candidates.append((c, int(norm.notna().sum()), norm))

    if not geo_candidates:
        raise ValueError(f"No usable tract geography column found. Columns:

    geo_candidates = sorted(geo_candidates, key=lambda x: x[1], reverse=True
    best_geo_col, best_nonnull, best_norm = geo_candidates[0]
    df["tractfips"] = best_norm

    for c in ["data_value", "low_confidence_limit", "high_confidence_limit",
        df[c] = pd.to_numeric(df[c], errors="coerce")
    df["year"] = pd.to_numeric(df["year"], errors="coerce").astype("Int64")

    print("Best geography column used:", best_geo_col)
```

```python
    print("Valid normalized tract GEOIDs:", best_nonnull)
    print("Rows pulled:", df.shape)
    print("Unique tract GEOIDs:", df["tractfips"].nunique())
    print("Years present:", sorted(df["year"].dropna().unique().tolist()))
    print("Data value types:", sorted(df["data_value_type"].dropna().astype(

    return df


places_shelby_long = fetch_places_tract_rows(
    state_abbr=TARGET_STATE_ABBR,
    county_fips=TARGET_COUNTY_FIPS,
    limit=PLACES_LIMIT
)

display(
    places_shelby_long[
        ["year", "tractfips", "measure", "short_question_text", "data_value_
    ].head(15)
)
```

```
Best geography column used: locationname
Valid normalized tract GEOIDs: 8085
Rows pulled: (8085, 23)
Unique tract GEOIDs: 245
Years present: [2022, 2023]
Data value types: ['Crude prevalence']
```

	year	tractfips	measure	short_question_text	data_value_type	data_va
0	2023	47157000400	Obesity among adults	Obesity	Crude prevalence	
1	2023	47157002500	Diagnosed diabetes among adults	Diabetes	Crude prevalence	
2	2023	47157005000	Obesity among adults	Obesity	Crude prevalence	
3	2023	47157007000	Stroke among adults	Stroke	Crude prevalence	
4	2023	47157007300	Obesity among adults	Obesity	Crude prevalence	
5	2023	47157008700	Stroke among adults	Stroke	Crude prevalence	
6	2023	47157009100	Obesity among adults	Obesity	Crude prevalence	
7	2023	47157010122	Obesity among adults	Obesity	Crude prevalence	
8	2023	47157011020	Stroke among adults	Stroke	Crude prevalence	
9	2023	47157011300	Stroke among adults	Stroke	Crude prevalence	
10	2023	47157020102	Obesity among adults	Obesity	Crude prevalence	
11	2023	47157020302	Obesity among adults	Obesity	Crude prevalence	
12	2023	47157020633	Diagnosed diabetes among adults	Diabetes	Crude prevalence	
13	2023	47157020656	Obesity among adults	Obesity	Crude prevalence	
14	2023	47157020700	Stroke among adults	Stroke	Crude prevalence	

Cell 7B — build tract-level healthcare proxies, not exact equivalents

In [8]:

```
# ============================================================
# Cell 7B — define tract-level healthcare proxy panels
# Panel A = care access / prevention proxies
# Panel B = broader health burden / need
# Panel C = barriers / risk
# ============================================================

# Keep crude prevalence rows when present, but do NOT force one single year.
places_work = places_shelby_long.copy()

dvt = places_work["data_value_type"].astype("string").str.lower().str.strip(
crude_mask = dvt.eq("crude prevalence") | dvt.str.contains(r"\bcrude\b", na=

if crude_mask.any():
    places_work = places_work.loc[crude_mask].copy()

print("Years present after crude filter:", sorted(places_work["year"].dropna
print("Rows after crude filter:", places_work.shape)
print("Unique tract GEOIDs after crude filter:", places_work["tractfips"].nu

# Broader panel design:
# A = access / prevention
# B = burden / need
# C = barriers / risk
MEASURE_CONFIG = {
    # ------------------------
    # Panel A — access / prevention
    # ------------------------
    "uninsured_adults_pct": {
        "panel": "A",
        "label": "Uninsured Adults",
        "patterns": [
            r"current lack of health insurance",
            r"lack of health insurance among adults aged 18.?64 years",
            r"lack of health insurance"
        ],
        "higher_is_worse": True,
        "required": True,
    },
    "routine_checkup_pct": {
        "panel": "A",
        "label": "Routine Checkup Gap",
        "patterns": [
            r"visits? to doctor for routine checkup within the past year",
            r"routine checkup within the past year",
            r"routine checkup"
        ],
        "higher_is_worse": False,  # convert to gap later
        "required": True,
    },
    "dental_visit_pct": {
```

```
        "panel": "A",
        "label": "Dental Visit Gap",
        "patterns": [
            r"visits? to dentist or dental clinic",
            r"dentist or dental clinic",
            r"dental clinic",
            r"dental visit"
        ],
        "higher_is_worse": False,  # convert to gap later
        "required": False,
    },
    "mammography_pct": {
        "panel": "A",
        "label": "Mammography Gap",
        "patterns": [
            r"mammography use among women aged 50.?74 years",
            r"mammography use",
            r"mammography"
        ],
        "higher_is_worse": False,  # convert to gap later
        "required": False,
    },
    "flu_vax_pct": {
        "panel": "A",
        "label": "Influenza Vaccination Gap",
        "patterns": [
            r"influenza vaccination",
            r"flu vaccination",
            r"flu shot"
        ],
        "higher_is_worse": False,  # convert to gap later
        "required": False,
    },

    # -------------------------
    # Panel B — broader health burden / need
    # -------------------------
    "fair_poor_health_pct": {
        "panel": "B",
        "label": "Poor/Fair Health",
        "patterns": [
            r"fair or poor health",
            r"poor or fair health"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "poor_physical_health_pct": {
        "panel": "B",
        "label": "Poor Physical Health",
        "patterns": [
            r"physical health not good",
            r"poor physical health",
            r"frequent physical distress"
        ],
        "higher_is_worse": True,
```

```
        "required": False,
    },
    "poor_mental_health_pct": {
        "panel": "B",
        "label": "Poor Mental Health",
        "patterns": [
            r"mental health not good",
            r"poor mental health",
            r"frequent mental distress"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "depression_pct": {
        "panel": "B",
        "label": "Depression",
        "patterns": [
            r"depression"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "diabetes_pct": {
        "panel": "B",
        "label": "Diabetes",
        "patterns": [
            r"diabetes"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "obesity_pct": {
        "panel": "B",
        "label": "Obesity",
        "patterns": [
            r"obesity"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "current_asthma_pct": {
        "panel": "B",
        "label": "Current Asthma",
        "patterns": [
            r"current asthma",
            r"\basthma\b"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "coronary_heart_disease_pct": {
        "panel": "B",
        "label": "Coronary Heart Disease",
        "patterns": [
            r"coronary heart disease"
        ],
```

```
        "higher_is_worse": True,
        "required": False,
    },
    "stroke_pct": {
        "panel": "B",
        "label": "Stroke",
        "patterns": [
            r"\bstroke\b"
        ],
        "higher_is_worse": True,
        "required": False,
    },

    # ------------------------
    # Panel C — barriers / risk
    # ------------------------
    "smoking_pct": {
        "panel": "C",
        "label": "Current Smoking",
        "patterns": [
            r"current smoking",
            r"\bsmoking\b"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "physical_inactivity_pct": {
        "panel": "C",
        "label": "Physical Inactivity",
        "patterns": [
            r"no leisure[- ]time physical activity",
            r"physical inactivity"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "binge_drinking_pct": {
        "panel": "C",
        "label": "Binge Drinking",
        "patterns": [
            r"binge drinking"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "sleep_lt7_pct": {
        "panel": "C",
        "label": "Sleeping <7 Hours",
        "patterns": [
            r"sleep.*less than 7 hours",
            r"sleep.*under 7 hours"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "food_insecurity_pct": {
```

```
        "panel": "C",
        "label": "Food Insecurity",
        "patterns": [
            r"food insecurity"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "housing_insecurity_pct": {
        "panel": "C",
        "label": "Housing Insecurity",
        "patterns": [
            r"housing insecurity"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "transportation_barriers_pct": {
        "panel": "C",
        "label": "Transportation Barriers",
        "patterns": [
            r"transportation barriers"
        ],
        "higher_is_worse": True,
        "required": False,
    },
    "social_isolation_pct": {
        "panel": "C",
        "label": "Social Isolation",
        "patterns": [
            r"social isolation"
        ],
        "higher_is_worse": True,
        "required": False,
    },
}

def extract_places_metric(long_df: pd.DataFrame, new_col: str, patterns: lis
    text = (
        long_df["measure"].fillna("").astype(str).str.lower() + " " +
        long_df["short_question_text"].fillna("").astype(str).str.lower()
    )

    mask = pd.Series(False, index=long_df.index)
    for pat in patterns:
        mask = mask | text.str.contains(pat, regex=True, na=False)

    out = long_df.loc[mask].copy()
    matched_measure_names = sorted(out["measure"].dropna().astype(str).uniqu

    if out.empty:
        return pd.DataFrame(columns=["geoid", new_col]), matched_measure_nam

    # Latest available year FOR THIS concept
    concept_year = out["year"].dropna().max()
    if pd.notna(concept_year):
```

```python
        out = out.loc[out["year"] == concept_year].copy()

    out = (
        out.sort_values(["tractfips", "measure"])
           .drop_duplicates(subset=["tractfips"], keep="first")
           [["tractfips", "data_value"]]
           .rename(columns={"tractfips": "geoid", "data_value": new_col})
           .copy()
    )

    out["geoid"] = normalize_geoid_series(out["geoid"])
    out[new_col] = pd.to_numeric(out[new_col], errors="coerce")
    return out, matched_measure_names, concept_year

metric_tables = []
audit_rows = []

for metric, cfg in MEASURE_CONFIG.items():
    metric_df, matched_names, concept_year = extract_places_metric(
        places_work,
        new_col=metric,
        patterns=cfg["patterns"]
    )
    metric_tables.append(metric_df)

    audit_rows.append({
        "metric": metric,
        "panel": cfg["panel"],
        "label": cfg["label"],
        "required": cfg["required"],
        "concept_year_used": concept_year,
        "matched_measure_names": " | ".join(matched_names) if matched_names
        "matched_tract_rows": int(len(metric_df))
    })

places_measure_audit = pd.DataFrame(audit_rows).sort_values(["panel", "metri

display(places_measure_audit)

# Helpful debug list of available measures in Shelby pull
available_measure_catalog = (
    places_work[["year", "measure", "short_question_text"]]
    .drop_duplicates()
    .sort_values(["measure", "year"])
    .reset_index(drop=True)
)

print("Sample of available PLACES measures in this Shelby pull:")
display(available_measure_catalog.head(60))
```

```
Years present after crude filter: [2022, 2023]
Rows after crude filter: (8085, 23)
Unique tract GEOIDs after crude filter: 245
```

	metric	panel	label	required	concept_year_used
0	dental_visit_pct	A	Dental Visit Gap	False	2022
1	flu_vax_pct	A	Influenza Vaccination Gap	False	<NA>
2	mammography_pct	A	Mammography Gap	False	2022
3	routine_checkup_pct	A	Routine Checkup Gap	True	2023
4	uninsured_adults_pct	A	Uninsured Adults	True	2023
5	coronary_heart_disease_pct	B	Coronary Heart Disease	False	2023
6	current_asthma_pct	B	Current Asthma	False	2023
7	depression_pct	B	Depression	False	2023
8	diabetes_pct	B	Diabetes	False	2023
9	fair_poor_health_pct	B	Poor/Fair Health	False	<NA>
10	obesity_pct	B	Obesity	False	2023
11	poor_mental_health_pct	B	Poor Mental Health	False	2023
12	poor_physical_health_pct	B	Poor Physical Health	False	2023
13	stroke_pct	B	Stroke	False	2023
14	binge_drinking_pct	C	Binge Drinking	False	2023
15	food_insecurity_pct	C	Food Insecurity	False	<NA>
16	housing_insecurity_pct	C	Housing Insecurity	False	<NA>
17	physical_inactivity_pct	C	Physical Inactivity	False	2023
18	sleep_lt7_pct	C	Sleeping <7 Hours	False	<NA>
19	smoking_pct	C	Current Smoking	False	2023
20	social_isolation_pct	C	Social Isolation	False	<NA>
21	transportation_barriers_pct	C	Transportation Barriers	False	<NA>

```
Sample of available PLACES measures in this Shelby pull:
```

	year	measure	short_question_text
0	2022	All teeth lost among adults aged >=65 years	All Teeth Lost
1	2023	Any disability among adults	Any Disability
2	2023	Arthritis among adults	Arthritis
3	2023	Binge drinking among adults	Binge Drinking
4	2023	Cancer (non-skin) or melanoma among adults	Cancer (non-skin) or Melanoma
5	2023	Cholesterol screening among adults	Cholesterol Screening
6	2023	Chronic obstructive pulmonary disease among ad...	COPD
7	2023	Cognitive disability among adults	Cognitive Disability
8	2022	Colorectal cancer screening among adults aged ...	Colorectal Cancer Screening
9	2023	Coronary heart disease among adults	Coronary Heart Disease
10	2023	Current asthma among adults	Current Asthma
11	2023	Current cigarette smoking among adults	Current Cigarette Smoking
12	2023	Current lack of health insurance among adults ...	Health Insurance
13	2023	Depression among adults	Depression
14	2023	Diagnosed diabetes among adults	Diabetes
15	2023	Fair or poor self-rated health status among ad...	General Health
16	2023	Frequent mental distress among adults	Frequent Mental Distress
17	2023	Frequent physical distress among adults	Frequent Physical Distress
18	2023	Hearing disability among adults	Hearing Disability
19	2023	High blood pressure among adults	High Blood Pressure
20	2023	High cholesterol among adults who have ever be...	High Cholesterol
21	2023	Independent living disability among adults	Independent Living Disability
22	2022	Mammography use among women aged 50-74 years	Mammography
23	2023	Mobility disability among adults	Mobility Disability
24	2023	No leisure-time physical activity among adults	Physical Inactivity
25	2023	Obesity among adults	Obesity
26	2023	Self-care disability among adults	Self-care Disability
27	2022	Short sleep duration among adults	Short Sleep Duration

	year	measure	short_question_text
28	2023	Stroke among adults	Stroke
29	2023	Taking medicine to control high blood pressure...	High Blood Pressure Medication
30	2023	Vision disability among adults	Vision Disability
31	2022	Visited dentist or dental clinic in the past y...	Dental Visit
32	2023	Visits to doctor for routine checkup within th...	Annual Checkup

Cell 7C — proxy visual only, no preventable stays claim

In [9]:

```
# ============================================================
# Cell 7C — selected Shelby community vs rest of Shelby County
# using tract-level PLACES measures
# ============================================================

FINAL_CLUSTER_ID = "Tennessee | Shelby County | cluster_77"
PANEL_YEAR = 2023   # population weights from your tract panel

# Get selected tract list
if "tract_cluster_map" in globals():
    selected_geoids = set(
        normalize_geoid_series(
            tract_cluster_map.loc[
                tract_cluster_map["cluster_id"].astype(str) == FINAL_CLUSTER
                "geoid"
            ]
        ).dropna().tolist()
    )
else:
    selected_geoids = set()

if not selected_geoids and "TARGET_GEOIDS" in globals():
    selected_geoids = set(normalize_geoid_series(pd.Series(TARGET_GEOIDS)).c

if not selected_geoids:
    raise ValueError("Could not determine selected Shelby community GEOIDs."

# Wide merge of all extracted metric tables
places_proxy_wide = None
for part in metric_tables:
    if places_proxy_wide is None:
        places_proxy_wide = part.copy()
    else:
        places_proxy_wide = places_proxy_wide.merge(part, on="geoid", how="c

if places_proxy_wide is None:
    raise ValueError("No CDC PLACES metrics were extracted.")

# Shelby tract base from your analysis panel
shelby_pop = (
```

```
    analysis_panel.loc[
        (analysis_panel["display_state"].astype(str) == "Tennessee") &
        (analysis_panel["display_county"].astype(str) == "Shelby County") &
        (analysis_panel["year"] == PANEL_YEAR),
        ["geoid", "pop_total", "display_state", "display_county", "display_r
    ]
    .drop_duplicates(subset=["geoid"])
    .copy()
)

if shelby_pop.empty:
    raise ValueError(f"No Shelby County tracts found in analysis_panel for y

shelby_places_base = (
    shelby_pop
    .merge(places_proxy_wide, on="geoid", how="left", validate="1:1")
    .copy()
)

shelby_places_base["is_selected_community"] = shelby_places_base["geoid"].is

selected_base = shelby_places_base.loc[shelby_places_base["is_selected_commu
rest_base = shelby_places_base.loc[~shelby_places_base["is_selected_communit

print("Selected community tracts:", selected_base["geoid"].nunique())
print("Rest-of-county tracts:", rest_base["geoid"].nunique())
print("Total Shelby tracts:", shelby_places_base["geoid"].nunique())

def weighted_mean_safe(values: pd.Series, weights: pd.Series) -> float:
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

# Build compare table for all matched metrics
matched_metrics = places_measure_audit.loc[
    places_measure_audit["matched_measure_names"] != "NO MATCH FOUND", "metr
].tolist()

compare_rows = []
for grp_name, grp in [
    ("Selected Shelby community", selected_base),
    ("Rest of Shelby County (excluding selected community)", rest_base),
]:
    row = {
        "group": grp_name,
        "n_tracts": int(grp["geoid"].nunique()),
        "population_sum": float(pd.to_numeric(grp["pop_total"], errors="coer
    }
    for metric in matched_metrics:
        row[metric] = weighted_mean_safe(grp[metric], grp["pop_total"])
```

```python
    compare_rows.append(row)

shelby_places_compare = pd.DataFrame(compare_rows)

# Convert "good" measures to gaps so higher = worse
positive_use_metrics = {
    "routine_checkup_pct": "routine_checkup_gap_pct",
    "dental_visit_pct": "dental_visit_gap_pct",
    "mammography_pct": "mammography_gap_pct",
    "flu_vax_pct": "flu_vax_gap_pct",
}

for src, dst in positive_use_metrics.items():
    if src in shelby_places_compare.columns:
        shelby_places_compare[dst] = 100 - shelby_places_compare[src]

# Proxy composites for Panel A
if "uninsured_adults_pct" in shelby_places_compare.columns:
    shelby_places_compare["uninsured_burden_proxy_pct"] = shelby_places_comp

provider_inputs = [c for c in ["routine_checkup_gap_pct", "dental_visit_gap_
if provider_inputs:
    shelby_places_compare["provider_access_proxy_gap_pct"] = shelby_places_c

preventive_inputs = [c for c in ["mammography_gap_pct", "flu_vax_gap_pct"] i
if preventive_inputs:
    shelby_places_compare["preventive_care_proxy_gap_pct"] = shelby_places_c

shelby_places_compare["provider_inputs_used"] = ", ".join(provider_inputs) i
shelby_places_compare["preventive_inputs_used"] = ", ".join(preventive_input

display(shelby_places_compare.round(2))

# Long summary table for easier analysis
summary_rows = []

left = shelby_places_compare.loc[shelby_places_compare["group"] == "Selectec
right = shelby_places_compare.loc[
    shelby_places_compare["group"] == "Rest of Shelby County (excluding sele
].iloc[0]

for metric, cfg in MEASURE_CONFIG.items():
    if metric not in shelby_places_compare.columns:
        continue

    sel_val = left.get(metric, np.nan)
    rest_val = right.get(metric, np.nan)

    summary_rows.append({
        "panel": cfg["panel"],
        "metric": metric,
        "label": cfg["label"],
        "selected_value": sel_val,
        "rest_value": rest_val,
        "selected_minus_rest": sel_val - rest_val if pd.notna(sel_val) and p
        "higher_is_worse_raw": cfg["higher_is_worse"]
```

```
    })

# Add panel A composites
for metric, label in [
    ("uninsured_burden_proxy_pct", "Uninsured Burden Proxy"),
    ("provider_access_proxy_gap_pct", "Provider Access Proxy Gap"),
    ("preventive_care_proxy_gap_pct", "Preventive Care Proxy Gap"),
]:
    if metric in shelby_places_compare.columns:
        summary_rows.append({
            "panel": "A",
            "metric": metric,
            "label": label,
            "selected_value": left.get(metric, np.nan),
            "rest_value": right.get(metric, np.nan),
            "selected_minus_rest": left.get(metric, np.nan) - right.get(metr
                if pd.notna(left.get(metric, np.nan)) and pd.notna(right.get
            "higher_is_worse_raw": True
        })

shelby_places_summary = pd.DataFrame(summary_rows).sort_values(["panel", "la
display(shelby_places_summary.round(2))

# Save
OUT_DIR.mkdir(parents=True, exist_ok=True)
places_measure_audit.to_csv(OUT_DIR / "cdc_places_measure_audit_expanded.csv
shelby_places_base.to_parquet(OUT_DIR / "shelby_places_tract_base_expanded.p
shelby_places_compare.to_csv(OUT_DIR / "shelby_places_compare_expanded.csv",
shelby_places_summary.to_csv(OUT_DIR / "shelby_places_summary_expanded.csv",

print("Saved:")
print("-", OUT_DIR / "cdc_places_measure_audit_expanded.csv")
print("-", OUT_DIR / "shelby_places_tract_base_expanded.parquet")
print("-", OUT_DIR / "shelby_places_compare_expanded.csv")
print("-", OUT_DIR / "shelby_places_summary_expanded.csv")
```

```
Selected community tracts: 96
Rest-of-county tracts: 153
Total Shelby tracts: 249
```

	group	n_tracts	population_sum	dental_visit_pct	mammography_pct	ro
0	Selected Shelby community	96	272665.0	44.16	77.14	
1	Rest of Shelby County (excluding selected comm...	153	649530.0	58.59	78.68	

	panel	metric	label	selected_value	rest_valu
0	A	dental_visit_pct	Dental Visit Gap	44.16	58.5
1	A	mammography_pct	Mammography Gap	77.14	78.6
2	A	preventive_care_proxy_gap_pct	Preventive Care Proxy Gap	22.86	21.3
3	A	provider_access_proxy_gap_pct	Provider Access Proxy Gap	36.91	30.5
4	A	routine_checkup_pct	Routine Checkup Gap	82.03	80.3
5	A	uninsured_adults_pct	Uninsured Adults	15.94	11.8
6	A	uninsured_burden_proxy_pct	Uninsured Burden Proxy	15.94	11.8
7	B	coronary_heart_disease_pct	Coronary Heart Disease	8.13	5.7
8	B	current_asthma_pct	Current Asthma	13.05	11.2
9	B	depression_pct	Depression	24.96	25.2
10	B	diabetes_pct	Diabetes	21.55	13.8
11	B	obesity_pct	Obesity	46.13	37.1
12	B	poor_mental_health_pct	Poor Mental Health	21.92	18.6
13	B	poor_physical_health_pct	Poor Physical Health	17.62	13.3
14	B	stroke_pct	Stroke	6.09	3.6
15	C	binge_drinking_pct	Binge Drinking	11.87	14.9
16	C	smoking_pct	Current Smoking	23.21	15.9
17	C	physical_inactivity_pct	Physical Inactivity	36.13	24.9

```
Saved:
- data\processed\cdc_places_measure_audit_expanded.csv
- data\processed\shelby_places_tract_base_expanded.parquet
- data\processed\shelby_places_compare_expanded.csv
- data\processed\shelby_places_summary_expanded.csv
```

Cell 7D — optional transparent raw-measure visual

In [10]:

```
# ============================================================
# Cell 7D — reusable panel plot helper
```

```python
# ============================================================

import matplotlib.pyplot as plt
import numpy as np

def plot_places_panel(
    summary_df,
    panel_code,
    title,
    outfile_name,
    extra_footer="",
    show_bar_labels=False
):
    panel_df = summary_df.loc[summary_df["panel"] == panel_code].copy()

    # Keep only rows with actual comparison values
    panel_df = panel_df.loc[
        panel_df["selected_value"].notna() & panel_df["rest_value"].notna()
    ].copy()

    if panel_df.empty:
        print(f"No matched metrics available for Panel {panel_code}.")
        return None

    # Sort by size of selected-minus-rest to make the story clearer
    panel_df = panel_df.sort_values("selected_minus_rest", ascending=False).

    x = np.arange(len(panel_df))
    bar_w = 0.36

    fig, ax = plt.subplots(figsize=(14, 7))

    bars1 = ax.bar(
        x - bar_w / 2,
        panel_df["selected_value"].values,
        width=bar_w,
        label="Selected Shelby community"
    )
    bars2 = ax.bar(
        x + bar_w / 2,
        panel_df["rest_value"].values,
        width=bar_w,
        label="Rest of Shelby County\n(excluding selected community)"
    )

    # Add percentage labels above bars if requested
    if show_bar_labels:
        y_max = max(
            panel_df["selected_value"].max(),
            panel_df["rest_value"].max()
        )
        label_pad = max(0.4, y_max * 0.015)

        for bars in [bars1, bars2]:
            for bar in bars:
                height = bar.get_height()
```

```
                if pd.notna(height):
                    ax.text(
                        bar.get_x() + bar.get_width() / 2,
                        height + label_pad,
                        f"{height:.1f}%",
                        ha="center",
                        va="bottom",
                        fontsize=9
                    )

    ax.set_xticks(x)
    ax.set_xticklabels(panel_df["label"], rotation=20, ha="right")
    ax.set_ylabel("Percent / gap percent (higher = worse)")
    ax.set_title(title)
    ax.grid(axis="y", alpha=0.25)
    ax.legend(frameon=True)

    # Give a little extra headroom when labels are on
    if show_bar_labels:
        current_top = max(
            panel_df["selected_value"].max(),
            panel_df["rest_value"].max()
        )
        ax.set_ylim(0, current_top * 1.14)

    footer = "CDC PLACES tract-level snapshot. Selected Shelby community vs
    if extra_footer:
        footer += "\n" + extra_footer


    fig.text(0.02, 0.02, footer, fontsize=10)

    plt.tight_layout(rect=[0, 0.10, 1, 0.95])
    plt.show()

    outpath = OUT_DIR / outfile_name
    fig.savefig(outpath, bbox_inches="tight", dpi=220)
    print("Saved:", outpath)

    return panel_df
```

In [11]:

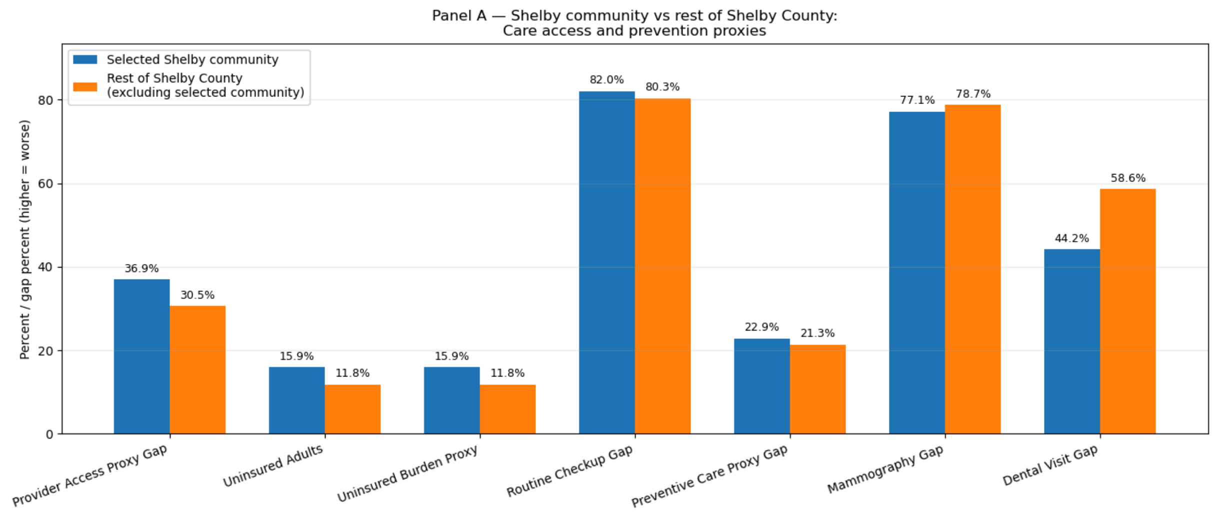
```
# ============================================================
# Cell 7E — Panel A: care access / prevention proxies
# ============================================================

panel_a_extra = (
    "Access/prevention proxies only. "
    f"Provider gap: {shelby_places_compare['provider_inputs_used'].iloc[0]}.
    f"Preventive gap: {shelby_places_compare['preventive_inputs_used'].iloc[
)


panel_a_df = plot_places_panel(
    summary_df=shelby_places_summary,
    panel_code="A",
    title="Panel A — Shelby community vs rest of Shelby County:\nCare access
```

```
    outfile_name="shelby_places_panel_a_access_prevention.png",
    extra_footer=panel_a_extra,
    show_bar_labels=True
)

display(panel_a_df.round(2) if panel_a_df is not None else pd.DataFrame())
```



CDC PLACES tract-level snapshot. Selected Shelby community vs rest of Shelby County.
Access/prevention proxies only. Provider gap: routine_checkup_gap_pct, dental_visit_gap_pct. Preventive gap: mammography_gap_pct.

```
Saved: data\processed\shelby_places_panel_a_access_prevention.png
```

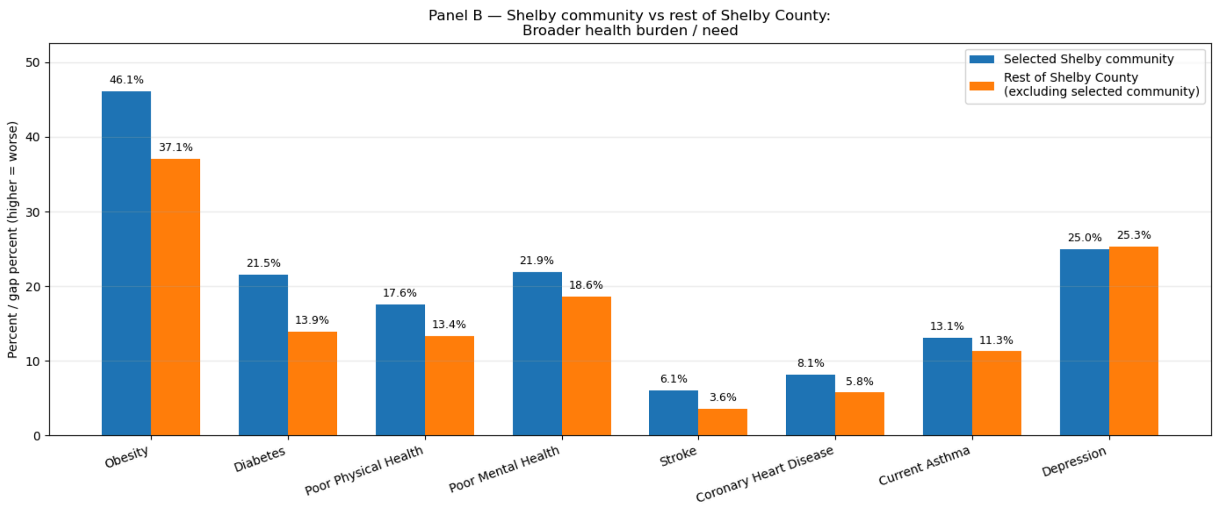
	panel	metric	label	selected_value	rest_value
0	A	provider_access_proxy_gap_pct	Provider Access Proxy Gap	36.91	30.53
1	A	uninsured_adults_pct	Uninsured Adults	15.94	11.82
2	A	uninsured_burden_proxy_pct	Uninsured Burden Proxy	15.94	11.82
3	A	routine_checkup_pct	Routine Checkup Gap	82.03	80.34
4	A	preventive_care_proxy_gap_pct	Preventive Care Proxy Gap	22.86	21.32
5	A	mammography_pct	Mammography Gap	77.14	78.68
6	A	dental_visit_pct	Dental Visit Gap	44.16	58.59

In [12]:
```
# ============================================================
# Cell 7F — Panel B: broader health burden / need
# ============================================================

panel_b_df = plot_places_panel(
    summary_df=shelby_places_summary,
    panel_code="B",
    title="Panel B — Shelby community vs rest of Shelby County:\nBroader hea
```

```
    outfile_name="shelby_places_panel_b_burden_need.png",
    extra_footer="Higher values indicate worse health burden or health statu
    show_bar_labels=True
)

display(panel_b_df.round(2) if panel_b_df is not None else pd.DataFrame())
```



CDC PLACES tract-level snapshot. Selected Shelby community vs rest of Shelby County.
Higher values indicate worse health burden or health status.

```
Saved: data\processed\shelby_places_panel_b_burden_need.png
```

	panel	metric	label	selected_value	rest_value	selec
0	B	obesity_pct	Obesity	46.13	37.11	
1	B	diabetes_pct	Diabetes	21.55	13.88	
2	B	poor_physical_health_pct	Poor Physical Health	17.62	13.38	
3	B	poor_mental_health_pct	Poor Mental Health	21.92	18.64	
4	B	stroke_pct	Stroke	6.09	3.60	
5	B	coronary_heart_disease_pct	Coronary Heart Disease	8.13	5.76	
6	B	current_asthma_pct	Current Asthma	13.05	11.28	
7	B	depression_pct	Depression	24.96	25.28	

In [13]:
```
# ============================================================
# SHELBY SPLIT HEALTH COMPARISON
# Chart 1: Access Gaps
# Chart 2: Health Burden
# Our community vs Rest of Shelby County vs High-growth comparator
# ============================================================

import pandas as pd
```

```python
import numpy as np
import matplotlib.pyplot as plt

# ----------------------------------------------------
# 1. Validate source table
# ----------------------------------------------------
if "shelby_cluster_input" not in globals():
    raise ValueError("`shelby_cluster_input` is not in memory.")

plot_df = shelby_cluster_input.copy()

# ----------------------------------------------------
# 2. Choose metrics
# ----------------------------------------------------
access_metrics = {
    "uninsured_adults_pct": "Uninsured Adults",
    "routine_checkup_gap_pct": "Routine Checkup Gap",
    "dental_visit_gap_pct": "Dental Visit Gap",
    "mammography_gap_pct": "Mammography Gap",
}

burden_metrics = {
    "diabetes_pct": "Diabetes",
    "obesity_pct": "Obesity",
    "depression_pct": "Depression",
    "current_asthma_pct": "Current Asthma",
}

access_metrics = {k: v for k, v in access_metrics.items() if k in plot_df.co
burden_metrics = {k: v for k, v in burden_metrics.items() if k in plot_df.co

if len(access_metrics) == 0 and len(burden_metrics) == 0:
    raise ValueError("None of the selected metrics were found in `shelby_clu

# ----------------------------------------------------
# 3. Build / recover 3-group membership
# ----------------------------------------------------
group_col = None

if "three_group_base" in globals():
    tg = three_group_base.copy()
    possible_group_cols = ["final_compare_group", "group"]
    found_group_col = next((c for c in possible_group_cols if c in tg.columr

    if "geoid" in tg.columns and found_group_col is not None:
        plot_df = plot_df.merge(
            tg[["geoid", found_group_col]].drop_duplicates("geoid"),
            on="geoid",
            how="left"
        )
        group_col = found_group_col

if group_col is None:
    for c in ["final_compare_group", "group"]:
        if c in plot_df.columns:
            group_col = c
```

```
            break

if group_col is None:
    if "shelby_full_cluster_id" not in plot_df.columns:
        raise ValueError(
            "Could not find 3-group labels. Run the earlier Shelby 3-group c
        )

    selected_candidates = [
        "SELECTED_SHELBY_CLUSTER_ID",
        "selected_shelby_cluster_id",
        "BEST_SHELBY_CLUSTER_ID",
        "best_shelby_cluster_id",
    ]
    comparator_candidates = [
        "BEST_SHELBY_COMPARATOR_ID",
        "best_shelby_comparator_id",
        "HIGH_GROWTH_SHELBY_CLUSTER_ID",
        "high_growth_shelby_cluster_id",
        "selected_high_growth_cluster_id",
    ]

    selected_id = next((globals()[n] for n in selected_candidates if n in gl
    comparator_id = next((globals()[n] for n in comparator_candidates if n i

    if selected_id is None or comparator_id is None:
        raise ValueError(
            "Could not reconstruct the 3 Shelby groups automatically. "
            "Run the earlier 3-group Shelby setup cell so `three_group_base`
        )

    def assign_group(cluster_id):
        if cluster_id == selected_id:
            return "Selected low-growth community"
        elif cluster_id == comparator_id:
            return "Selected high-growth comparator"
        else:
            return "Rest of Shelby County (excluding both selected communiti

    plot_df["final_compare_group"] = plot_df["shelby_full_cluster_id"].apply
    group_col = "final_compare_group"

# --------------------------------------------------
# 4. Labels / colors
# --------------------------------------------------
group_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}
```

```
color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

# ---------------------------------------------------
# 5. Weighted average helper
# ---------------------------------------------------
def safe_weighted_mean(values, weights):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")

    mask = values.notna() & weights.notna() & (weights > 0)

    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return values.loc[valid].mean()

    return np.average(values.loc[mask], weights=weights.loc[mask])

if "pop_total_recovery" not in plot_df.columns:
    raise ValueError("`pop_total_recovery` is missing from `shelby_cluster_i

# ---------------------------------------------------
# 6. Build summary helper
# ---------------------------------------------------
def build_summary(metric_dict):
    rows = []

    for grp in group_order:
        grp_df = plot_df.loc[plot_df[group_col] == grp].copy()
        weights = grp_df["pop_total_recovery"]

        row = {"group": grp}
        for col, pretty in metric_dict.items():
            row[pretty] = safe_weighted_mean(grp_df[col], weights)
        rows.append(row)

    out = pd.DataFrame(rows).set_index("group")
    return out.loc[group_order]

access_compare = build_summary(access_metrics) if len(access_metrics) > 0 el
burden_compare = build_summary(burden_metrics) if len(burden_metrics) > 0 el

if access_compare is not None:
    display(access_compare)

if burden_compare is not None:
    display(burden_compare)

# ---------------------------------------------------
# 7. Plot helper
```

```python
# -------------------------------------------------------
def plot_grouped_bars(ax, compare_df, title, ylabel):
    metric_names = list(compare_df.columns)
    x = np.arange(len(metric_names))
    bar_w = 0.24

    for i, grp in enumerate(group_order):
        vals = compare_df.loc[grp, metric_names].values
        ax.bar(
            x + (i - 1) * bar_w,
            vals,
            width=bar_w,
            color=color_map[grp],
            label=label_map[grp],
            edgecolor="none"
        )

    ax.set_xticks(x)
    ax.set_xticklabels(metric_names, rotation=0, ha="center")
    ax.yaxis.set_major_formatter(lambda val, pos: f"{val:.0f}%")
    ax.grid(axis="y", alpha=0.25)
    ax.set_axisbelow(True)
    ax.set_title(title, fontsize=16, pad=12)
    ax.set_ylabel(ylabel)
    ax.set_xlabel("Metric")

    for i, grp in enumerate(group_order):
        vals = compare_df.loc[grp, metric_names].values
        for j, v in enumerate(vals):
            if pd.notna(v):
                ax.text(
                    x[j] + (i - 1) * bar_w,
                    v + 0.35,
                    f"{v:.1f}%",
                    ha="center",
                    va="bottom",
                    fontsize=9
                )

# -------------------------------------------------------
# 8. Create figure
# -------------------------------------------------------
n_panels = int(access_compare is not None) + int(burden_compare is not None)

if n_panels == 1:
    fig, ax = plt.subplots(figsize=(14, 8))
    axes = [ax]
else:
    fig, axes = plt.subplots(1, 2, figsize=(18, 8))

if not isinstance(axes, (list, np.ndarray)):
    axes = [axes]

plot_idx = 0

if access_compare is not None:
```

```python
    plot_grouped_bars(
        axes[plot_idx],
        access_compare,
        "Shelby healthcare access gaps",
        "Population-weighted gap / access burden"
    )
    plot_idx += 1

if burden_compare is not None:
    plot_grouped_bars(
        axes[plot_idx],
        burden_compare,
        "Shelby health burden proxies",
        "Population-weighted prevalence / burden"
    )

handles, labels = axes[0].get_legend_handles_labels()
fig.legend(
    handles,
    labels,
    loc="upper center",
    bbox_to_anchor=(0.5, 1.02),
    ncol=3,
    frameon=True
)

fig.text(
    0.02, 0.02,
    "Recovery-period tract-level proxy measures, population-weighted within
    fontsize=10
)

plt.tight_layout(rect=[0, 0.04, 1, 0.95])
plt.show()

# ---------------------------------------------------
# 9. Save
# ---------------------------------------------------
out_path = OUT_DIR / "shelby_split_access_burden_comparison.png"
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[13], line 16
     12 # ---------------------------------------------------
     13 # 1. Validate source table
     14 # ---------------------------------------------------
     15 if "shelby_cluster_input" not in globals():
---> 16     raise ValueError("`shelby_cluster_input` is not in memory.")
     18 plot_df = shelby_cluster_input.copy()
     20 # ---------------------------------------------------
     21 # 2. Choose metrics
     22 # ---------------------------------------------------

ValueError: `shelby_cluster_input` is not in memory.
```

```
In [ ]: # ============================================================
        # Cell 7G — Panel C: barriers / risk
        # ============================================================

        panel_c_df = plot_places_panel(
            summary_df=shelby_places_summary,
            panel_code="C",
            title="Panel C — Shelby community vs rest of Shelby County:\nBarriers ar
            outfile_name="shelby_places_panel_c_barriers_risk.png",
            extra_footer="Higher values indicate more risk behaviors or more barrier
        )

        display(panel_c_df.round(2) if panel_c_df is not None else pd.DataFrame())
```

```
In [ ]: # ============================================================
        # Cell 7H — ranked gap table
        # ============================================================

        ranked_gaps = (
            shelby_places_summary.loc[shelby_places_summary["selected_minus_rest"].r
            .sort_values("selected_minus_rest", ascending=False)
            .reset_index(drop=True)
        )

        print("Top metrics where the selected Shelby community is worse than the res
        display(ranked_gaps.head(20).round(2))

        ranked_gaps.to_csv(OUT_DIR / "shelby_places_ranked_gaps.csv", index=False)
        print("Saved:", OUT_DIR / "shelby_places_ranked_gaps.csv")
```

Cell 8A — Shelby tract-level base for comparison-community search

```
In [ ]: # ============================================================
        # Cell 8A — Shelby tract-level base for comparison-community search
        # TRACT-LEVEL ONLY:
        # - recovery economics from analysis_panel (tract-year -> tract average)
        # - CDC PLACES from shelby_places_base (tract snapshot)
        # ============================================================

        import pandas as pd
        import numpy as np

        RECOVERY_YEARS = [2022, 2023, 2024]

        # ------------------------------
        # 1) tract-level recovery economics
        # ------------------------------
        shelby_recovery = (
            analysis_panel.loc[
                (analysis_panel["display_state"].astype(str) == "Tennessee") &
                (analysis_panel["display_county"].astype(str) == "Shelby County") &
                (analysis_panel["year"].isin(RECOVERY_YEARS)),
                [
```

```
            "geoid", "year", "pop_total",
            "igs_total", "igs_economy",
            "poverty_rate", "unemp_rate",
            "median_household_income", "lfpr_16p"
        ]
    ]
    .copy()
)

shelby_recovery["geoid"] = normalize_geoid_series(shelby_recovery["geoid"])

shelby_tract_recovery = (
    shelby_recovery
    .groupby("geoid", as_index=False)
    .agg(
        pop_total_recovery=("pop_total", "mean"),
        igs_total_recovery=("igs_total", "mean"),
        igs_economy_recovery=("igs_economy", "mean"),
        poverty_rate_recovery=("poverty_rate", "mean"),
        unemp_rate_recovery=("unemp_rate", "mean"),
        median_household_income_recovery=("median_household_income", "mean")
        lfpr_16p_recovery=("lfpr_16p", "mean"),
    )
)

print("Shelby tract recovery rows:", shelby_tract_recovery.shape)
display(shelby_tract_recovery.head())

# -----------------------------
# 2) tract-level CDC PLACES features
# -----------------------------
shelby_cdc_tract = shelby_places_base.copy()
shelby_cdc_tract["geoid"] = normalize_geoid_series(shelby_cdc_tract["geoid"]

# Create tract-level gap versions where possible
positive_use_metrics = {
    "routine_checkup_pct": "routine_checkup_gap_pct",
    "dental_visit_pct": "dental_visit_gap_pct",
    "mammography_pct": "mammography_gap_pct",
    "flu_vax_pct": "flu_vax_gap_pct",
}

for src, dst in positive_use_metrics.items():
    if src in shelby_cdc_tract.columns:
        shelby_cdc_tract[dst] = 100 - pd.to_numeric(shelby_cdc_tract[src], e

cdc_metric_candidates = [
    # Panel A
    "uninsured_adults_pct",
    "routine_checkup_gap_pct",
    "dental_visit_gap_pct",
    "mammography_gap_pct",
    "flu_vax_gap_pct",

    # Panel B
    "fair_poor_health_pct",
```

```
    "poor_physical_health_pct",
    "poor_mental_health_pct",
    "depression_pct",
    "diabetes_pct",
    "obesity_pct",
    "current_asthma_pct",
    "coronary_heart_disease_pct",
    "stroke_pct",

    # Panel C
    "smoking_pct",
    "physical_inactivity_pct",
    "binge_drinking_pct",
    "sleep_lt7_pct",
    "food_insecurity_pct",
    "housing_insecurity_pct",
    "transportation_barriers_pct",
    "social_isolation_pct",
]

available_cdc_metrics = [
    c for c in cdc_metric_candidates
    if c in shelby_cdc_tract.columns and shelby_cdc_tract[c].notna().any()
]

shelby_cdc_tract = shelby_cdc_tract[["geoid"] + available_cdc_metrics].drop_

print("Available tract-level CDC metrics:", available_cdc_metrics)
print("Shelby CDC tract rows:", shelby_cdc_tract.shape)
display(shelby_cdc_tract.head())

# -----------------------------
# 3) full Shelby tract-level base
# -----------------------------
shelby_full_tract_base = (
    shelby_tract_recovery
    .merge(shelby_cdc_tract, on="geoid", how="left", validate="1:1")
    .copy()
)

print("Shelby full tract-level base:", shelby_full_tract_base.shape)
display(shelby_full_tract_base.head())
```

Cell 8B — load all Shelby tract geometries + tract adjacency

In [ ]:

```
# ============================================================
# Cell 8B — load all Shelby tract geometries + tract adjacency
# ============================================================

import geopandas as gpd
from pathlib import Path

# Reuse repo path discovery pattern
ROOT_CANDIDATES = [
```

```
    Path.cwd(),
    Path.cwd() / "src",
    Path.cwd().parent,
]

RAW_DIR = None
PROCESSED_DIR = None

for root in ROOT_CANDIDATES:
    if (root / "src/data/raw").exists():
        RAW_DIR = root / "src/data/raw"
        PROCESSED_DIR = root / "src/data/processed"
        break
    if (root / "data/raw").exists():
        RAW_DIR = root / "data/raw"
        PROCESSED_DIR = root / "data/processed"
        break

if RAW_DIR is None:
    raise FileNotFoundError("Could not find data/raw or src/data/raw.")

OUT_DIR = PROCESSED_DIR / "presentation_ready"
OUT_DIR.mkdir(parents=True, exist_ok=True)

# Find Tennessee tract shapefile
shp_candidates = list(RAW_DIR.rglob("*47*tract*.shp")) + list(RAW_DIR.rglob(

tract_shp = None
for p in shp_candidates:
    name = p.name.lower()
    parent = str(p.parent).lower()
    if "tract" in name and ("47" in name or "tennessee" in name or "state_tr
        tract_shp = p
        break

if tract_shp is None:
    raise FileNotFoundError("Could not find a Tennessee tract shapefile in c

print("Using tract shapefile:", tract_shp)

tracts = gpd.read_file(tract_shp).copy()

# Standardize GEOID
possible_geoid_cols = ["GEOID", "geoid", "GEOIDFQ", "TRACTCE"]
geoid_col = None
for c in possible_geoid_cols:
    if c in tracts.columns:
        geoid_col = c
        break

if geoid_col is None:
    raise ValueError(f"Could not find a GEOID-like column. Columns: {list(tr

if geoid_col == "TRACTCE":
    if not all(c in tracts.columns for c in ["STATEFP", "COUNTYFP", "TRACTCE
        raise ValueError("TRACTCE found, but STATEFP/COUNTYFP/TRACTCE not al
```

```
    tracts["geoid"] = (
        tracts["STATEFP"].astype(str).str.zfill(2)
        + tracts["COUNTYFP"].astype(str).str.zfill(3)
        + tracts["TRACTCE"].astype(str).str.zfill(6)
    )
else:
    tracts["geoid"] = tracts[geoid_col].astype(str).str.extract(r"(\d{11})",
    if tracts["geoid"].isna().all():
        tracts["geoid"] = tracts[geoid_col].astype(str).str.zfill(11)

# Keep Shelby County only
if "STATEFP" in tracts.columns and "COUNTYFP" in tracts.columns:
    shelby_geo = tracts.loc[
        (tracts["STATEFP"].astype(str).str.zfill(2) == "47") &
        (tracts["COUNTYFP"].astype(str).str.zfill(3) == "157")
    ].copy()
else:
    shelby_geo = tracts.loc[tracts["geoid"].str[:5] == "47157"].copy()

# Keep only tracts that exist in the tract-level base
shelby_geo["geoid"] = normalize_geoid_series(shelby_geo["geoid"])
shelby_geo = shelby_geo.loc[shelby_geo["geoid"].isin(shelby_full_tract_base[

print("Shelby geometry rows kept:", shelby_geo.shape)

# Build adjacency from touching polygons
left = shelby_geo[["geoid", "geometry"]].rename(columns={"geoid": "geoid_lef
right = shelby_geo[["geoid", "geometry"]].rename(columns={"geoid": "geoid_ri

adj = gpd.sjoin(left, right, how="inner", predicate="touches")
adj = adj.loc[adj["geoid_left"] != adj["geoid_right"], ["geoid_left", "geoic

print("Shelby adjacency edges:", adj.shape)
display(adj.head())
```

Cell 8C — full-county Shelby community map

In [ ]:

```
# ============================================================
# Cell 8C — full-county Shelby community map
# IMPORTANT:
# We do NOT use connected components alone, because Shelby is contiguous.
# We use spatially constrained clustering:
# - adjacency enforces contiguity
# - tract-level recovery + CDC metrics define similarity
# ============================================================

from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.sparse import coo_matrix

# Number of Shelby communities to create
# You can tune this later. 6 is a good first pass.
N_SHELBY_COMMUNITIES = 6
```

```python
# Merge tract base with geometry
shelby_cluster_input = (
    shelby_geo[["geoid", "geometry"]]
    .merge(shelby_full_tract_base, on="geoid", how="inner", validate="1:1")
    .copy()
)

# Tract-level features only
feature_candidates = [
    # Recovery economics
    "igs_total_recovery",
    "igs_economy_recovery",
    "poverty_rate_recovery",
    "unemp_rate_recovery",
    "median_household_income_recovery",
    "lfpr_16p_recovery",

    # Panel A
    "uninsured_adults_pct",
    "routine_checkup_gap_pct",
    "dental_visit_gap_pct",
    "mammography_gap_pct",
    "flu_vax_gap_pct",

    # Panel B
    "fair_poor_health_pct",
    "poor_physical_health_pct",
    "poor_mental_health_pct",
    "depression_pct",
    "diabetes_pct",
    "obesity_pct",
    "current_asthma_pct",
    "coronary_heart_disease_pct",
    "stroke_pct",

    # Panel C
    "smoking_pct",
    "physical_inactivity_pct",
    "binge_drinking_pct",
    "sleep_lt7_pct",
    "food_insecurity_pct",
    "housing_insecurity_pct",
    "transportation_barriers_pct",
    "social_isolation_pct",
]

available_features = [
    c for c in feature_candidates
    if c in shelby_cluster_input.columns and shelby_cluster_input[c].notna()
]

if not available_features:
    raise ValueError("No tract-level features available for Shelby clusterir

print("Features used for Shelby community map:")
print(available_features)
```

```python
X = shelby_cluster_input[available_features].copy()

# Flip higher-is-better economics so higher values consistently mean "more v
for col in ["igs_total_recovery", "igs_economy_recovery", "median_household_
    if col in X.columns:
        X[col] = -pd.to_numeric(X[col], errors="coerce")

# Impute + scale
X = SimpleImputer(strategy="median").fit_transform(X)
X = StandardScaler().fit_transform(X)

# Sparse adjacency matrix
geoid_order = shelby_cluster_input["geoid"].tolist()
idx_map = {g: i for i, g in enumerate(geoid_order)}

edge_df = adj.loc[
    adj["geoid_left"].isin(idx_map) & adj["geoid_right"].isin(idx_map)
].copy()

rows = edge_df["geoid_left"].map(idx_map).to_numpy()
cols = edge_df["geoid_right"].map(idx_map).to_numpy()
data = np.ones(len(edge_df), dtype=int)

connectivity = coo_matrix((data, (rows, cols)), shape=(len(geoid_order), len
connectivity = connectivity.maximum(connectivity.T)

# Spatially constrained clustering
try:
    cluster_model = AgglomerativeClustering(
        n_clusters=N_SHELBY_COMMUNITIES,
        linkage="ward",
        metric="euclidean",
        connectivity=connectivity
    )
except TypeError:
    cluster_model = AgglomerativeClustering(
        n_clusters=N_SHELBY_COMMUNITIES,
        linkage="ward",
        affinity="euclidean",
        connectivity=connectivity
    )

labels = cluster_model.fit_predict(X)

shelby_cluster_input["shelby_full_cluster_label"] = labels
shelby_cluster_input["shelby_full_cluster_id"] = [
    f"Shelby County full community {int(x)+1}"
    for x in labels
]

print("Shelby full-community tract counts:")
display(
    shelby_cluster_input["shelby_full_cluster_id"]
    .value_counts()
    .rename_axis("community")
```

```
    .reset_index(name="tract_count")
)

# Save tract-level community map
shelby_full_tract_cluster_map = shelby_cluster_input[["geoid", "geometry", "
shelby_full_tract_cluster_map.to_file(
    OUT_DIR / "shelby_full_tract_cluster_map.geojson",
    driver="GeoJSON"
)

print("Saved:", OUT_DIR / "shelby_full_tract_cluster_map.geojson")
```

Cell 8D — aggregate Shelby full communities

In [ ]:

```
# ============================================================
# Cell 8D — aggregate Shelby full communities
# TRACT-LEVEL ONLY aggregation
# ============================================================

def weighted_mean_safe(values: pd.Series, weights: pd.Series) -> float:
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

community_metric_cols = [c for c in available_features if c in shelby_cluste

community_rows = []
for community_id, grp in shelby_cluster_input.groupby("shelby_full_cluster_i
    weights = grp["pop_total_recovery"] if "pop_total_recovery" in grp.colum

    row = {
        "community_id": community_id,
        "n_tracts": int(grp["geoid"].nunique()),
        "population_sum": float(pd.to_numeric(grp["pop_total_recovery"], err
    }

    for col in community_metric_cols:
        row[col] = weighted_mean_safe(grp[col], weights)

    community_rows.append(row)

shelby_full_community_compare = pd.DataFrame(community_rows)

# Useful tract-level proxy summaries
access_cols = [
    c for c in [
        "uninsured_adults_pct",
        "routine_checkup_gap_pct",
        "dental_visit_gap_pct",
```

```
        "mammography_gap_pct",
        "flu_vax_gap_pct",
    ] if c in shelby_full_community_compare.columns
]

burden_cols = [
    c for c in [
        "fair_poor_health_pct",
        "poor_physical_health_pct",
        "poor_mental_health_pct",
        "depression_pct",
        "diabetes_pct",
        "obesity_pct",
        "current_asthma_pct",
        "coronary_heart_disease_pct",
        "stroke_pct",
    ] if c in shelby_full_community_compare.columns
]

barrier_cols = [
    c for c in [
        "smoking_pct",
        "physical_inactivity_pct",
        "binge_drinking_pct",
        "sleep_lt7_pct",
        "food_insecurity_pct",
        "housing_insecurity_pct",
        "transportation_barriers_pct",
        "social_isolation_pct",
    ] if c in shelby_full_community_compare.columns
]

if access_cols:
    shelby_full_community_compare["access_proxy_mean_pct"] = shelby_full_com
if burden_cols:
    shelby_full_community_compare["burden_proxy_mean_pct"] = shelby_full_com
if barrier_cols:
    shelby_full_community_compare["barrier_proxy_mean_pct"] = shelby_full_co

display_cols = [
    "community_id", "n_tracts", "population_sum",
    "igs_total_recovery", "igs_economy_recovery",
    "poverty_rate_recovery", "unemp_rate_recovery",
    "median_household_income_recovery", "lfpr_16p_recovery",
    "access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy_mean_pc
]
display_cols = [c for c in display_cols if c in shelby_full_community_compar

display(
    shelby_full_community_compare[display_cols]
    .sort_values(["igs_total_recovery", "igs_economy_recovery"], ascending=[
    .round(2)
)

shelby_full_community_compare.to_csv(OUT_DIR / "shelby_full_community_compar
print("Saved:", OUT_DIR / "shelby_full_community_compare.csv")
```

Cell 8E — selected Shelby community benchmark

In [ ]:

```python
# ============================================================
# Cell 8E — selected Shelby community benchmark
# Uses the SAME tract-level base as the full-county communities
# ============================================================

SELECTED_CLUSTER_ID = "Tennessee | Shelby County | cluster_77"

selected_geoids = set()

if "tract_cluster_map" in globals():
    tmp = tract_cluster_map.copy()
    if "geoid" in tmp.columns and "cluster_id" in tmp.columns:
        tmp["geoid"] = normalize_geoid_series(tmp["geoid"])
        selected_geoids = set(
            tmp.loc[tmp["cluster_id"].astype(str) == SELECTED_CLUSTER_ID, "g
            .dropna()
            .tolist()
        )

if not selected_geoids and "TARGET_GEOIDS" in globals():
    selected_geoids = set(normalize_geoid_series(pd.Series(TARGET_GEOIDS)).d

if not selected_geoids:
    raise ValueError("Could not recover the selected Shelby community GEOIDs

selected_benchmark_tracts = shelby_cluster_input.loc[
    shelby_cluster_input["geoid"].isin(selected_geoids)
].copy()

if selected_benchmark_tracts.empty:
    raise ValueError("Selected Shelby community has no overlap with the Shel

weights = selected_benchmark_tracts["pop_total_recovery"]

selected_benchmark = {
    "community_id": "Selected Shelby community",
    "n_tracts": int(selected_benchmark_tracts["geoid"].nunique()),
    "population_sum": float(pd.to_numeric(selected_benchmark_tracts["pop_tot
}

for col in community_metric_cols:
    selected_benchmark[col] = weighted_mean_safe(selected_benchmark_tracts[c

selected_benchmark = pd.Series(selected_benchmark)

# proxy summaries
if access_cols:
    selected_benchmark["access_proxy_mean_pct"] = np.nanmean([selected_bench
if burden_cols:
    selected_benchmark["burden_proxy_mean_pct"] = np.nanmean([selected_bench
if barrier_cols:
    selected_benchmark["barrier_proxy_mean_pct"] = np.nanmean([selected_benc
```

```
display(pd.DataFrame([selected_benchmark]).round(2))
```

Cell 8F — identify a better Shelby comparison community

In [ ]:

```
# ============================================================
# Cell 8F — identify a better Shelby comparison community
# Rules:
# - same county
# - no overlap with selected Shelby community
# - tract-level recovery IGS > 45
# - stronger recovery economics
# - lower access/burden/barrier proxy means where available
# ============================================================

# Count overlap with selected community
selected_geoid_set = set(selected_benchmark_tracts["geoid"].tolist())

overlap_rows = []
for community_id, grp in shelby_cluster_input.groupby("shelby_full_cluster_i
    overlap_rows.append({
        "community_id": community_id,
        "selected_overlap_tracts": int(grp["geoid"].isin(selected_geoid_set)
    })

overlap_df = pd.DataFrame(overlap_rows)

candidate_df = (
    shelby_full_community_compare
    .merge(overlap_df, on="community_id", how="left", validate="1:1")
    .copy()
)

# Hard filters
eligibility = pd.Series(True, index=candidate_df.index)

# Must not overlap selected cluster
eligibility &= candidate_df["selected_overlap_tracts"].fillna(0).eq(0)

# Avoid tiny communities
eligibility &= candidate_df["n_tracts"] >= 5
eligibility &= candidate_df["population_sum"] >= 10000

# Must be clearly less vulnerable on tract-level recovery IGS
if "igs_total_recovery" in candidate_df.columns and pd.notna(selected_benchm
    eligibility &= candidate_df["igs_total_recovery"] > 45
    eligibility &= candidate_df["igs_total_recovery"] > selected_benchmark["

# Better tract-level economics
for col, op in [
    ("igs_economy_recovery", "gt"),
    ("poverty_rate_recovery", "lt"),
    ("unemp_rate_recovery", "lt"),
    ("median_household_income_recovery", "gt"),
```

```
    ("lfpr_16p_recovery", "gt"),
]:
    if col in candidate_df.columns and pd.notna(selected_benchmark.get(col,
        if op == "gt":
            eligibility &= candidate_df[col] > selected_benchmark[col]
        else:
            eligibility &= candidate_df[col] < selected_benchmark[col]

# Better tract-level access / burden / barriers where available
for col in ["access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy
    if col in candidate_df.columns and pd.notna(selected_benchmark.get(col,
        eligibility &= candidate_df[col] < selected_benchmark[col]

eligible_candidates = candidate_df.loc[eligibility].copy()

print("Eligible better Shelby communities found:", len(eligible_candidates))

display_cols = [
    "community_id", "n_tracts", "population_sum",
    "igs_total_recovery", "igs_economy_recovery",
    "poverty_rate_recovery", "unemp_rate_recovery",
    "median_household_income_recovery", "lfpr_16p_recovery",
    "access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy_mean_pc
]
display_cols = [c for c in display_cols if c in eligible_candidates.columns]

display(
    eligible_candidates[display_cols]
    .sort_values(["igs_total_recovery", "igs_economy_recovery"], ascending=[
    .round(2)
)

if eligible_candidates.empty:
    print("No community passed the strict screen. Relax one filter and rerun
```

Cell 8G — rank the eligible candidates

In [ ]:

```
# ============================================================
# Cell 8G — rank the eligible candidates
# ============================================================

rank_df = eligible_candidates.copy()

if rank_df.empty:
    raise ValueError("No eligible candidates to rank. Relax the rules in Cel

score_cols = []

def add_rank_component(df, col, ascending, new_col):
    if col in df.columns:
        df[new_col] = df[col].rank(pct=True, ascending=ascending)
        score_cols.append(new_col)

# better if larger
add_rank_component(rank_df, "igs_total_recovery", ascending=True, new_col="r
```

```
add_rank_component(rank_df, "igs_economy_recovery", ascending=True, new_col=
add_rank_component(rank_df, "median_household_income_recovery", ascending=Tr
add_rank_component(rank_df, "lfpr_16p_recovery", ascending=True, new_col="ra

# better if smaller
add_rank_component(rank_df, "poverty_rate_recovery", ascending=False, new_co
add_rank_component(rank_df, "unemp_rate_recovery", ascending=False, new_col=
add_rank_component(rank_df, "access_proxy_mean_pct", ascending=False, new_co
add_rank_component(rank_df, "burden_proxy_mean_pct", ascending=False, new_co
add_rank_component(rank_df, "barrier_proxy_mean_pct", ascending=False, new_c

rank_df["comparison_rank_score"] = rank_df[score_cols].mean(axis=1, skipna=T

rank_df = rank_df.sort_values(
    ["comparison_rank_score", "igs_total_recovery", "igs_economy_recovery"],
    ascending=[False, False, False]
).reset_index(drop=True)

display(
    rank_df[
        [c for c in [
            "community_id", "comparison_rank_score",
            "n_tracts", "population_sum",
            "igs_total_recovery", "igs_economy_recovery",
            "poverty_rate_recovery", "unemp_rate_recovery",
            "median_household_income_recovery", "lfpr_16p_recovery",
            "access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy
        ] if c in rank_df.columns]
    ].head(10).round(3)
)

BEST_SHELBY_COMPARATOR_ID = rank_df.iloc[0]["community_id"]
print("Best Shelby comparator selected:", BEST_SHELBY_COMPARATOR_ID)

rank_df.to_csv(OUT_DIR / "shelby_better_community_candidates_ranked.csv", in
print("Saved:", OUT_DIR / "shelby_better_community_candidates_ranked.csv")
```

Cell 8H — selected vs better Shelby comparison table

In [ ]:

```
# ============================================================
# Cell 8H — selected vs better Shelby comparison table
# ============================================================

best_row = rank_df.loc[rank_df["community_id"] == BEST_SHELBY_COMPARATOR_ID]

selected_vs_better = pd.DataFrame([
    selected_benchmark.to_dict(),
    best_row.to_dict()
])

display_cols = [
    "community_id", "n_tracts", "population_sum",
    "igs_total_recovery", "igs_economy_recovery",
    "poverty_rate_recovery", "unemp_rate_recovery",
    "median_household_income_recovery", "lfpr_16p_recovery",
```

```
    "access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy_mean_pc
]
display_cols = [c for c in display_cols if c in selected_vs_better.columns]

display(selected_vs_better[display_cols].round(2))

selected_vs_better.to_csv(OUT_DIR / "selected_vs_better_shelby_community.csv
print("Saved:", OUT_DIR / "selected_vs_better_shelby_community.csv")
```

Cell 8I — map the 6 Shelby full communities

In [ ]:

```
# ============================================================
# Cell 8I — map the 6 Shelby full communities
# ============================================================

import geopandas as gpd
import pandas as pd
import matplotlib.pyplot as plt
import re

# ----------------------------------------------------
# 1) Recover community geometry table
# ----------------------------------------------------
if "shelby_full_tract_cluster_map" in globals():
    map_gdf = shelby_full_tract_cluster_map.copy()
elif "shelby_cluster_input" in globals():
    map_gdf = shelby_cluster_input[["geoid", "geometry", "shelby_full_cluste
else:
    raise ValueError(
        "Could not find shelby_full_tract_cluster_map or shelby_cluster_inpu
        "Run Cells 8B–8C first."
    )

if "shelby_full_cluster_id" not in map_gdf.columns:
    raise ValueError("Expected column 'shelby_full_cluster_id' not found.")

map_gdf = gpd.GeoDataFrame(map_gdf, geometry="geometry", crs=getattr(map_gdf
map_gdf["geoid"] = normalize_geoid_series(map_gdf["geoid"])

# Use Shelby geometry CRS if available
if "shelby_geo" in globals():
    county_base = shelby_geo.copy()
else:
    county_base = map_gdf.copy()

# Reproject for cleaner plotting
if map_gdf.crs is not None:
    map_gdf = map_gdf.to_crs(3857)
    county_base = county_base.to_crs(3857)

county_outline = county_base.dissolve()

# ----------------------------------------------------
# 2) Sort community IDs in numeric order
# ----------------------------------------------------
```

```python
def extract_num(s):
    m = re.search(r"(\d+)$", str(s))
    return int(m.group(1)) if m else 9999

community_order = sorted(map_gdf["shelby_full_cluster_id"].dropna().unique()
map_gdf["shelby_full_cluster_id"] = pd.Categorical(
    map_gdf["shelby_full_cluster_id"],
    categories=community_order,
    ordered=True
)

# --------------------------------------------------
# 3) Plot
# --------------------------------------------------
fig, ax = plt.subplots(figsize=(12, 10))

# Fill tracts by community
map_gdf.plot(
    ax=ax,
    column="shelby_full_cluster_id",
    categorical=True,
    cmap="tab20",
    edgecolor="white",
    linewidth=0.45,
    legend=True,
    legend_kwds={
        "title": "Shelby full communities",
        "loc": "upper left",
        "bbox_to_anchor": (1.02, 1.0)
    }
)

# County outline
county_outline.boundary.plot(
    ax=ax,
    color="black",
    linewidth=1.2,
    zorder=3
)

# --------------------------------------------------
# 4) Add labels at representative points
# --------------------------------------------------
community_shapes = map_gdf.dissolve(by="shelby_full_cluster_id").reset_index
community_shapes["label_pt"] = community_shapes.representative_point()

for _, row in community_shapes.iterrows():
    x, y = row["label_pt"].x, row["label_pt"].y
    label = str(row["shelby_full_cluster_id"]).replace("Shelby County full c
    ax.text(
        x, y, label,
        ha="center", va="center",
        fontsize=10,
        bbox=dict(boxstyle="round,pad=0.25", facecolor="white", edgecolor="b
        zorder=4
    )
```

```
ax.set_title("Shelby County full-community map", fontsize=17)
ax.set_axis_off()

plt.tight_layout()
plt.show()

# --------------------------------------------------
# 5) Save
# --------------------------------------------------
map_path = OUT_DIR / "shelby_full_communities_map.png"
fig.savefig(map_path, bbox_inches="tight", dpi=220)
print("Saved:", map_path)
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

```
# ============================================================
# Cell 8K — Shelby adult education comparison using Bachelor's+
# Uses acs_features.parquet -> ba_plus_share_25p
# ============================================================

import numpy as np
import pandas as pd
from pathlib import Path

if "three_group_base" not in globals():
    raise ValueError("Run Cell 8I first so three_group_base exists.")

# ------------------------------
# helpers
# ------------------------------
def _normalize_geoid(series: pd.Series) -> pd.Series:
    s = series.astype("string").str.strip()
    numeric = pd.to_numeric(s, errors="coerce")
    numeric_mask = numeric.notna()
    s = s.copy()
    s.loc[numeric_mask] = numeric.loc[numeric_mask].astype("Int64").astype("
    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    return s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

def _weighted_mean(values: pd.Series, weights: pd.Series) -> float:
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
```

```python
    if mask.sum() == 0:
        valid = values.notna()
        return float(values.loc[valid].mean()) if valid.sum() > 0 else np.na
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

def _to_share(s: pd.Series) -> pd.Series:
    s = pd.to_numeric(s, errors="coerce")
    if s.dropna().empty:
        return s
    if s.dropna().max() > 1.5:
        return s / 100.0
    return s

# ------------------------------
# load ACS features
# ------------------------------
acs_candidates = [
    Path("data/processed/acs_features.parquet"),
    Path("src/data/processed/acs_features.parquet"),
]

acs_path = next((p for p in acs_candidates if p.exists()), None)
if acs_path is None:
    raise FileNotFoundError("Could not find acs_features.parquet")

acs_edu = pd.read_parquet(acs_path).copy()
print("Using education source:", acs_path)

required_cols = ["geoid", "year", "pop_total", "ba_plus_share_25p"]
missing = [c for c in required_cols if c not in acs_edu.columns]
if missing:
    raise ValueError(f"acs_features.parquet is missing required columns: {mi

acs_edu["geoid"] = _normalize_geoid(acs_edu["geoid"])
acs_edu["year"] = pd.to_numeric(acs_edu["year"], errors="coerce").astype("In
acs_edu = acs_edu.loc[acs_edu["year"].isin(RECOVERY_YEARS)].copy()
acs_edu["year"] = acs_edu["year"].astype(int)

acs_edu["Bachelor's+"] = _to_share(acs_edu["ba_plus_share_25p"])

# ------------------------------
# 3-group lookup
# ------------------------------
group_lookup = (
    three_group_base[["geoid", "final_compare_group"]]
    .drop_duplicates()
    .copy()
)
group_lookup["geoid"] = _normalize_geoid(group_lookup["geoid"])

group_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]
```

```python
label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

acs_edu = acs_edu.merge(
    group_lookup,
    on="geoid",
    how="inner",
    validate="many_to_one"
)

# ----------------------------
# tract-level recovery average
# ----------------------------
education_tract = (
    acs_edu[
        ["geoid", "final_compare_group", "pop_total", "Bachelor's+"]
    ]
    .groupby(["geoid", "final_compare_group"], as_index=False)
    .agg({
        "pop_total": "mean",
        "Bachelor's+": "mean",
    })
    .copy()
)

# ----------------------------
# group-level summary
# ----------------------------
rows = []
for grp in group_order:
    sub = education_tract.loc[
        education_tract["final_compare_group"] == grp
    ].copy()

    row = {
        "group": grp,
        "group_label": label_map[grp],
        "n_tracts": int(sub["geoid"].nunique()),
        "population_sum": float(pd.to_numeric(sub["pop_total"], errors="coer
        "Bachelor's+": _weighted_mean(sub["Bachelor's+"], sub["pop_total"]),
    }
    rows.append(row)

education_compare = pd.DataFrame(rows)
education_compare["group"] = pd.Categorical(
    education_compare["group"],
    categories=group_order,
```

```
    ordered=True
)
education_compare = education_compare.sort_values("group").reset_index(drop=

education_mode = "adult_attainment_single_metric"

# ----------------------------
# save
# ----------------------------
if "OUT_DIR" not in globals():
    OUT_DIR = Path("data/processed")

DEMO_OUT_DIR = OUT_DIR if Path(OUT_DIR).name == "presentation_ready" else (F
DEMO_OUT_DIR.mkdir(parents=True, exist_ok=True)

education_compare.to_csv(DEMO_OUT_DIR / "shelby_education_compare.csv", inde

print("\nSaved:", DEMO_OUT_DIR / "shelby_education_compare.csv")
display(education_compare.round(4))
```

In [ ]:

```
# ============================================================
# Cell 8L — Shelby demographic comparison chart (vertical)
# ============================================================

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.ticker import PercentFormatter

if "demographic_compare" not in globals():
    raise ValueError("Run Cell 8K first.")

plot_groups = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

meta_cols = {"group", "group_label", "n_tracts", "population_sum"}
plot_categories = [c for c in demographic_compare.columns if c not in meta_c

demo_plot_df = demographic_compare.copy()
demo_plot_df = demo_plot_df.set_index("group").loc[plot_groups].reset_index(

fig, ax = plt.subplots(figsize=(13, 8))

x = np.arange(len(plot_categories))
bar_w = 0.24
offsets = [-bar_w, 0, bar_w]

for grp, off in zip(plot_groups, offsets):
    row = demo_plot_df.loc[demo_plot_df["group"] == grp].iloc[0]
    vals = [row.get(cat, np.nan) for cat in plot_categories]

    ax.bar(
        x + off,
        vals,
```

```python
        width=bar_w,
        color=color_map[grp],
        label=label_map[grp]
    )

    for i, v in enumerate(vals):
        if pd.notna(v) and v >= 0.05 and v <= 1.05:
            ax.text(
                x[i] + off,
                v + 0.015,
                f"{v:.0%}",
                ha="center",
                va="bottom",
                fontsize=10
            )

ax.set_xticks(x)
ax.set_xticklabels(plot_categories, fontsize=11, rotation=15)
ymax = max(0.65, float(np.nanmax(demo_plot_df[plot_categories].to_numpy())))
ax.set_ylim(0, min(ymax, 1.0))
ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
ax.grid(axis="y", alpha=0.25)
ax.set_title("Shelby demographic comparison", fontsize=18, pad=14)
ax.legend(loc="upper center", bbox_to_anchor=(0.5, 1.12), ncol=3, frameon=Tr

fig.text(
    0.02, 0.02,
    "Recovery-period averages (2022–2024), aggregated from tract-level race
    fontsize=10
)

plt.tight_layout(rect=[0, 0.05, 1, 0.93])
plt.show()

out_path = DEMO_OUT_DIR / "shelby_visual_demographic_comparison.png"
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:

```python
# ============================================================
# Cell 8M — Shelby adult education comparison (vertical)
# Bachelor's+ only
# ============================================================

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.ticker import PercentFormatter

if "education_compare" not in globals() or education_compare is None:
    raise ValueError("Run the Bachelor's+ education build cell first.")

plot_groups = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]
```

```
edu_plot_df = education_compare.copy()
edu_plot_df = edu_plot_df.set_index("group").loc[plot_groups].reset_index()

fig, ax = plt.subplots(figsize=(10, 7))

x = np.arange(len(plot_groups))
vals = [edu_plot_df.loc[edu_plot_df["group"] == grp, "Bachelor's+"].iloc[0]
colors = [color_map[grp] for grp in plot_groups]
labels = [label_map[grp] for grp in plot_groups]

bars = ax.bar(x, vals, color=colors, width=0.65)

for i, v in enumerate(vals):
    if pd.notna(v):
        ax.text(
            x[i],
            v + 0.01,
            f"{v:.1%}",
            ha="center",
            va="bottom",
            fontsize=11
        )

ax.set_xticks(x)
ax.set_xticklabels(labels, fontsize=11, rotation=0)
ax.set_ylim(0, min(max(vals) + 0.08, 1.0))
ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
ax.grid(axis="y", alpha=0.25)

ax.set_title("Shelby adult education attainment comparison", fontsize=18, pa

fig.text(
    0.02, 0.02,
    "Share of adults age 25+ with a bachelor's degree or higher, using recov
    fontsize=10
)

plt.tight_layout(rect=[0, 0.05, 1, 0.95])
plt.show()

out_path = DEMO_OUT_DIR / "shelby_visual_education_comparison.png"
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:

Cell 8I — final 3-group tract-level comparison

In [ ]:

```
# ============================================================
# Cell 8I — final 3-group tract-level comparison
# Groups:
# 1) Selected low-growth Shelby community (original 96-tract community)
# 2) Selected high-growth Shelby comparator (chosen from full-county partiti
# 3) Rest of Shelby County excluding both selected communities
```

```python
# TRACT-LEVEL ONLY
# ============================================================

import pandas as pd
import numpy as np

required_objs = [
    "shelby_cluster_input",
    "selected_benchmark_tracts",
    "BEST_SHELBY_COMPARATOR_ID",
    "community_metric_cols"
]
for obj in required_objs:
    if obj not in globals():
        raise ValueError(f"Missing required object: {obj}")

# ----------------------------------------------------
# 1) tract sets
# ----------------------------------------------------
all_shelby_geoids = set(shelby_cluster_input["geoid"].dropna().tolist())

low_growth_geoids = set(selected_benchmark_tracts["geoid"].dropna().tolist()

high_growth_geoids = set(
    shelby_cluster_input.loc[
        shelby_cluster_input["shelby_full_cluster_id"].astype(str) == str(BE
        "geoid"
    ].dropna().tolist()
)

# Remove any accidental overlap
high_growth_geoids = high_growth_geoids - low_growth_geoids

rest_geoids = all_shelby_geoids - low_growth_geoids - high_growth_geoids

print("Low-growth selected community tracts:", len(low_growth_geoids))
print("High-growth selected comparator tracts:", len(high_growth_geoids))
print("Rest of Shelby County tracts:", len(rest_geoids))
print("All Shelby tracts:", len(all_shelby_geoids))
print("Check sum:", len(low_growth_geoids) + len(high_growth_geoids) + len(r

assert len(low_growth_geoids) + len(high_growth_geoids) + len(rest_geoids) =
    "The 3-group partition does not sum to all Shelby tracts."

# ----------------------------------------------------
# 2) assign final 3-group labels
# ----------------------------------------------------
three_group_base = shelby_cluster_input.copy()

def assign_group(g):
    if g in low_growth_geoids:
        return "Selected low-growth community"
    elif g in high_growth_geoids:
        return "Selected high-growth comparator"
    else:
        return "Rest of Shelby County (excluding both selected communities)"
```

```python
three_group_base["final_compare_group"] = three_group_base["geoid"].map(assi

print(three_group_base["final_compare_group"].value_counts(dropna=False))

# ---------------------------------------------------
# 3) weighted aggregation helper
# ---------------------------------------------------
def weighted_mean_safe(values: pd.Series, weights: pd.Series) -> float:
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

# ---------------------------------------------------
# 4) aggregate tract-level metrics into the 3 groups
# ---------------------------------------------------
three_group_rows = []

for grp_name, grp in three_group_base.groupby("final_compare_group", dropna=
    weights = grp["pop_total_recovery"] if "pop_total_recovery" in grp.colum

    row = {
        "group": grp_name,
        "n_tracts": int(grp["geoid"].nunique()),
        "population_sum": float(pd.to_numeric(grp["pop_total_recovery"], err
    }

    for col in community_metric_cols:
        if col in grp.columns:
            row[col] = weighted_mean_safe(grp[col], weights)

    three_group_rows.append(row)

three_group_compare = pd.DataFrame(three_group_rows)

# ---------------------------------------------------
# 5) tract-level proxy summaries
# ---------------------------------------------------
access_cols = [
    c for c in [
        "uninsured_adults_pct",
        "routine_checkup_gap_pct",
        "dental_visit_gap_pct",
        "mammography_gap_pct",
        "flu_vax_gap_pct",
    ] if c in three_group_compare.columns
]

burden_cols = [
    c for c in [
```

```
        "fair_poor_health_pct",
        "poor_physical_health_pct",
        "poor_mental_health_pct",
        "depression_pct",
        "diabetes_pct",
        "obesity_pct",
        "current_asthma_pct",
        "coronary_heart_disease_pct",
        "stroke_pct",
    ] if c in three_group_compare.columns
]

barrier_cols = [
    c for c in [
        "smoking_pct",
        "physical_inactivity_pct",
        "binge_drinking_pct",
        "sleep_lt7_pct",
        "food_insecurity_pct",
        "housing_insecurity_pct",
        "transportation_barriers_pct",
        "social_isolation_pct",
    ] if c in three_group_compare.columns
]

if access_cols:
    three_group_compare["access_proxy_mean_pct"] = three_group_compare[acces
if burden_cols:
    three_group_compare["burden_proxy_mean_pct"] = three_group_compare[burde
if barrier_cols:
    three_group_compare["barrier_proxy_mean_pct"] = three_group_compare[barr

display_cols = [
    "group", "n_tracts", "population_sum",
    "igs_total_recovery", "igs_economy_recovery",
    "poverty_rate_recovery", "unemp_rate_recovery",
    "median_household_income_recovery", "lfpr_16p_recovery",
    "access_proxy_mean_pct", "burden_proxy_mean_pct", "barrier_proxy_mean_pc
]
display_cols = [c for c in display_cols if c in three_group_compare.columns]

display(three_group_compare[display_cols].round(2))

three_group_compare.to_csv(OUT_DIR / "three_group_selected_vs_comparator_vs_
print("Saved:", OUT_DIR / "three_group_selected_vs_comparator_vs_rest.csv")
```

Cell 8J— 3-group Shelby map

In [ ]:

```
# ================================================================
# Cell 8J — 3-group Shelby map
# Groups:
# 1) Selected low-growth community
# 2) Selected high-growth comparator
# 3) Rest of Shelby County excluding both
# ================================================================
```

```
import geopandas as gpd
import matplotlib.pyplot as plt
import pandas as pd

if "three_group_base" not in globals():
    raise ValueError("Run Cell 8I first so three_group_base exists.")

map_gdf = gpd.GeoDataFrame(three_group_base.copy(), geometry="geometry", crs

if map_gdf.crs is not None:
    map_gdf = map_gdf.to_crs(3857)

group_order = [
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
    "Selected low-growth community",
]

color_map = {
    "Rest of Shelby County (excluding both selected communities)": "#B05EF1"
    "Selected high-growth comparator": "#231971",  # dark blue/purple
    "Selected low-growth community": "#910D0D",    # dark red
}

map_gdf["final_compare_group"] = pd.Categorical(
    map_gdf["final_compare_group"],
    categories=group_order,
    ordered=True
)

fig, ax = plt.subplots(figsize=(12, 10))

# Plot each group separately so colors stay controlled
for grp in group_order:
    sub = map_gdf.loc[map_gdf["final_compare_group"] == grp].copy()
    if not sub.empty:
        sub.plot(
            ax=ax,
            color=color_map[grp],
            edgecolor="white",
            linewidth=0.45,
            label=grp,
            zorder=2
        )

# Outer county outline
county_outline = map_gdf.dissolve()
county_outline.boundary.plot(ax=ax, color="black", linewidth=1.2, zorder=3)

# Add labels
label_shapes = map_gdf.dissolve(by="final_compare_group").reset_index()
label_shapes["label_pt"] = label_shapes.representative_point()

label_text_map = {
    "Rest of Shelby County (excluding both selected communities)": "Rest of
```

```
    "Selected high-growth comparator": "High-growth comparator",
    "Selected low-growth community": "Low-growth selected community",
}

for _, row in label_shapes.iterrows():
    x, y = row["label_pt"].x, row["label_pt"].y
    label = label_text_map.get(row["final_compare_group"], row["final_compar
    ax.text(
        x, y, label,
        ha="center", va="center",
        fontsize=10,
        bbox=dict(boxstyle="round,pad=0.25", facecolor="white", edgecolor="b
        zorder=4
    )

ax.set_title("Shelby County: low-growth selected community vs high-growth co
ax.set_axis_off()
ax.legend(title="3-group comparison", loc="upper left", bbox_to_anchor=(1.02

plt.tight_layout()
plt.show()

map_path = OUT_DIR / "shelby_three_group_map.png"
fig.savefig(map_path, bbox_inches="tight", dpi=220)
print("Saved:", map_path)
```

Cell 8K — 3-group bar chart for IGS / economic metrics

In [ ]:

```
# ============================================================
# Cell 8K — individual 3-group comparison charts for IGS/economic metrics
# Updated:
# - one chart per metric
# - removes x-axis labels/ticks
# - keeps legend/key
# ============================================================

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from matplotlib.patches import Patch

if "three_group_compare" not in globals():
    raise ValueError("Run Cell 8I first so three_group_compare exists.")

plot_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

colors = {
    "Selected low-growth community": "#fb6a4a",
    "Rest of Shelby County (excluding both selected communities)": "#bdbdbd"
    "Selected high-growth comparator": "#74c476",
}
```

```
metric_specs = {
    "igs_total_recovery": {
        "label": "IGS Total",
        "mult": 1,
        "fmt": lambda v: f"{v:.1f}",
        "ylabel": "Score"
    },
    "poverty_rate_recovery": {
        "label": "Poverty Rate",
        "mult": 100,
        "fmt": lambda v: f"{v:.1f}%",
        "ylabel": "Percent"
    },
    "unemp_rate_recovery": {
        "label": "Unemployment Rate",
        "mult": 100,
        "fmt": lambda v: f"{v:.1f}%",
        "ylabel": "Percent"
    },
    "median_household_income_recovery": {
        "label": "Median Household Income",
        "mult": 1,
        "fmt": lambda v: f"${v:,.0f}",
        "ylabel": "Dollars"
    },
    "lfpr_16p_recovery": {
        "label": "Labor Force Participation",
        "mult": 100,
        "fmt": lambda v: f"{v:.1f}%",
        "ylabel": "Percent"
    },
}

legend_handles = [
    Patch(facecolor=colors["Selected low-growth community"], label="Our comm
    Patch(facecolor=colors["Selected high-growth comparator"], label="High-g
    Patch(facecolor=colors["Rest of Shelby County (excluding both selected c
]

for metric, spec in metric_specs.items():
    if metric not in three_group_compare.columns:
        continue

    plot_df = (
        three_group_compare
        .set_index("group")
        .loc[plot_order, [metric]]
        .copy()
    )

    vals = plot_df[metric].astype(float) * spec["mult"]
    x = np.arange(len(plot_order))

    fig, ax = plt.subplots(figsize=(8, 6))
```

```
    bars = ax.bar(
        x,
        vals.values,
        color=[colors[g] for g in plot_order],
        width=0.6
    )

    ymax = np.nanmax(vals.values)
    pad = max(ymax * 0.03, 0.3)

    for bar, val in zip(bars, vals.values):
        if pd.notna(val):
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + pad,
                spec["fmt"](val),
                ha="center",
                va="bottom",
                fontsize=10
            )

    # Remove x-axis completely
    ax.set_xticks([])
    ax.set_xlabel("")
    ax.tick_params(axis="x", length=0)

    ax.set_ylabel(spec["ylabel"])
    ax.set_title(spec["label"])
    ax.grid(axis="y", alpha=0.25)
    ax.set_ylim(0, ymax * 1.15 if ymax > 0 else 1)

    ax.legend(handles=legend_handles, frameon=True, loc="upper left")

    plt.tight_layout()
    plt.show()

    out_path = OUT_DIR / f"shelby_three_group_{metric}_individual.png"
    fig.savefig(out_path, bbox_inches="tight", dpi=220)
    print("Saved:", out_path)
```

Cell 8L — helper for 3-group health charts

In [ ]:

```
# ============================================================
# Cell 8L — helper for individual 3-group health charts
# ============================================================

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from matplotlib.patches import Patch

if "three_group_compare" not in globals():
    raise ValueError("Run Cell 8I first so three_group_compare exists.")

plot_order = [
```

```
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

health_colors = {
    "Selected low-growth community": "#fb6a4a",
    "Rest of Shelby County (excluding both selected communities)": "#bdbdbd"
    "Selected high-growth comparator": "#74c476",
}

health_legend_handles = [
    Patch(facecolor=health_colors["Selected low-growth community"], label="C
    Patch(facecolor=health_colors["Selected high-growth comparator"], label=
    Patch(facecolor=health_colors["Rest of Shelby County (excluding both sel
]

def plot_three_group_metric(
    df,
    metric,
    title,
    ylabel="Percent",
    fmt_func=None,
    filename=None,
    footnote=None
):
    if metric not in df.columns:
        print(f"Skipping {metric}: not found in three_group_compare.")
        return

    plot_df = (
        df.set_index("group")
          .loc[plot_order, [metric]]
          .copy()
    )

    vals = pd.to_numeric(plot_df[metric], errors="coerce")
    x = np.arange(len(plot_order))

    fig, ax = plt.subplots(figsize=(8, 6))

    bars = ax.bar(
        x,
        vals.values,
        color=[health_colors[g] for g in plot_order],
        width=0.6
    )

    ymax = np.nanmax(vals.values)
    if not np.isfinite(ymax):
        ymax = 1.0
    pad = max(ymax * 0.03, 0.3)

    if fmt_func is None:
        fmt_func = lambda v: f"{v:.1f}%"
```

```python
    for bar, val in zip(bars, vals.values):
        if pd.notna(val):
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + pad,
                fmt_func(val),
                ha="center",
                va="bottom",
                fontsize=10
            )

    # remove x-axis labels/ticks
    ax.set_xticks([])
    ax.set_xlabel("")
    ax.tick_params(axis="x", length=0)

    ax.set_ylabel(ylabel)
    ax.set_title(title)
    ax.grid(axis="y", alpha=0.25)
    ax.set_ylim(0, ymax * 1.15 if ymax > 0 else 1)

    ax.legend(handles=health_legend_handles, frameon=True, loc="upper left")

    if footnote:
        fig.text(0.02, 0.02, footnote, fontsize=9)

    plt.tight_layout(rect=[0, 0.04 if footnote else 0, 1, 1])
    plt.show()

    if filename:
        out_path = OUT_DIR / filename
        fig.savefig(out_path, bbox_inches="tight", dpi=220)
        print("Saved:", out_path)
```

Cell 8M — summary health proxy visuals

In [ ]:

```python
# ============================================================
# Cell 8M — 3-group summary health proxy visuals
# Higher = worse
# ============================================================

summary_health_specs = {
    "access_proxy_mean_pct": {
        "title": "Healthcare Access Proxy Mean",
        "ylabel": "Percent / proxy mean"
    },
    "burden_proxy_mean_pct": {
        "title": "Broader Health Burden Proxy Mean",
        "ylabel": "Percent / proxy mean"
    },
    "barrier_proxy_mean_pct": {
        "title": "Health Barrier / Risk Proxy Mean",
        "ylabel": "Percent / proxy mean"
    },
}
```

```
for metric, spec in summary_health_specs.items():
    if metric in three_group_compare.columns:
        plot_three_group_metric(
            df=three_group_compare,
            metric=metric,
            title=spec["title"],
            ylabel=spec["ylabel"],
            fmt_func=lambda v: f"{v:.1f}",
            filename=f"three_group_{metric}.png",
            footnote="Higher values indicate worse access gaps, burden, or b
        )
```

Cell 8N — detailed healthcare access / prevention visuals

In [ ]:
```
# ============================================================
# Cell 8N — detailed 3-group healthcare access / prevention visuals
# Higher = worse for all gap metrics
# ============================================================

access_specs = {
    "uninsured_adults_pct": {
        "title": "Uninsured Adults",
        "ylabel": "Percent"
    },
    "routine_checkup_gap_pct": {
        "title": "Routine Checkup Gap",
        "ylabel": "Gap percent"
    },
    "dental_visit_gap_pct": {
        "title": "Dental Visit Gap",
        "ylabel": "Gap percent"
    },
    "mammography_gap_pct": {
        "title": "Mammography Gap",
        "ylabel": "Gap percent"
    },
    "flu_vax_gap_pct": {
        "title": "Influenza Vaccination Gap",
        "ylabel": "Gap percent"
    },
}

for metric, spec in access_specs.items():
    if metric in three_group_compare.columns and three_group_compare[metric]
        plot_three_group_metric(
            df=three_group_compare,
            metric=metric,
            title=spec["title"],
            ylabel=spec["ylabel"],
            fmt_func=lambda v: f"{v:.1f}%",
            filename=f"three_group_{metric}.png",
```

```
            footnote="Selected low-growth community vs high-growth Shelby cc
        )
```

Cell 8O — detailed broader health burden visuals

In [ ]:
```
# ============================================================
# Cell 8O — detailed 3-group broader health burden visuals
# Higher = worse
# ============================================================

burden_specs = {
    "fair_poor_health_pct": "Poor / Fair Health",
    "poor_physical_health_pct": "Poor Physical Health",
    "poor_mental_health_pct": "Poor Mental Health",
    "depression_pct": "Depression",
    "diabetes_pct": "Diabetes",
    "obesity_pct": "Obesity",
    "current_asthma_pct": "Current Asthma",
    "coronary_heart_disease_pct": "Coronary Heart Disease",
    "stroke_pct": "Stroke",
}

for metric, title in burden_specs.items():
    if metric in three_group_compare.columns and three_group_compare[metric]
        plot_three_group_metric(
            df=three_group_compare,
            metric=metric,
            title=title,
            ylabel="Percent",
            fmt_func=lambda v: f"{v:.1f}%",
            filename=f"three_group_{metric}.png",
            footnote="Higher values indicate worse health burden."
        )
```

Cell 8P — detailed barriers / risk visuals

In [ ]:
```
# ============================================================
# Cell 8P — detailed 3-group barriers / risk visuals
# Higher = worse
# ============================================================

barrier_specs = {
    "smoking_pct": "Current Smoking",
    "physical_inactivity_pct": "Physical Inactivity",
    "binge_drinking_pct": "Binge Drinking",
    "sleep_lt7_pct": "Sleeping Less Than 7 Hours",
    "food_insecurity_pct": "Food Insecurity",
    "housing_insecurity_pct": "Housing Insecurity",
    "transportation_barriers_pct": "Transportation Barriers",
    "social_isolation_pct": "Social Isolation",
}

for metric, title in barrier_specs.items():
```

```
    if metric in three_group_compare.columns and three_group_compare[metric]
        plot_three_group_metric(
            df=three_group_compare,
            metric=metric,
            title=title,
            ylabel="Percent",
            fmt_func=lambda v: f"{v:.1f}%",
            filename=f"three_group_{metric}.png",
            footnote="Higher values indicate worse barriers or risk conditio
        )
```

In [ ]:

```

```

Cell 7 — load County Health Rankings 2023 healthcare data from analytic_data2023_0.csv

In [ ]:

```
# # Cell 7 — load and join multi-year County Health Rankings healthcare data

# from pathlib import Path
# import pandas as pd
# import numpy as np

# HEALTHCARE_DIR = RAW_DIR / "healthcare"
# HEALTHCARE_DIR.mkdir(parents=True, exist_ok=True)

# # Only load actual County Health Rankings analytic files
# healthcare_files = sorted(HEALTHCARE_DIR.glob("analytic_data*.csv"))

# if len(healthcare_files) == 0:
#     raise FileNotFoundError(
#         f"No County Health Rankings analytic_data*.csv files found in {HEA
#     )

# print("Healthcare files found:")
# for p in healthcare_files:
#     print("-", p.name)

# # --------------------------------------------------
# # A. Helpers
# # --------------------------------------------------
# def normalize_county_fips(series):
#     s = series.astype("string").str.strip()
#     s = s.str.replace(r"\.0$", "", regex=True)
#     s = s.str.replace(r"\D", "", regex=True)
#     s = s.str.zfill(5)
#     return s.where(s.str.fullmatch(r"\d{5}"), pd.NA)

# def find_first_matching_column(df, candidates):
#     for c in candidates:
#         if c in df.columns:
#             return c
#     return None

# def find_first_startswith_column(df, prefixes):
```

```
#     lowered = {str(c).strip().lower(): c for c in df.columns}
#     for prefix in prefixes:
#         prefix_l = str(prefix).strip().lower()
#         for col_l, original_col in lowered.items():
#             if col_l.startswith(prefix_l):
#                 return original_col
#     return None

# def standardize_chr_file(chr_raw, source_name):
#     chr_df = chr_raw.copy()

#     fips_col = find_first_matching_column(chr_df, [
#         "5-digit FIPS Code",
#         "5 digit FIPS Code",
#         "5-digit FIPS code",
#         "FIPS Code",
#     ])
#     if fips_col is None:
#         raise ValueError(f"{source_name}: could not find county FIPS colum

#     year_col = find_first_matching_column(chr_df, [
#         "Release Year",
#         "release year",
#         "Year",
#     ])

#     state_col = find_first_matching_column(chr_df, [
#         "State Abbreviation",
#         "State",
#     ])

#     name_col = find_first_matching_column(chr_df, [
#         "Name",
#         "County",
#     ])

#     chr_df["county_fips"] = normalize_county_fips(chr_df[fips_col])

#     if year_col is not None:
#         chr_df["year"] = pd.to_numeric(chr_df[year_col], errors="coerce").
#     else:
#         chr_df["year"] = pd.NA

#     chr_df["chr_state"] = chr_df[state_col].astype("string") if state_col
#     chr_df["chr_county"] = chr_df[name_col].astype("string") if name_col i

#     health_measure_candidates = {
#         "uninsured_rate": ["Uninsured raw value"],
#         "uninsured_adults_rate": ["Uninsured Adults raw value"],
#         "uninsured_children_rate": ["Uninsured Children raw value"],
#         "primary_care_physicians_rate": ["Primary Care Physicians raw valu
#         "dentists_rate": ["Dentists raw value"],
#         "mental_health_providers_rate": ["Mental Health Providers raw valu
#         "other_primary_care_providers_rate": ["Other Primary Care Provider
#         "preventable_hospital_stays_rate": ["Preventable Hospital Stays ra
#         "mammography_screening_rate": ["Mammography Screening raw value"],
```

```
#         "flu_vaccinations_rate": ["Flu Vaccinations raw value"],
#     }

#     resolved_measure_cols = {}

#     for clean_name, candidates in health_measure_candidates.items():
#         exact_col = find_first_matching_column(chr_df, candidates)
#         if exact_col is not None:
#             resolved_measure_cols[exact_col] = clean_name
#             continue

#         prefix_col = find_first_startswith_column(chr_df, candidates)
#         if prefix_col is not None:
#             resolved_measure_cols[prefix_col] = clean_name

#     if len(resolved_measure_cols) == 0:
#         possible_cols = [c for c in chr_df.columns if "raw value" in str(c
#         raise ValueError(
#             f"{source_name}: could not find expected healthcare measure co
#             f"Example raw-value columns: {possible_cols[:25]}"
#         )

#     keep_cols = ["county_fips", "year", "chr_state", "chr_county"] + list(

#     out = (
#         chr_df[keep_cols]
#         .copy()
#         .rename(columns=resolved_measure_cols)
#     )

#     for col in out.columns:
#         if col in ["county_fips", "year", "chr_state", "chr_county"]:
#             continue
#         out[col] = pd.to_numeric(out[col], errors="coerce")

#     out["source_file"] = source_name

#     out = (
#         out
#         .dropna(subset=["county_fips", "year"])
#         .reset_index(drop=True)
#     )

#     return out

# # --------------------------------------------------
# # B. Load and standardize all CHR files
# # --------------------------------------------------
# healthcare_frames = []

# for fp in healthcare_files:
#     try:
#         df_raw = pd.read_csv(fp, low_memory=False)
#         # print (df_raw)
#         df_std = standardize_chr_file(df_raw, fp.name)
#         healthcare_frames.append(df_std)
```

```
#         print(f"Loaded {fp.name}: {df_std.shape}")
#     except Exception as e:
#         print(f"Skipped {fp.name}: {e}")

# if len(healthcare_frames) == 0:
#     raise RuntimeError("No healthcare CSVs could be standardized successfu

# healthcare_standardized = pd.concat(healthcare_frames, ignore_index=True)

# # Drop U.S. total and state rollup rows
# healthcare_standardized = healthcare_standardized.loc[
#     healthcare_standardized["county_fips"].notna()
#     & (healthcare_standardized["county_fips"] != "00000")
#     & (~healthcare_standardized["county_fips"].astype(str).str.endswith("0
# ].copy()

# # Keep the last-loaded version if a county-year repeats
# healthcare_standardized = (
#     healthcare_standardized
#     .sort_values(["county_fips", "year", "source_file"])
#     .drop_duplicates(subset=["county_fips", "year"], keep="last")
#     .reset_index(drop=True)
# )

# healthcare_join_level = "county"

# healthcare_metric_cols = [
#     c for c in healthcare_standardized.columns
#     if c not in ["county_fips", "year", "chr_state", "chr_county", "source
# ]

# available_healthcare_years = sorted(
#     healthcare_standardized["year"].dropna().astype(int).unique().tolist()
# )

# print("\nCombined healthcare shape:", healthcare_standardized.shape)
# print("Available healthcare years:", available_healthcare_years)
# print("Healthcare metric cols:", healthcare_metric_cols)

# # -----------------------------------------------------
# # C. Validate enrichment inputs from Cell 6
# # -----------------------------------------------------
# if "enrichment_clusters" not in globals() or len(enrichment_clusters) == 0
#     raise ValueError("enrichment_clusters is empty or missing. Rerun Cell

# if "enrichment_tract_year_panel" not in globals() or len(enrichment_tract_
#     raise ValueError("enrichment_tract_year_panel is empty or missing. Rer

# if "enrichment_cluster_year_panel" not in globals() or len(enrichment_clus
#     raise ValueError("enrichment_cluster_year_panel is empty or missing. R

# # -----------------------------------------------------
# # D. Build YEAR-SPECIFIC county weights from shortlisted clusters
# # FIX:
# # - use actual year-specific tract populations
# # - sum tract populations to county-year totals
```

```
# # - compute county share within cluster-year
# # ------------------------------------------------------
# cluster_county_weights = (
#     enrichment_tract_year_panel[
#         ["cluster_id", "year", "geoid", "pop_total"]
#     ]
#     .copy()
# )

# cluster_county_weights["geoid"] = cluster_county_weights["geoid"].astype("
# cluster_county_weights["county_fips"] = normalize_county_fips(
#     cluster_county_weights["geoid"].str[:5]
# )
# cluster_county_weights["pop_total"] = pd.to_numeric(
#     cluster_county_weights["pop_total"], errors="coerce"
# )

# cluster_county_weights = cluster_county_weights.loc[
#     cluster_county_weights["year"].isin(available_healthcare_years)
# ].copy()

# if len(cluster_county_weights) == 0:
#     raise ValueError("No tract-year rows remain after restricting to healt

# cluster_county_weights = (
#     cluster_county_weights
#     .dropna(subset=["cluster_id", "year", "county_fips", "pop_total"])
#     .groupby(["cluster_id", "year", "county_fips"], as_index=False)
#     .agg(cluster_county_pop=("pop_total", "sum"))
# )

# cluster_county_weights["cluster_total_pop_for_weights"] = (
#     cluster_county_weights.groupby(["cluster_id", "year"])["cluster_county
# )

# cluster_county_weights["cluster_county_weight"] = np.where(
#     cluster_county_weights["cluster_total_pop_for_weights"] > 0,
#     cluster_county_weights["cluster_county_pop"] / cluster_county_weights[
#     np.nan
# )

# cluster_county_weight_panel = cluster_county_weights.copy()

# if len(cluster_county_weight_panel) == 0:
#     raise ValueError("cluster_county_weight_panel is empty after building

# # ------------------------------------------------------
# # E. Coverage audit before join
# # ------------------------------------------------------
# shortlist_fips = set(cluster_county_weight_panel["county_fips"].dropna().a
# healthcare_fips = set(healthcare_standardized["county_fips"].dropna().asty
# overlap_fips = shortlist_fips & healthcare_fips

# print("\nCoverage audit:")
# print("Shortlist county_fips:", len(shortlist_fips))
# print("Healthcare county_fips:", len(healthcare_fips))
```

```
# print("Overlap county_fips:", len(overlap_fips))

# missing_fips = sorted(list(shortlist_fips - healthcare_fips))
# print("\nSample missing shortlist county_fips:")
# print(missing_fips[:25])

# healthcare_overlap_audit = pd.DataFrame({
#     "metric": [
#         "shortlist_county_fips",
#         "healthcare_county_fips",
#         "overlap_county_fips"
#     ],
#     "value": [
#         len(shortlist_fips),
#         len(healthcare_fips),
#         len(overlap_fips)
#     ]
# })

# # ---------------------------------------------------
# # F. Join county healthcare into shortlisted clusters
# # ---------------------------------------------------
# healthcare_county_joined = (
#     cluster_county_weight_panel
#     .merge(
#         healthcare_standardized,
#         on=["county_fips", "year"],
#         how="left",
#         indicator=True
#     )
#     .copy()
# )

# healthcare_county_joined["matched_chr_county_flag"] = healthcare_county_jo
# healthcare_county_joined["county_has_any_healthcare_metric"] = (
#     healthcare_county_joined[healthcare_metric_cols].notna().any(axis=1)
# )

# print("\nJoin result:")
# print(healthcare_county_joined["_merge"].value_counts(dropna=False))

# unmatched_counties = (
#     healthcare_county_joined.loc[
#         healthcare_county_joined["_merge"] == "left_only",
#         ["cluster_id", "year", "county_fips"]
#     ]
#     .drop_duplicates()
#     .sort_values(["cluster_id", "year", "county_fips"])
#     .reset_index(drop=True)
# )

# print("\nUnmatched county-year examples:")
# print(unmatched_counties.head(25))

# # ---------------------------------------------------
# # G. Aggregate county healthcare to cluster-year
```

```
# # ---------------------------------------------------
# cluster_year_meta = (
#     enrichment_cluster_year_panel[
#         ["cluster_id", "year", "display_state", "display_county", "n_clust
#     ]
#     .drop_duplicates(subset=["cluster_id", "year"])
#     .copy()
# )

# cluster_rows = []

# for (cluster_id, year), grp in healthcare_county_joined.groupby(["cluster_
#     row = {
#         "cluster_id": cluster_id,
#         "year": int(year) if pd.notna(year) else pd.NA,
#     }

#     meta = cluster_year_meta.loc[
#         (cluster_year_meta["cluster_id"] == cluster_id) &
#         (cluster_year_meta["year"] == year)
#     ]

#     if len(meta) > 0:
#         row["display_state"] = meta["display_state"].iloc[0]
#         row["display_county"] = meta["display_county"].iloc[0]
#         row["n_cluster_tracts"] = meta["n_cluster_tracts"].iloc[0]
#         row["cluster_pop_total"] = meta["cluster_pop_total"].iloc[0]
#     else:
#         row["display_state"] = pd.NA
#         row["display_county"] = pd.NA
#         row["n_cluster_tracts"] = pd.NA
#         row["cluster_pop_total"] = pd.NA

#     weights = pd.to_numeric(grp["cluster_county_weight"], errors="coerce")

#     row["cluster_county_count"] = grp["county_fips"].nunique()
#     row["matched_chr_county_count"] = grp.loc[grp["matched_chr_county_flag
#     row["matched_chr_weight_share"] = grp.loc[
#         grp["matched_chr_county_flag"] & weights.notna(),
#         "cluster_county_weight"
#     ].sum()

#     row["any_metric_weight_share"] = grp.loc[
#         grp["county_has_any_healthcare_metric"] & weights.notna(),
#         "cluster_county_weight"
#     ].sum()

#     for col in healthcare_metric_cols:
#         values = pd.to_numeric(grp[col], errors="coerce")
#         mask = values.notna() & weights.notna() & (weights > 0)

#         if mask.sum() > 0:
#             row[col] = np.average(values.loc[mask], weights=weights.loc[ma
#         elif values.notna().sum() > 0:
#             row[col] = values.mean()
#         else:
```

```
#             row[col] = np.nan

#     row["healthcare_any_data_flag"] = pd.Series(
#         [row.get(col, np.nan) for col in healthcare_metric_cols]
#     ).notna().any()

#     cluster_rows.append(row)

# if len(cluster_rows) == 0:
#     raise ValueError("No cluster-year rows were produced for healthcare ag

# healthcare_cluster_panel = (
#     pd.DataFrame(cluster_rows)
#     .sort_values(["cluster_id", "year"])
#     .reset_index(drop=True)
# )

# # Keep only clusters with actual healthcare data in summary
# if healthcare_cluster_panel["healthcare_any_data_flag"].any():
#     healthcare_cluster_summary = (
#         healthcare_cluster_panel
#         .loc[healthcare_cluster_panel["healthcare_any_data_flag"] == True]
#         .groupby("cluster_id", as_index=False)
#         .agg({
#             "display_state": "first",
#             "display_county": "first",
#             "n_cluster_tracts": "max",
#             "cluster_pop_total": "mean",
#             "cluster_county_count": "max",
#             "matched_chr_county_count": "max",
#             "matched_chr_weight_share": "mean",
#             "any_metric_weight_share": "mean",
#             "year": "nunique",
#             **{col: "mean" for col in healthcare_metric_cols}
#         })
#         .rename(columns={"year": "n_healthcare_years_with_any_data"})
#     )
# else:
#     healthcare_cluster_summary = pd.DataFrame(
#         columns=[
#             "cluster_id",
#             "display_state",
#             "display_county",
#             "n_cluster_tracts",
#             "cluster_pop_total",
#             "cluster_county_count",
#             "matched_chr_county_count",
#             "matched_chr_weight_share",
#             "any_metric_weight_share",
#             "n_healthcare_years_with_any_data",
#         ] + healthcare_metric_cols
#     )

# print("\nHealthcare cluster panel rows with any data:")
# print(healthcare_cluster_panel["healthcare_any_data_flag"].value_counts(dr
# print("Healthcare cluster summary shape:", healthcare_cluster_summary.shap
```

```
# # ---------------------------------------------------
# # H. Save outputs
# # ---------------------------------------------------
# OUT_DIR.mkdir(parents=True, exist_ok=True)

# healthcare_standardized.to_parquet(OUT_DIR / "healthcare_standardized.parq
# healthcare_county_joined.to_parquet(OUT_DIR / "healthcare_county_joined.pa
# healthcare_cluster_panel.to_parquet(OUT_DIR / "healthcare_cluster_panel.pa
# healthcare_cluster_summary.to_parquet(OUT_DIR / "healthcare_cluster_summar
# unmatched_counties.to_parquet(OUT_DIR / "healthcare_unmatched_counties.par
# healthcare_overlap_audit.to_csv(OUT_DIR / "healthcare_overlap_audit.csv",

# healthcare_standardized.to_csv(OUT_DIR / "healthcare_standardized.csv", in
# healthcare_cluster_summary.to_csv(OUT_DIR / "healthcare_cluster_summary.cs
# unmatched_counties.to_csv(OUT_DIR / "healthcare_unmatched_counties.csv", i

# print("\nSaved:")
# print("-", OUT_DIR / "healthcare_standardized.parquet")
# print("-", OUT_DIR / "healthcare_county_joined.parquet")
# print("-", OUT_DIR / "healthcare_cluster_panel.parquet")
# print("-", OUT_DIR / "healthcare_cluster_summary.parquet")
# print("-", OUT_DIR / "healthcare_unmatched_counties.parquet")
# print("-", OUT_DIR / "healthcare_overlap_audit.csv")
# print("-", OUT_DIR / "healthcare_standardized.csv")
# print("-", OUT_DIR / "healthcare_cluster_summary.csv")
# print("-", OUT_DIR / "healthcare_unmatched_counties.csv")

# print("\nHealthcare cluster panel preview:")
# print(
#     healthcare_cluster_panel[
#         [
#             "cluster_id",
#             "year",
#             "display_state",
#             "display_county",
#             "cluster_pop_total",
#             "cluster_county_count",
#             "matched_chr_county_count",
#             "matched_chr_weight_share",
#             "any_metric_weight_share",
#             "healthcare_any_data_flag",
#         ] + healthcare_metric_cols[:5]
#     ].head(20)
# )
```

In [ ]:
```
### Cell 7.5 — healthcare feasibility decision checkpoint
```

In [ ]:
```
# # Cell 7.5 — healthcare feasibility decision checkpoint
# # Purpose:
# # make the healthcare-source mismatch explicit before any healthcare scori

# # ----------------------------------
# # A. cluster-level healthcare feasibility summary
# # ----------------------------------
```

```
# healthcare_feasibility = (
#     healthcare_cluster_panel
#     .groupby("cluster_id", as_index=False)
#     .agg(
#         display_state=("display_state", "first"),
#         display_county=("display_county", "first"),
#         n_cluster_tracts=("n_cluster_tracts", "max"),
#         cluster_pop_latest_proxy=("cluster_pop_total", "max"),
#         n_years_in_panel=("year", "nunique"),
#         n_years_with_any_healthcare_data=("healthcare_any_data_flag", lamb
#         mean_matched_chr_weight_share=("matched_chr_weight_share", "mean")
#         mean_any_metric_weight_share=("any_metric_weight_share", "mean"),
#     )
#     .copy()
# )

# healthcare_feasibility["healthcare_usable_flag"] = (
#     (healthcare_feasibility["n_years_with_any_healthcare_data"] >= 2) &
#     (healthcare_feasibility["mean_any_metric_weight_share"] > 0)
# )

# healthcare_feasibility = (
#     healthcare_feasibility
#     .merge(
#         enrichment_clusters[
#             ["cluster_id", "enrichment_rank", "rank", "cluster_shortlist_s
#         ].drop_duplicates(),
#         on="cluster_id",
#         how="left",
#         validate="1:1"
#     )
#     .sort_values(["enrichment_rank", "cluster_id"])
#     .reset_index(drop=True)
# )

# # ---------------------------------------
# # B. overall decision stats
# # ---------------------------------------
# n_shortlisted_clusters = len(enrichment_clusters)
# n_usable_healthcare_clusters = int(healthcare_feasibility["healthcare_usab

# decision_summary = pd.DataFrame({
#     "metric": [
#         "shortlisted_clusters",
#         "clusters_with_usable_healthcare_data",
#         "clusters_without_usable_healthcare_data",
#         "share_usable"
#     ],
#     "value": [
#         n_shortlisted_clusters,
#         n_usable_healthcare_clusters,
#         n_shortlisted_clusters - n_usable_healthcare_clusters,
#         (n_usable_healthcare_clusters / n_shortlisted_clusters) if n_short
#     ]
# })
```

```
# # ----------------------------------
# # C. explicit notebook decision
# # ----------------------------------
# if n_usable_healthcare_clusters <= 1:
#     healthcare_decision_message = (
#         "County Health Rankings is not sufficient for cross-cluster health
#         "on the current shortlist. The shortlist is mostly Puerto Rico, wh
#         "matches one shortlisted cluster. Do NOT build the final healthcar
#         "this source as-is."
#     )
#     recommended_next_step = (
#         "Choose one of two branches: "
#         "(1) keep the current shortlist and switch to a healthcare source
#         "(2) keep CHR and create a mainland-compatible healthcare branch f
#     )
# else:
#     healthcare_decision_message = (
#         "County Health Rankings has enough overlap to support preliminary
#         "across multiple shortlisted clusters."
#     )
#     recommended_next_step = (
#         "Proceed to Cell 8 using recovery-period healthcare metrics only,
#     )

# print("Healthcare decision checkpoint")
# print("-" * 60)
# print(healthcare_decision_message)
# print()
# print("Recommended next step:")
# print(recommended_next_step)

# print("\nDecision summary:")
# print(decision_summary)

# print("\nCluster-level healthcare feasibility:")
# print(
#     healthcare_feasibility[
#         [
#             "enrichment_rank",
#             "cluster_id",
#             "display_state",
#             "display_county",
#             "n_years_in_panel",
#             "n_years_with_any_healthcare_data",
#             "mean_matched_chr_weight_share",
#             "mean_any_metric_weight_share",
#             "healthcare_usable_flag",
#         ]
#     ]
# )

# # ----------------------------------
# # D. Save outputs
# # ----------------------------------
# OUT_DIR.mkdir(parents=True, exist_ok=True)
```

```
# healthcare_feasibility.to_parquet(OUT_DIR / "healthcare_feasibility.parque
# healthcare_feasibility.to_csv(OUT_DIR / "healthcare_feasibility.csv", inde
# decision_summary.to_csv(OUT_DIR / "healthcare_decision_summary.csv", index

# print("\nSaved:")
# print("-", OUT_DIR / "healthcare_feasibility.parquet")
# print("-", OUT_DIR / "healthcare_feasibility.csv")
# print("-", OUT_DIR / "healthcare_decision_summary.csv")
```

In [ ]:

```
# # Cell 8 — score healthcare need on the healthcare-covered subset (recover
# # IMPORTANT:
# # - Uses County Health Rankings only where coverage is actually usable
# # - Uses recovery-period overlap years only (given CHR availability, this
# # - Scores metrics within-year, then averages across recovery years
# #   so cross-year raw-value scale changes do not distort the comparison

# import pandas as pd
# import numpy as np

# # ---------------------------------------------------
# # A. Helpers
# # ---------------------------------------------------
# def bad_when_high(series: pd.Series) -> pd.Series:
#     s = pd.to_numeric(series, errors="coerce")
#     return s.rank(pct=True, method="average")

# def bad_when_low(series: pd.Series) -> pd.Series:
#     s = pd.to_numeric(series, errors="coerce")
#     return (-s).rank(pct=True, method="average")

# # ---------------------------------------------------
# # B. Basic validation
# # ---------------------------------------------------
# if "healthcare_cluster_panel" not in globals() or len(healthcare_cluster_p
#     raise ValueError("healthcare_cluster_panel is empty or missing. Rerun

# if "cluster_shortlist" not in globals() or len(cluster_shortlist) == 0:
#     raise ValueError("cluster_shortlist is empty or missing. Rerun Cell 5

# if "healthcare_metric_cols" not in globals() or len(healthcare_metric_cols
#     raise ValueError("healthcare_metric_cols is empty or missing. Rerun Ce

# # ---------------------------------------------------
# # C. Determine usable recovery years for healthcare scoring
# # ---------------------------------------------------
# recovery_years_from_main = sorted(
#     cluster_year_panel.loc[
#         cluster_year_panel["analysis_period"] == "recovery",
#         "year"
#     ].dropna().astype(int).unique().tolist()
# )

# healthcare_years_available = sorted(
#     healthcare_cluster_panel["year"].dropna().astype(int).unique().tolist(
# )
```

```
# healthcare_recovery_years = sorted(set(recovery_years_from_main) & set(hea

# if len(healthcare_recovery_years) == 0:
#     raise ValueError(
#         "No overlap between recovery years in the main panel and available
#     )

# # With CHR through 2023 and recovery = 2022–2024, this should usually be [
# print("Main recovery years:", recovery_years_from_main)
# print("Healthcare years available:", healthcare_years_available)
# print("Healthcare recovery years used for scoring:", healthcare_recovery_y

# # ------------------------------------------------------
# # D. Metric directions
# # higher is worse vs lower is worse
# # ------------------------------------------------------
# healthcare_metric_direction_map = {
#     "uninsured_rate": "high_is_bad",
#     "uninsured_adults_rate": "high_is_bad",
#     "uninsured_children_rate": "high_is_bad",
#     "primary_care_physicians_rate": "low_is_bad",
#     "dentists_rate": "low_is_bad",
#     "mental_health_providers_rate": "low_is_bad",
#     "other_primary_care_providers_rate": "low_is_bad",
#     "preventable_hospital_stays_rate": "high_is_bad",
#     "mammography_screening_rate": "low_is_bad",
#     "flu_vaccinations_rate": "low_is_bad",
# }

# healthcare_scoring_metrics = [
#     c for c in healthcare_metric_cols
#     if c in healthcare_metric_direction_map
# ]

# if len(healthcare_scoring_metrics) == 0:
#     raise ValueError("No healthcare scoring metrics were found in healthca

# healthcare_metric_directions = pd.DataFrame({
#     "metric": healthcare_scoring_metrics,
#     "direction": [healthcare_metric_direction_map[c] for c in healthcare_s
# })

# # ------------------------------------------------------
# # E. Build recovery-period healthcare base
# # ------------------------------------------------------
# MIN_MATCHED_WEIGHT_SHARE = 0.50
# MIN_ANY_METRIC_WEIGHT_SHARE = 0.50
# MIN_METRICS_PRESENT_PER_YEAR = 4
# MIN_RECOVERY_HEALTHCARE_YEARS = min(2, len(healthcare_recovery_years))

# healthcare_recovery_panel = (
#     healthcare_cluster_panel.loc[
#         healthcare_cluster_panel["year"].isin(healthcare_recovery_years)
#     ]
#     .copy()
#     .sort_values(["cluster_id", "year"])
```

```
#     .reset_index(drop=True)
# )

# healthcare_recovery_panel["n_nonmissing_healthcare_metrics"] = (
#     healthcare_recovery_panel[healthcare_scoring_metrics].notna().sum(axis
# )

# healthcare_recovery_panel["recovery_healthcare_row_usable_flag"] = (
#     healthcare_recovery_panel["healthcare_any_data_flag"].fillna(False)
#     & (pd.to_numeric(healthcare_recovery_panel["matched_chr_weight_share"]
#     & (pd.to_numeric(healthcare_recovery_panel["any_metric_weight_share"],
#     & (pd.to_numeric(healthcare_recovery_panel["n_nonmissing_healthcare_me
# )

# # cluster-level coverage summary over recovery years
# healthcare_recovery_coverage = (
#     healthcare_recovery_panel
#     .groupby("cluster_id", as_index=False)
#     .agg(
#         display_state=("display_state", "first"),
#         display_county=("display_county", "first"),
#         n_cluster_tracts=("n_cluster_tracts", "max"),
#         n_recovery_healthcare_years=("year", "nunique"),
#         n_usable_recovery_healthcare_years=("recovery_healthcare_row_usabl
#         mean_matched_chr_weight_share=("matched_chr_weight_share", "mean")
#         mean_any_metric_weight_share=("any_metric_weight_share", "mean"),
#         mean_nonmissing_healthcare_metrics=("n_nonmissing_healthcare_metri
#     )
#     .copy()
# )

# healthcare_recovery_coverage["healthcare_eligible_flag"] = (
#     (healthcare_recovery_coverage["n_usable_recovery_healthcare_years"] >=
#     & (healthcare_recovery_coverage["mean_any_metric_weight_share"] >= MIN
# )

# eligible_healthcare_cluster_ids = set(
#     healthcare_recovery_coverage.loc[
#         healthcare_recovery_coverage["healthcare_eligible_flag"],
#         "cluster_id"
#     ].astype(str).tolist()
# )

# print("\nHealthcare recovery coverage summary:")
# print("Recovery healthcare rows:", healthcare_recovery_panel.shape[0])
# print("Eligible healthcare clusters:", len(eligible_healthcare_cluster_ids
# print("Min usable recovery years required:", MIN_RECOVERY_HEALTHCARE_YEARS
# print()

# print(
#     healthcare_recovery_coverage.sort_values(
#         ["healthcare_eligible_flag", "n_usable_recovery_healthcare_years",
#         ascending=[False, False, False, True]
#     )
# )
```

```
# # --------------------------------------------------------
# # F. Score healthcare need within each year
# #    (rank within year, then average across years)
# # --------------------------------------------------------
# healthcare_scored_yearly = (
#     healthcare_recovery_panel.loc[
#         healthcare_recovery_panel["cluster_id"].astype(str).isin(eligible_
#         & healthcare_recovery_panel["recovery_healthcare_row_usable_flag"]
#     ]
#     .copy()
# )

# if len(healthcare_scored_yearly) == 0:
#     print("\nNo clusters met healthcare eligibility for scoring.")
#     healthcare_cluster_scored = pd.DataFrame(columns=[
#         "cluster_id",
#         "display_state",
#         "display_county",
#         "n_cluster_tracts",
#         "n_recovery_healthcare_years_scored",
#         "mean_matched_chr_weight_share",
#         "mean_any_metric_weight_share",
#         "mean_nonmissing_healthcare_metrics",
#         "healthcare_need_score",
#         "healthcare_need_rank",
#     ] + healthcare_scoring_metrics)
# else:
#     metric_score_cols = []

#     for metric in healthcare_scoring_metrics:
#         direction = healthcare_metric_direction_map[metric]
#         score_col = f"{metric}_need_score"
#         metric_score_cols.append(score_col)

#         if direction == "high_is_bad":
#             healthcare_scored_yearly[score_col] = (
#                 healthcare_scored_yearly
#                 .groupby("year")[metric]
#                 .transform(bad_when_high)
#             )
#         elif direction == "low_is_bad":
#             healthcare_scored_yearly[score_col] = (
#                 healthcare_scored_yearly
#                 .groupby("year")[metric]
#                 .transform(bad_when_low)
#             )
#         else:
#             raise ValueError(f"Unexpected direction for {metric}: {directi

#     healthcare_scored_yearly["healthcare_need_score_year"] = (
#         healthcare_scored_yearly[metric_score_cols].mean(axis=1, skipna=Tr
#     )

#     # --------------------------------------------------------
#     # G. Aggregate to cluster-level healthcare score
#     # --------------------------------------------------------
```

```
#     healthcare_cluster_scored = (
#         healthcare_scored_yearly
#         .groupby("cluster_id", as_index=False)
#         .agg(
#             display_state=("display_state", "first"),
#             display_county=("display_county", "first"),
#             n_cluster_tracts=("n_cluster_tracts", "max"),
#             n_recovery_healthcare_years_scored=("year", "nunique"),
#             mean_matched_chr_weight_share=("matched_chr_weight_share", "me
#             mean_any_metric_weight_share=("any_metric_weight_share", "mean
#             mean_nonmissing_healthcare_metrics=("n_nonmissing_healthcare_m
#             healthcare_need_score=("healthcare_need_score_year", "mean"),
#             **{metric: (metric, "mean") for metric in healthcare_scoring_m
#         )
#         .copy()
#     )

#     healthcare_cluster_scored["healthcare_need_rank"] = (
#         healthcare_cluster_scored["healthcare_need_score"]
#         .rank(method="first", ascending=False)
#         .astype(int)
#     )

#     healthcare_cluster_scored = healthcare_cluster_scored.sort_values(
#         ["healthcare_need_score", "cluster_id"],
#         ascending=[False, True]
#     ).reset_index(drop=True)

# # --------------------------------------------------
# # H. Merge healthcare back into the shortlist
# # --------------------------------------------------
# cluster_shortlist_with_healthcare = (
#     cluster_shortlist
#     .merge(
#         healthcare_recovery_coverage[
#             [
#                 "cluster_id",
#                 "n_recovery_healthcare_years",
#                 "n_usable_recovery_healthcare_years",
#                 "mean_matched_chr_weight_share",
#                 "mean_any_metric_weight_share",
#                 "mean_nonmissing_healthcare_metrics",
#                 "healthcare_eligible_flag",
#             ]
#         ],
#         on="cluster_id",
#         how="left",
#         validate="1:1"
#     )
#     .merge(
#         healthcare_cluster_scored[
#             ["cluster_id", "healthcare_need_score", "healthcare_need_rank"
#         ] if len(healthcare_cluster_scored) > 0 else pd.DataFrame(columns=
#         on="cluster_id",
#         how="left",
#         validate="1:1"
```

```
#     )
#     .copy()
# )

# cluster_shortlist_with_healthcare["healthcare_eligible_flag"] = (
#     cluster_shortlist_with_healthcare["healthcare_eligible_flag"].fillna(F
# )

# # preliminary combined score before business enrichment
# # keep the shortlist foundation primary, then let healthcare refine among
# SHORTLIST_WEIGHT = 0.70
# HEALTHCARE_WEIGHT = 0.30

# cluster_shortlist_with_healthcare["preliminary_combined_score"] = np.where
#     cluster_shortlist_with_healthcare["healthcare_eligible_flag"]
#     & cluster_shortlist_with_healthcare["healthcare_need_score"].notna(),
#     (SHORTLIST_WEIGHT * cluster_shortlist_with_healthcare["cluster_shortli
#     + (HEALTHCARE_WEIGHT * cluster_shortlist_with_healthcare["healthcare_n
#     np.nan
# )

# healthcare_comparison_set = (
#     cluster_shortlist_with_healthcare
#     .loc[cluster_shortlist_with_healthcare["healthcare_eligible_flag"]]
#     .sort_values(
#         [
#             "preliminary_combined_score",
#             "healthcare_need_score",
#             "cluster_shortlist_score",
#             "latest_cluster_pop_total",
#         ],
#         ascending=[False, False, False, False]
#     )
#     .reset_index(drop=True)
# )

# if len(healthcare_comparison_set) > 0:
#     healthcare_comparison_set["healthcare_comparison_rank"] = np.arange(1,

# # ------------------------------------------------------
# # I. Diagnostics
# # ------------------------------------------------------
# print("\nHealthcare metric directions:")
# print(healthcare_metric_directions)

# print("\nHealthcare-scored yearly panel shape:", healthcare_scored_yearly.
# print("Healthcare cluster scored shape:", healthcare_cluster_scored.shape
# print("Shortlist with healthcare shape:", cluster_shortlist_with_healthcar
# print("Healthcare comparison set shape:", healthcare_comparison_set.shape)

# print("\nTop healthcare-covered clusters:")
# view_cols = [
#     "rank",
#     "cluster_id",
#     "display_state",
#     "display_county",
```

```
#     "n_cluster_tracts",
#     "latest_cluster_pop_total",
#     "cluster_shortlist_score",
#     "n_usable_recovery_healthcare_years",
#     "mean_matched_chr_weight_share",
#     "mean_any_metric_weight_share",
#     "healthcare_need_score",
#     "preliminary_combined_score",
# ]

# if len(healthcare_comparison_set) > 0:
#     print(healthcare_comparison_set[view_cols].head(20))
# else:
#     print("No healthcare-eligible clusters were available for comparison."

# print("\nShortlist clusters without usable healthcare coverage:")
# print(
#     cluster_shortlist_with_healthcare.loc[
#         ~cluster_shortlist_with_healthcare["healthcare_eligible_flag"],
#         ["rank", "cluster_id", "display_state", "display_county", "cluster
#     ].sort_values(["rank", "cluster_id"]).reset_index(drop=True)
# )

# # ---------------------------------------------------
# # J. Save outputs
# # ---------------------------------------------------
# OUT_DIR.mkdir(parents=True, exist_ok=True)

# healthcare_metric_directions.to_csv(OUT_DIR / "healthcare_metric_direction
# healthcare_recovery_panel.to_parquet(OUT_DIR / "healthcare_recovery_panel.
# healthcare_recovery_coverage.to_parquet(OUT_DIR / "healthcare_recovery_cov
# healthcare_scored_yearly.to_parquet(OUT_DIR / "healthcare_scored_yearly.pa
# healthcare_cluster_scored.to_parquet(OUT_DIR / "healthcare_cluster_scored.
# cluster_shortlist_with_healthcare.to_parquet(OUT_DIR / "cluster_shortlist_
# healthcare_comparison_set.to_parquet(OUT_DIR / "healthcare_comparison_set.

# cluster_shortlist_with_healthcare.to_csv(OUT_DIR / "cluster_shortlist_with
# healthcare_comparison_set.to_csv(OUT_DIR / "healthcare_comparison_set.csv"

# print("\nSaved:")
# print("-", OUT_DIR / "healthcare_metric_directions.csv")
# print("-", OUT_DIR / "healthcare_recovery_panel.parquet")
# print("-", OUT_DIR / "healthcare_recovery_coverage.parquet")
# print("-", OUT_DIR / "healthcare_scored_yearly.parquet")
# print("-", OUT_DIR / "healthcare_cluster_scored.parquet")
# print("-", OUT_DIR / "cluster_shortlist_with_healthcare.parquet")
# print("-", OUT_DIR / "healthcare_comparison_set.parquet")
# print("-", OUT_DIR / "cluster_shortlist_with_healthcare.csv")
# print("-", OUT_DIR / "healthcare_comparison_set.csv")

# #12 eligible out of the 15 enrichment clusters
# # not “12 eligible out of the full 67 shortlist clusters”
```

Cell 10F — provisional solution-fit visual

In [ ]:
```python
# ============================================================
# Georgia-only candidate search:
# find the highest-scoring adjacency-based cluster in Georgia
# ============================================================

import warnings
warnings.filterwarnings("ignore")

from pathlib import Path
import numpy as np
import pandas as pd
import geopandas as gpd
import networkx as nx

# ------------------------------------------------------------
# 0. Helpers
# ------------------------------------------------------------
def bad_when_high(series):
    s = pd.to_numeric(series, errors="coerce")
    return s.rank(pct=True, method="average")

def bad_when_low(series):
    s = pd.to_numeric(series, errors="coerce")
    return (-s).rank(pct=True, method="average")

def weighted_avg(df, value_col, weight_col):
    tmp = df[[value_col, weight_col]].copy()
    tmp[value_col] = pd.to_numeric(tmp[value_col], errors="coerce")
    tmp[weight_col] = pd.to_numeric(tmp[weight_col], errors="coerce")
    tmp = tmp.dropna()
    tmp = tmp[tmp[weight_col] > 0]
    if tmp.empty:
        return np.nan
    return np.average(tmp[value_col], weights=tmp[weight_col])

def assign_period(year):
    year = int(year)
    if year in [2017, 2018, 2019]:
        return "pre_covid"
    if year in [2020, 2021]:
        return "covid"
    if year in [2022, 2023, 2024]:
        return "recovery"
    return np.nan

def normalize_geoid(series):
    return series.astype(str).str.extract(r"(\d{11})", expand=False).fillna(

# ------------------------------------------------------------
# 1. Locate folders + load analysis panel
# ------------------------------------------------------------
PROC_CANDIDATES = [
    Path("src/data/processed"),
    Path("data/processed"),
    Path.cwd() / "src/data/processed",
```

```python
    Path.cwd() / "data/processed",
]
RAW_CANDIDATES = [
    Path("src/data/raw"),
    Path("data/raw"),
    Path.cwd() / "src/data/raw",
    Path.cwd() / "data/raw",
]

PROCESSED_DIR = next((p for p in PROC_CANDIDATES if p.exists()), None)
RAW_DIR = next((p for p in RAW_CANDIDATES if p.exists()), None)

if PROCESSED_DIR is None:
    raise FileNotFoundError("Could not find processed data directory.")
if RAW_DIR is None:
    raise FileNotFoundError("Could not find raw data directory.")

OUT_DIR = PROCESSED_DIR / "presentation_ready"
OUT_DIR.mkdir(parents=True, exist_ok=True)

if "analysis_panel" in globals() and isinstance(analysis_panel, pd.DataFrame
    panel = analysis_panel.copy()
else:
    panel = pd.read_parquet(PROCESSED_DIR / "eda_panel_clean.parquet").copy(

panel["geoid"] = normalize_geoid(panel["geoid"])
panel["state_fips"] = panel["geoid"].str[:2]
panel["county_fips"] = panel["geoid"].str[:5]

if "analysis_period" not in panel.columns:
    panel["analysis_period"] = panel["year"].map(assign_period)

panel = panel.loc[panel["year"].between(2017, 2024)].copy()

# Pick a tract population column
POP_CANDIDATES = ["pop_total", "population", "pop", "total_population"]
pop_col = next((c for c in POP_CANDIDATES if c in panel.columns), None)
if pop_col is None:
    raise ValueError(f"Could not find a tract population column. Tried: {POP

# Keep Georgia only
ga_panel = panel.loc[panel["state_fips"] == "13"].copy()
if ga_panel.empty:
    raise ValueError("No Georgia rows found in analysis panel.")

print("Georgia tract-year rows:", len(ga_panel))
print("Georgia unique tracts:", ga_panel["geoid"].nunique())

# --------------------------------------------------------------
# 2. Build Georgia tract recovery means + tract signal score
# --------------------------------------------------------------
tract_recovery = (
    ga_panel.loc[ga_panel["analysis_period"] == "recovery"]
    .groupby("geoid", as_index=False)
    .agg(
        igs_total=("igs_total", "mean"),
```

```
        igs_economy=("igs_economy", "mean"),
        poverty_rate=("poverty_rate", "mean"),
        unemp_rate=("unemp_rate", "mean"),
        median_household_income=("median_household_income", "mean"),
        lfpr_16p=("lfpr_16p", "mean"),
        county_fips=("county_fips", "first"),
        pop_total=(pop_col, "mean"),
        display_county=("display_county", "first") if "display_county" in ga
        display_state=("display_state", "first") if "display_state" in ga_pa
    )
)

tract_recovery["tract_need_score"] = (
    bad_when_low(tract_recovery["igs_total"]) +
    bad_when_low(tract_recovery["igs_economy"]) +
    bad_when_high(tract_recovery["poverty_rate"]) +
    bad_when_high(tract_recovery["unemp_rate"]) +
    bad_when_low(tract_recovery["median_household_income"]) +
    bad_when_low(tract_recovery["lfpr_16p"])
) / 6

# Signal tracts = top ~8% statewide
signal_cut = tract_recovery["tract_need_score"].quantile(0.92)
signal_tracts = tract_recovery.loc[tract_recovery["tract_need_score"] >= sig

# Optional county concentration filter so we don't chase single-tract noise
county_counts = signal_tracts["county_fips"].value_counts()
eligible_counties = county_counts.loc[county_counts >= 3].index.tolist()
if len(eligible_counties) > 0:
    signal_tracts = signal_tracts.loc[signal_tracts["county_fips"].isin(elig

signal_geoids = set(signal_tracts["geoid"])
print("Georgia signal tracts used:", len(signal_geoids))
print("Georgia signal counties:", signal_tracts["county_fips"].nunique())

if len(signal_geoids) == 0:
    raise ValueError("No Georgia signal tracts survived the county filter. L

# ------------------------------------------------------------
# 3. Load Georgia tract geometries robustly
# ------------------------------------------------------------
def load_georgia_tracts(raw_dir):
    shp_candidates = []
    shp_candidates += list(raw_dir.rglob("tl_2024_13_tract/*.shp"))
    shp_candidates += list(raw_dir.rglob("tl_2024_state_tracts/*.shp"))
    shp_candidates += list(raw_dir.rglob("*tract*.shp"))

    tested = []
    for shp in shp_candidates:
        try:
            gdf = gpd.read_file(shp)
            cols = set(gdf.columns)

            if {"STATEFP", "COUNTYFP", "TRACTCE"}.issubset(cols):
                gdf["geoid"] = (
                    gdf["STATEFP"].astype(str).str.zfill(2)
```

```python
                    + gdf["COUNTYFP"].astype(str).str.zfill(3)
                    + gdf["TRACTCE"].astype(str).str.zfill(6)
                )
            elif "GEOID" in cols:
                gdf["geoid"] = normalize_geoid(gdf["GEOID"])
            elif "geoid" in cols:
                gdf["geoid"] = normalize_geoid(gdf["geoid"])
            elif "GEOIDFQ" in cols:
                gdf["geoid"] = normalize_geoid(gdf["GEOIDFQ"])
            else:
                tested.append((str(shp), "no usable geoid fields"))
                continue

            gdf = gdf.loc[gdf["geoid"].str[:2] == "13"].copy()
            if len(gdf) >= 100:
                return gdf, shp, tested

            tested.append((str(shp), f"usable but only {len(gdf)} Georgia ro
        except Exception as e:
            tested.append((str(shp), str(e)))

    raise FileNotFoundError(
        "Could not find a usable Georgia tract shapefile.\n"
        + "\n".join([f"{p} -> {msg}" for p, msg in tested[:20]])
    )

ga_tracts, tract_shp_used, tract_tests = load_georgia_tracts(RAW_DIR)
ga_tracts = ga_tracts[["geoid", "geometry"]].drop_duplicates("geoid").copy()

if ga_tracts.crs is None:
    raise ValueError("Georgia tract shapefile has no CRS metadata.")
if str(ga_tracts.crs) != "EPSG:3857":
    ga_tracts = ga_tracts.to_crs(3857)

print("Using Georgia tract shapefile:", tract_shp_used)
print("Georgia tract geometries loaded:", len(ga_tracts))

# Join recovery signal onto geometries
ga_geo = ga_tracts.merge(
    tract_recovery.drop(columns=["display_state", "display_county"], errors=
    on="geoid",
    how="inner",
    validate="1:1"
)

# ------------------------------------------------------------
# 4. Build adjacency graph across Georgia tracts
# ------------------------------------------------------------
# Spatial-index touches graph
ga_geo = ga_geo.reset_index(drop=True)
sindex = ga_geo.sindex

G = nx.Graph()
G.add_nodes_from(ga_geo["geoid"].tolist())

for i, geom in enumerate(ga_geo.geometry):
```

```python
    hits = list(sindex.query(geom, predicate="touches"))
    src = ga_geo.loc[i, "geoid"]
    for j in hits:
        if j <= i:
            continue
        dst = ga_geo.loc[j, "geoid"]
        G.add_edge(src, dst)

print("Georgia adjacency graph nodes:", G.number_of_nodes())
print("Georgia adjacency graph edges:", G.number_of_edges())

# ------------------------------------------------------------
# 5. Seed -> one-hop expansion -> connected components
# ------------------------------------------------------------
expanded_geoids = set(signal_geoids)
for geoid in list(signal_geoids):
    if geoid in G:
        expanded_geoids.update(G.neighbors(geoid))

subG = G.subgraph(expanded_geoids).copy()
components = [sorted(list(c)) for c in nx.connected_components(subG)]

# keep only reasonably sized candidate communities
components = [c for c in components if len(c) >= 3]

if len(components) == 0:
    raise ValueError("No Georgia connected components found after expansion.

cluster_rows = []
for idx, comp in enumerate(components, start=1):
    cluster_id = f"Georgia | candidate_cluster_{idx}"
    for geoid in comp:
        cluster_rows.append({"cluster_id": cluster_id, "geoid": geoid})

ga_cluster_map = pd.DataFrame(cluster_rows)
print("Georgia candidate clusters found:", ga_cluster_map["cluster_id"].nuni

# ------------------------------------------------------------
# 6. Build Georgia cluster-year panel
# ------------------------------------------------------------
ga_cluster_year = ga_cluster_map.merge(
    ga_panel,
    on="geoid",
    how="left",
    validate="m:m"
)

metrics = [
    "igs_total", "igs_economy", "poverty_rate",
    "unemp_rate", "median_household_income", "lfpr_16p"
]

agg_rows = []
for (cluster_id, year), df_ in ga_cluster_year.groupby(["cluster_id", "year"
    row = {
        "cluster_id": cluster_id,
```

```
        "year": int(year),
        "cluster_pop_total": pd.to_numeric(df_[pop_col], errors="coerce").su
        "n_cluster_tracts": df_["geoid"].nunique(),
    }
    for m in metrics:
        row[m] = weighted_avg(df_, m, pop_col)
    agg_rows.append(row)

ga_cluster_year_panel = pd.DataFrame(agg_rows)
ga_cluster_year_panel["analysis_period"] = ga_cluster_year_panel["year"].map

# ------------------------------------------------------------
# 7. Period summary
# ------------------------------------------------------------
period_wide_rows = []

for cluster_id, df_ in ga_cluster_year_panel.groupby("cluster_id"):
    row = {"cluster_id": cluster_id}

    for period in ["pre_covid", "covid", "recovery"]:
        tmp = df_.loc[df_["analysis_period"] == period]
        if tmp.empty:
            continue
        row[f"n_cluster_tracts"] = int(tmp["n_cluster_tracts"].max())
        row[f"cluster_pop_total_latest"] = float(tmp.loc[tmp["year"] == tmp[

        for m in metrics:
            row[f"{m}_{period}"] = tmp[m].mean()

    period_wide_rows.append(row)

ga_period = pd.DataFrame(period_wide_rows)

# Attach primary county / county count for interpretability
tract_meta = tract_recovery[["geoid", "county_fips", "display_county"]].copy
cluster_meta = ga_cluster_map.merge(tract_meta, on="geoid", how="left")
meta_rows = []
for cluster_id, df_ in cluster_meta.groupby("cluster_id"):
    county_counts = df_["county_fips"].value_counts(dropna=True)
    primary_county_fips = county_counts.index[0] if len(county_counts) else
    primary_county_name = (
        df_.loc[df_["county_fips"] == primary_county_fips, "display_county"]
        if len(county_counts) and df_.loc[df_["county_fips"] == primary_coun
        else primary_county_fips
    )
    meta_rows.append({
        "cluster_id": cluster_id,
        "primary_county_fips": primary_county_fips,
        "primary_county": primary_county_name,
        "n_counties": int(df_["county_fips"].nunique()),
    })
cluster_meta_summary = pd.DataFrame(meta_rows)

ga_period = ga_period.merge(cluster_meta_summary, on="cluster_id", how="left

# ------------------------------------------------------------
```

```python
# 8. Score Georgia clusters
# ------------------------------------------------------------
# Low IGS in recovery
ga_period["score_low_igs_recovery"] = bad_when_low(ga_period["igs_total_reco

# Economic vulnerability in recovery
econ_parts = []
for col, fn in [
    ("igs_economy_recovery", bad_when_low),
    ("poverty_rate_recovery", bad_when_high),
    ("unemp_rate_recovery", bad_when_high),
    ("median_household_income_recovery", bad_when_low),
    ("lfpr_16p_recovery", bad_when_low),
]:
    if col in ga_period.columns:
        econ_parts.append(fn(ga_period[col]))
ga_period["score_econ_vulnerability_recovery"] = pd.concat(econ_parts, axis=

# COVID shock
ga_period["igs_drop_pre_to_covid"] = ga_period["igs_total_pre_covid"] - ga_p
ga_period["poverty_jump_pre_to_covid"] = ga_period["poverty_rate_covid"] - g
ga_period["unemp_jump_pre_to_covid"] = ga_period["unemp_rate_covid"] - ga_pe
ga_period["income_drop_pre_to_covid"] = ga_period["median_household_income_p
ga_period["lfpr_drop_pre_to_covid"] = ga_period["lfpr_16p_pre_covid"] - ga_p

shock_parts = [
    bad_when_high(ga_period["igs_drop_pre_to_covid"]),
    bad_when_high(ga_period["poverty_jump_pre_to_covid"]),
    bad_when_high(ga_period["unemp_jump_pre_to_covid"]),
    bad_when_high(ga_period["income_drop_pre_to_covid"]),
    bad_when_high(ga_period["lfpr_drop_pre_to_covid"]),
]
ga_period["score_covid_shock"] = pd.concat(shock_parts, axis=1).mean(axis=1)

# Weak recovery
ga_period["igs_rebound_covid_to_recovery"] = ga_period["igs_total_recovery"]
ga_period["poverty_improve_covid_to_recovery"] = ga_period["poverty_rate_cov
ga_period["unemp_improve_covid_to_recovery"] = ga_period["unemp_rate_covid"]
ga_period["income_rebound_covid_to_recovery"] = ga_period["median_household_
ga_period["lfpr_rebound_covid_to_recovery"] = ga_period["lfpr_16p_recovery"]

recovery_parts = [
    bad_when_low(ga_period["igs_rebound_covid_to_recovery"]),
    bad_when_low(ga_period["poverty_improve_covid_to_recovery"]),
    bad_when_low(ga_period["unemp_improve_covid_to_recovery"]),
    bad_when_low(ga_period["income_rebound_covid_to_recovery"]),
    bad_when_low(ga_period["lfpr_rebound_covid_to_recovery"]),
]
ga_period["score_weak_recovery"] = pd.concat(recovery_parts, axis=1).mean(ax

# Combined Georgia candidate score
ga_period["ga_candidate_score"] = (
    ga_period["score_low_igs_recovery"] +
    ga_period["score_econ_vulnerability_recovery"] +
    ga_period["score_covid_shock"] +
    ga_period["score_weak_recovery"]
```

```python
) / 4

ga_period = ga_period.sort_values("ga_candidate_score", ascending=False).res
ga_period["ga_rank"] = np.arange(1, len(ga_period) + 1)

# ------------------------------------------------------------
# 9. Show top Georgia candidates
# ------------------------------------------------------------
show_cols = [
    "ga_rank", "cluster_id", "primary_county", "n_counties",
    "n_cluster_tracts", "cluster_pop_total_latest",
    "igs_total_recovery", "igs_economy_recovery",
    "poverty_rate_recovery", "unemp_rate_recovery",
    "median_household_income_recovery", "lfpr_16p_recovery",
    "ga_candidate_score"
]
show_cols = [c for c in show_cols if c in ga_period.columns]

print("\nTop Georgia candidate clusters:")
print(ga_period[show_cols].head(10).to_string(index=False))

# ------------------------------------------------------------
# 10. Pull GEOIDs for the top Georgia cluster
# ------------------------------------------------------------
top_cluster_id = ga_period.iloc[0]["cluster_id"]
top_cluster_tracts = (
    ga_cluster_map.loc[ga_cluster_map["cluster_id"] == top_cluster_id]
    .copy()
    .sort_values("geoid")
    .reset_index(drop=True)
)
top_cluster_tracts["tract_code"] = top_cluster_tracts["geoid"].str[5:]

print("\nTop Georgia cluster selected:")
print("cluster_id:", top_cluster_id)
print("primary_county:", ga_period.iloc[0]["primary_county"])
print("n_cluster_tracts:", int(ga_period.iloc[0]["n_cluster_tracts"]))
print("ga_candidate_score:", round(float(ga_period.iloc[0]["ga_candidate_sco

print("\nTop cluster GEOIDs:")
print(top_cluster_tracts[["geoid", "tract_code"]].to_string(index=False))

# ------------------------------------------------------------
# 11. Save outputs
# ------------------------------------------------------------
ga_period.to_csv(OUT_DIR / "georgia_candidate_clusters_ranked.csv", index=Fa
top_cluster_tracts.to_csv(OUT_DIR / "georgia_top_candidate_cluster_tracts.cs

print("\nSaved:")
print("-", OUT_DIR / "georgia_candidate_clusters_ranked.csv")
print("-", OUT_DIR / "georgia_top_candidate_cluster_tracts.csv")
```

In [ ]:

```python
# ============================================================
# SHELBY FINALIST VISUALS — SHARED SETUP
# Run this once before the 3 visual cells below
# ============================================================
```

```python
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

plt.rcParams["figure.dpi"] = 120
plt.rcParams["savefig.dpi"] = 180

# ------------------------------------------------------------
# 1. Locate processed data
# ------------------------------------------------------------
PROC_CANDIDATES = [
    Path("src/data/processed"),
    Path("data/processed"),
    Path.cwd() / "src/data/processed",
    Path.cwd() / "data/processed",
]

PROCESSED_DIR = next((p for p in PROC_CANDIDATES if p.exists()), None)
if PROCESSED_DIR is None:
    raise FileNotFoundError("Could not find src/data/processed or data/proce

OUT_DIR = PROCESSED_DIR / "presentation_ready"
OUT_DIR.mkdir(parents=True, exist_ok=True)

# ------------------------------------------------------------
# 2. Load core data (reuse globals if already loaded)
# ------------------------------------------------------------
def assign_period(year):
    year = int(year)
    if year in [2017, 2018, 2019]:
        return "pre_covid"
    if year in [2020, 2021]:
        return "covid"
    if year in [2022, 2023, 2024]:
        return "recovery"
    return np.nan

def normalize_geoid(series):
    return (
        series.astype(str)
        .str.extract(r"(\d{11})", expand=False)
        .fillna(series.astype(str))
        .str.zfill(11)
    )

def weighted_avg(df, value_col, weight_col):
    tmp = df[[value_col, weight_col]].copy()
    tmp[value_col] = pd.to_numeric(tmp[value_col], errors="coerce")
    tmp[weight_col] = pd.to_numeric(tmp[weight_col], errors="coerce")
    tmp = tmp.dropna()
    tmp = tmp[tmp[weight_col] > 0]
    if tmp.empty:
        return np.nan
    return np.average(tmp[value_col], weights=tmp[weight_col])
```

```
def bad_when_high(series):
    s = pd.to_numeric(series, errors="coerce")
    return s.rank(pct=True, method="average")

def bad_when_low(series):
    s = pd.to_numeric(series, errors="coerce")
    return (-s).rank(pct=True, method="average")

analysis_panel_local = globals().get("analysis_panel", None)
if analysis_panel_local is None or len(analysis_panel_local) == 0:
    analysis_panel_local = pd.read_parquet(PROCESSED_DIR / "eda_panel_clean.
else:
    analysis_panel_local = analysis_panel_local.copy()

cluster_period_summary_local = globals().get("cluster_period_summary", None)
if cluster_period_summary_local is None or len(cluster_period_summary_local)
    cluster_period_summary_local = pd.read_parquet(PROCESSED_DIR / "cluster_
else:
    cluster_period_summary_local = cluster_period_summary_local.copy()

healthcare_standardized_local = globals().get("healthcare_standardized", Nor
if healthcare_standardized_local is None or len(healthcare_standardized_loca
    healthcare_standardized_local = pd.read_parquet(PROCESSED_DIR / "healthc
else:
    healthcare_standardized_local = healthcare_standardized_local.copy()

# optional but helpful
cluster_year_panel_local = globals().get("cluster_year_panel", None)
if cluster_year_panel_local is None:
    try:
        cluster_year_panel_local = pd.read_parquet(PROCESSED_DIR / "cluster_
    except Exception:
        cluster_year_panel_local = None
else:
    cluster_year_panel_local = cluster_year_panel_local.copy()

# ------------------------------------------------------------
# 3. Standardize panel fields
# ------------------------------------------------------------
analysis_panel_local["geoid"] = normalize_geoid(analysis_panel_local["geoid"
analysis_panel_local["state_fips"] = analysis_panel_local["geoid"].str[:2]
analysis_panel_local["county_fips"] = analysis_panel_local["geoid"].str[:5]

if "analysis_period" not in analysis_panel_local.columns:
    analysis_panel_local["analysis_period"] = analysis_panel_local["year"].m

healthcare_standardized_local["county_fips"] = (
    healthcare_standardized_local["county_fips"].astype(str).str.zfill(5)
)
healthcare_standardized_local["state_fips"] = healthcare_standardized_local[

# pick population column
POP_CANDIDATES = ["pop_total", "population", "pop", "total_population"]
POP_COL = next((c for c in POP_CANDIDATES if c in analysis_panel_local.colum
if POP_COL is None:
```

```python
    raise ValueError(f"Could not find tract population column. Tried: {POP_C

# ------------------------------------------------------------
# 4. Shelby finalist selection
# ------------------------------------------------------------
SHELBY_CLUSTER_ID = "Tennessee | Shelby County | cluster_77"
SHELBY_COUNTY_FIPS = "47157"
TN_STATE_FIPS = "47"

# if exact cluster id is not present, fall back to best Shelby row
if SHELBY_CLUSTER_ID not in cluster_period_summary_local["cluster_id"].astyp
    shelby_rows = cluster_period_summary_local.loc[
        (cluster_period_summary_local["display_state"] == "Tennessee") &
        (cluster_period_summary_local["display_county"] == "Shelby County")
    ].copy()
    if shelby_rows.empty:
        raise ValueError("Could not find any Shelby County cluster in cluste
    shelby_rows = shelby_rows.sort_values("igs_total_recovery", ascending=Tr
    SHELBY_CLUSTER_ID = shelby_rows.iloc[0]["cluster_id"]

print("Using Shelby finalist cluster:", SHELBY_CLUSTER_ID)

# ------------------------------------------------------------
# 5. Annual economic series builder
# ------------------------------------------------------------
econ_metrics = {
    "igs_total": "IGS Total",
    "igs_economy": "IGS Economy",
    "poverty_rate": "Poverty Rate",
    "unemp_rate": "Unemployment Rate",
    "median_household_income": "Median Income",
    "lfpr_16p": "Labor Force Participation",
}

def build_weighted_year_series(df, label):
    rows = []
    for year, tmp in df.groupby("year"):
        row = {"year": int(year), "group": label}
        row["analysis_period"] = assign_period(year)
        row["cluster_pop_total"] = pd.to_numeric(tmp[POP_COL], errors="coerc
        for m in econ_metrics.keys():
            row[m] = weighted_avg(tmp, m, POP_COL)
        rows.append(row)
    return pd.DataFrame(rows)

# Shelby cluster annual series
if cluster_year_panel_local is not None and "cluster_id" in cluster_year_par
    shelby_cluster_year = cluster_year_panel_local.loc[
        cluster_year_panel_local["cluster_id"].astype(str) == SHELBY_CLUSTER
    ].copy()

    if shelby_cluster_year.empty:
        raise ValueError(f"No annual rows found for {SHELBY_CLUSTER_ID} in c

    shelby_cluster_year["group"] = "Selected Shelby Community"
    if "analysis_period" not in shelby_cluster_year.columns:
```

```python
        shelby_cluster_year["analysis_period"] = shelby_cluster_year["year"]

else:
    raise ValueError("cluster_year_panel is required for the time-story visu

# Shelby county annual series (all Shelby tracts)
shelby_panel = analysis_panel_local.loc[
    analysis_panel_local["county_fips"] == SHELBY_COUNTY_FIPS
].copy()
if shelby_panel.empty:
    raise ValueError("No Shelby County tract rows found in analysis panel.")

shelby_county_year = build_weighted_year_series(shelby_panel, "Shelby County

# Tennessee annual series
tn_panel = analysis_panel_local.loc[
    analysis_panel_local["state_fips"] == TN_STATE_FIPS
].copy()
tn_year = build_weighted_year_series(tn_panel, "Tennessee")

# U.S. annual series
us_year = build_weighted_year_series(analysis_panel_local, "U.S.")

# ------------------------------------------------------------
# 6. Recovery-period benchmark values
# ------------------------------------------------------------
def recovery_weighted_row(df, label):
    tmp = df.loc[df["analysis_period"] == "recovery"].copy()
    row = {"group": label}
    for m in econ_metrics.keys():
        row[m] = weighted_avg(tmp, m, POP_COL)
    return row

bench_cluster = cluster_period_summary_local.loc[
    cluster_period_summary_local["cluster_id"].astype(str) == SHELBY_CLUSTER
].copy()

if bench_cluster.empty:
    raise ValueError(f"No recovery benchmark row found for {SHELBY_CLUSTER_I

bench_cluster = bench_cluster.iloc[0]

econ_benchmark_raw = pd.DataFrame([
    {
        "group": "Selected Shelby Community",
        "IGS Total": bench_cluster["igs_total_recovery"],
        "IGS Economy": bench_cluster["igs_economy_recovery"],
        "Poverty": bench_cluster["poverty_rate_recovery"],
        "Unemployment": bench_cluster["unemp_rate_recovery"],
        "Median Income": bench_cluster["median_household_income_recovery"],
        "LFPR": bench_cluster["lfpr_16p_recovery"],
    },
    {
        "group": "Shelby County Overall",
        "IGS Total": recovery_weighted_row(shelby_panel, "Shelby County Over
        "IGS Economy": recovery_weighted_row(shelby_panel, "Shelby County Ov
```

```
        "Poverty": recovery_weighted_row(shelby_panel, "Shelby County Overal
        "Unemployment": recovery_weighted_row(shelby_panel, "Shelby County C
        "Median Income": recovery_weighted_row(shelby_panel, "Shelby County 
        "LFPR": recovery_weighted_row(shelby_panel, "Shelby County Overall")
    },
    {
        "group": "Tennessee",
        "IGS Total": recovery_weighted_row(tn_panel, "Tennessee")["igs_total
        "IGS Economy": recovery_weighted_row(tn_panel, "Tennessee")["igs_eco
        "Poverty": recovery_weighted_row(tn_panel, "Tennessee")["poverty_rat
        "Unemployment": recovery_weighted_row(tn_panel, "Tennessee")["unemp_
        "Median Income": recovery_weighted_row(tn_panel, "Tennessee")["media
        "LFPR": recovery_weighted_row(tn_panel, "Tennessee")["lfpr_16p"],
    },
    {
        "group": "U.S.",
        "IGS Total": recovery_weighted_row(analysis_panel_local, "U.S.")["ig
        "IGS Economy": recovery_weighted_row(analysis_panel_local, "U.S.")["
        "Poverty": recovery_weighted_row(analysis_panel_local, "U.S.")["pove
        "Unemployment": recovery_weighted_row(analysis_panel_local, "U.S.")[
        "Median Income": recovery_weighted_row(analysis_panel_local, "U.S.")
        "LFPR": recovery_weighted_row(analysis_panel_local, "U.S.")["lfpr_16
    },
])

# -------------------------------------------------------------
# 7. Healthcare benchmark prep
# NOTE: cluster inherits Shelby County county-level healthcare environment
# -------------------------------------------------------------
health_yrs = healthcare_standardized_local.loc[
    healthcare_standardized_local["year"].isin([2022, 2023])
].copy()

uninsured_cols = [
    "uninsured_rate",
    "uninsured_adults_rate",
    "uninsured_children_rate",
]
provider_cols = [
    "primary_care_physicians_rate",
    "dentists_rate",
    "mental_health_providers_rate",
    "other_primary_care_providers_rate",
]
preventive_cols = [
    "mammography_screening_rate",
    "flu_vaccinations_rate",
]

def mean_of_cols(df, cols):
    vals = df[cols].apply(pd.to_numeric, errors="coerce").stack().dropna()
    if len(vals) == 0:
        return np.nan
    return float(vals.mean())

def percentile_of_value(series, value, high_is_bad=True):
```

```
    s = pd.to_numeric(series, errors="coerce").dropna()
    if s.empty or pd.isna(value):
        return np.nan
    if high_is_bad:
        return float((s <= value).mean())
    else:
        return float((s >= value).mean())

shelby_health = health_yrs.loc[health_yrs["county_fips"] == SHELBY_COUNTY_FI
tn_health = health_yrs.loc[health_yrs["state_fips"] == TN_STATE_FIPS].copy()
us_health = health_yrs.copy()

def build_health_story_row(df, label, reference_df):
    uninsured_burden = np.nanmean([
        percentile_of_value(reference_df[c], mean_of_cols(df, [c]), high_is_
        for c in uninsured_cols
    ])
    provider_access_gap = np.nanmean([
        percentile_of_value(reference_df[c], mean_of_cols(df, [c]), high_is_
        for c in provider_cols
    ])
    preventive_care_gap = np.nanmean([
        percentile_of_value(reference_df[c], mean_of_cols(df, [c]), high_is_
        for c in preventive_cols
    ])
    preventable_stays = percentile_of_value(
        reference_df["preventable_hospital_stays_rate"],
        mean_of_cols(df, ["preventable_hospital_stays_rate"]),
        high_is_bad=True
    )
    overall_need = np.nanmean([
        uninsured_burden,
        provider_access_gap,
        preventive_care_gap,
        preventable_stays
    ])
    return {
        "group": label,
        "Overall Need": overall_need,
        "Uninsured Burden": uninsured_burden,
        "Provider Access Gap": provider_access_gap,
        "Preventive Care Gap": preventive_care_gap,
        "Preventable Stays": preventable_stays,
    }

health_story_scores = pd.DataFrame([
    build_health_story_row(shelby_health, "Shelby County / Selected Communit
    build_health_story_row(tn_health, "Tennessee", health_yrs),
    build_health_story_row(us_health, "U.S.", health_yrs),
])

print("Setup complete.")
print("Output folder:", OUT_DIR)
```

In [ ]:

```
# ============================================================
# VISUAL 1 — Shelby story over time (3-group version, simplified)
```

```python
# Our community vs Rest of Shelby County vs High-growth comparator
# Assumes `three_group_year` has already been created
# ============================================================

from matplotlib.ticker import PercentFormatter, FuncFormatter
import matplotlib.pyplot as plt

plot_metrics = [
    ("igs_total", "IGS Total"),
    ("poverty_rate", "Poverty Rate"),
    ("median_household_income", "Median Income"),
    ("lfpr_16p", "Labor Force Participation"),
]

group_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

fig, axes = plt.subplots(2, 2, figsize=(16, 9))
axes = axes.flatten()

for ax, (col, title) in zip(axes, plot_metrics):
    for grp in group_order:
        line_df = (
            three_group_year.loc[three_group_year["group"] == grp]
            .sort_values("year")
            .copy()
        )

        if line_df.empty:
            continue

        ax.plot(
            line_df["year"],
            line_df[col],
            marker="o",
            linewidth=2.5,
            color=color_map[grp],
            label=label_map[grp],
        )

    # period shading
```

```
    ax.axvspan(2017 - 0.5, 2019 + 0.5, alpha=0.08, color="gray")
    ax.axvspan(2020 - 0.5, 2021 + 0.5, alpha=0.12, color="gray")
    ax.axvspan(2022 - 0.5, 2024 + 0.5, alpha=0.08, color="gray")

    ax.set_title(title, fontsize=15)
    ax.set_xticks(sorted(three_group_year["year"].unique()))
    ax.grid(alpha=0.25)

    if col in ["poverty_rate", "lfpr_16p"]:
        ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
    elif col == "median_household_income":
        ax.yaxis.set_major_formatter(FuncFormatter(lambda x, pos: f"${x:,.0f

# move legend to top
handles, labels = axes[0].get_legend_handles_labels()
fig.legend(
    handles,
    labels,
    loc="upper center",
    ncol=3,
    frameon=True,
    bbox_to_anchor=(0.5, 0.98)
)

fig.suptitle("Shelby community story over time", fontsize=20, y=1.03)

fig.text(
    0.02, 0.02,
    "Shaded periods: pre-COVID (2017–2019), COVID (2020–2021), recovery (202
    fontsize=10
)

plt.tight_layout(rect=[0, 0.05, 1, 0.92])
plt.show()

out_path = OUT_DIR / "shelby_visual_1_story_over_time_four_panel.png"
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:

```
# ============================================================
# VISUAL 2 — Shelby healthcare outlook (3-group CDC tract snapshot)
# Our community vs Rest of Shelby County vs High-growth comparator
# Uses tract-level CDC PLACES metrics already aggregated into three_group_cc
# ============================================================

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from matplotlib.patches import Patch

if "three_group_compare" not in globals():
    raise ValueError("Run the 3-group comparison cells first so `three_group

health_specs = {
    "uninsured_adults_pct": {
        "title": "Uninsured Adults",
```

```
        "ylabel": "Percent",
        "fmt": lambda v: f"{v:.1f}%"
    },
    "routine_checkup_gap_pct": {
        "title": "Routine Checkup Gap",
        "ylabel": "Percent",
        "fmt": lambda v: f"{v:.1f}%"
    },
    "poor_physical_health_pct": {
        "title": "Poor Physical Health",
        "ylabel": "Percent",
        "fmt": lambda v: f"{v:.1f}%"
    },
    "poor_mental_health_pct": {
        "title": "Poor Mental Health",
        "ylabel": "Percent",
        "fmt": lambda v: f"{v:.1f}%"
    },
}

plot_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

legend_handles = [
    Patch(facecolor=color_map["Selected low-growth community"], label="Our S
    Patch(facecolor=color_map["Rest of Shelby County (excluding both selecte
    Patch(facecolor=color_map["Selected high-growth comparator"], label="Hig
]

fig, axes = plt.subplots(2, 2, figsize=(16, 9))
axes = axes.flatten()

for ax, (metric, spec) in zip(axes, health_specs.items()):
    if metric not in three_group_compare.columns:
        ax.axis("off")
        continue

    plot_df = (
        three_group_compare
        .set_index("group")
        .reindex(plot_order)
```

```
        [[metric]]
        .copy()
    )

    vals = pd.to_numeric(plot_df[metric], errors="coerce")
    x = np.arange(len(plot_order))

    bars = ax.bar(
        x,
        vals.values,
        color=[color_map[g] for g in plot_order],
        width=0.6
    )

    ymax = np.nanmax(vals.values) if np.isfinite(np.nanmax(vals.values)) els
    pad = max(ymax * 0.03, 0.3)

    for bar, val in zip(bars, vals.values):
        if pd.notna(val):
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + pad,
                spec["fmt"](val),
                ha="center",
                va="bottom",
                fontsize=10
            )

    ax.set_title(spec["title"], fontsize=15)
    ax.set_ylabel(spec["ylabel"])
    ax.set_xticks([])
    ax.set_xlabel("")
    ax.tick_params(axis="x", length=0)
    ax.grid(axis="y", alpha=0.25)
    ax.set_ylim(0, ymax * 1.18 if ymax > 0 else 1)

fig.legend(
    handles=legend_handles,
    loc="upper center",
    ncol=3,
    frameon=True,
    bbox_to_anchor=(0.5, 0.98)
)

fig.suptitle("Shelby healthcare outlook", fontsize=20, y=1.03)

fig.text(
    0.02, 0.02,
    "CDC PLACES tract-level snapshot aggregated to the 3 Shelby comparison g
    fontsize=10
)

plt.tight_layout(rect=[0, 0.05, 1, 0.92])
plt.show()

out_path = OUT_DIR / "shelby_healthcare_outlook_three_group.png"
```

```
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:
```
# ============================================================
# VISUAL 2 — Shelby healthcare story
# NOTE: healthcare is county-level in this project, so the selected
# Shelby cluster inherits Shelby County's healthcare environment.
# ============================================================

health_plot = health_story_scores.set_index("group").loc[
    ["Shelby County / Selected Community", "Tennessee", "U.S."],
    ["Overall Need", "Uninsured Burden", "Provider Access Gap", "Preventive
]

x = np.arange(len(health_plot.columns))
bar_w = 0.22
groups = health_plot.index.tolist()

fig, ax = plt.subplots(figsize=(14, 7))

for i, g in enumerate(groups):
    ax.bar(x + (i - 1) * bar_w, health_plot.loc[g].values, width=bar_w, labe

ax.set_xticks(x)
ax.set_xticklabels(health_plot.columns, rotation=20, ha="right")
ax.set_ylim(0, 1.05)
ax.set_ylabel("Relative severity score (higher = worse)")
ax.set_title("Visual 2 — Shelby healthcare story vs Tennessee and U.S.")
ax.grid(axis="y", alpha=0.25)
ax.legend(frameon=True)

fig.text(
    0.02, 0.02,
    "Healthcare uses 2022–2023 County Health Rankings overlap years.\n"
    "Because healthcare is joined at the county level, the selected Shelby c
    fontsize=10
)

plt.tight_layout(rect=[0, 0.08, 1, 0.95])
plt.show()

fig.savefig(OUT_DIR / "shelby_visual_2_health_story.png", bbox_inches="tight
print("Saved:", OUT_DIR / "shelby_visual_2_health_story.png")
```

In [ ]:
```
# ============================================================
# VISUAL 3 — Shelby economic benchmark (raw-value version, simplified)
# Our community vs Rest of Shelby County vs High-growth Shelby
# vs Tennessee vs U.S.
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import PercentFormatter, FuncFormatter
from matplotlib.patches import Patch
```

```python
# ------------------------------
# settings
# ------------------------------
recovery_years = [2022, 2023, 2024]

plot_metrics = [
    ("igs_total", "IGS Total"),
    ("poverty_rate", "Poverty Rate"),
    ("unemp_rate", "Unemployment Rate"),
    ("median_household_income", "Median Income"),
]

group_order = [
    "Our Shelby Community",
    "Rest of Shelby County",
    "High-Growth Shelby Community",
    "Tennessee",
    "U.S.",
]

color_map = {
    "Our Shelby Community": "#fb8072",
    "Rest of Shelby County": "#bebada",
    "High-Growth Shelby Community": "#8dd3c7",
    "Tennessee": "#2ca02c",
    "U.S.": "#d62728",
}

comparison_label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

# ------------------------------
# helpers
# ------------------------------
def safe_weighted_mean(values: pd.Series, weights: pd.Series):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)

    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())

    return float(np.average(values.loc[mask], weights=weights.loc[mask]))


def normalize_geoids(series):
    return (
        series.astype(str)
        .str.replace(r"\D", "", regex=True)
```

```
        .str.zfill(11)
    )


def format_value_for_label(metric, val):
    if pd.isna(val):
        return ""
    if metric in ["poverty_rate", "unemp_rate", "lfpr_16p"]:
        return f"{val:.1%}"
    if metric == "median_household_income":
        return f"${val:,.0f}"
    return f"{val:.1f}"


def apply_axis_formatter(ax, metric):
    if metric in ["poverty_rate", "unemp_rate", "lfpr_16p"]:
        ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
    elif metric == "median_household_income":
        ax.yaxis.set_major_formatter(FuncFormatter(lambda x, pos: f"${x:,.0f


# ----------------------------
# build 3-group lookup
# ----------------------------
group_lookup = (
    three_group_base[["geoid", "final_compare_group"]]
    .drop_duplicates(subset=["geoid"])
    .copy()
)

if "normalize_geoid_series" in globals():
    group_lookup["geoid"] = normalize_geoid_series(group_lookup["geoid"])
else:
    group_lookup["geoid"] = normalize_geoids(group_lookup["geoid"])

group_lookup["group"] = group_lookup["final_compare_group"].map(comparison_l

# ----------------------------
# Shelby recovery-period panel
# ----------------------------
shelby_panel = analysis_panel.loc[
    (analysis_panel["display_state"].astype(str) == "Tennessee") &
    (analysis_panel["display_county"].astype(str) == "Shelby County") &
    (analysis_panel["year"].isin(recovery_years))
].copy()

if "normalize_geoid_series" in globals():
    shelby_panel["geoid"] = normalize_geoid_series(shelby_panel["geoid"])
else:
    shelby_panel["geoid"] = normalize_geoids(shelby_panel["geoid"])

shelby_panel = shelby_panel.merge(
    group_lookup[["geoid", "group"]],
    on="geoid",
    how="inner",
    validate="many_to_one"
```

```
)

# ------------------------------
# aggregate recovery-period values
# ------------------------------
rows = []

# Shelby 3 groups
for grp, df_grp in shelby_panel.groupby("group", dropna=False):
    weights = pd.to_numeric(df_grp["pop_total"], errors="coerce")

    row = {"group": grp}
    for col, _ in plot_metrics:
        row[col] = safe_weighted_mean(df_grp[col], weights)
    rows.append(row)

# Tennessee
tn_panel = analysis_panel.loc[
    (analysis_panel["display_state"].astype(str) == "Tennessee") &
    (analysis_panel["year"].isin(recovery_years))
].copy()

tn_weights = pd.to_numeric(tn_panel["pop_total"], errors="coerce")
tn_row = {"group": "Tennessee"}
for col, _ in plot_metrics:
    tn_row[col] = safe_weighted_mean(tn_panel[col], tn_weights)
rows.append(tn_row)

# U.S.
us_panel = analysis_panel.loc[
    analysis_panel["year"].isin(recovery_years)
].copy()

us_weights = pd.to_numeric(us_panel["pop_total"], errors="coerce")
us_row = {"group": "U.S."}
for col, _ in plot_metrics:
    us_row[col] = safe_weighted_mean(us_panel[col], us_weights)
rows.append(us_row)

benchmark_raw = pd.DataFrame(rows)

benchmark_raw = (
    benchmark_raw.set_index("group")
    .reindex(group_order)
    .reset_index()
)

# ------------------------------
# plot
# ------------------------------
fig, axes = plt.subplots(2, 2, figsize=(16, 10))
axes = axes.flatten()

x = np.arange(len(group_order))

for ax, (metric, title) in zip(axes, plot_metrics):
```

```python
    vals = pd.to_numeric(benchmark_raw[metric], errors="coerce").values

    bars = ax.bar(
        x,
        vals,
        color=[color_map[g] for g in benchmark_raw["group"]],
        width=0.72
    )

    ax.set_title(title, fontsize=17)
    ax.set_xticks(x)
    ax.set_xticklabels(
        ["Our\nCommunity", "Rest of\nShelby", "High-Growth\nShelby", "Tennes
        fontsize=11
    )
    ax.grid(axis="y", alpha=0.25)
    apply_axis_formatter(ax, metric)

    ymax = np.nanmax(vals) if np.isfinite(np.nanmax(vals)) else 1
    ymin = np.nanmin(vals) if np.isfinite(np.nanmin(vals)) else 0
    yrange = ymax - ymin

    if yrange == 0:
        upper = ymax * 1.15 if ymax != 0 else 1
    else:
        upper = ymax + yrange * 0.18

    ax.set_ylim(0, upper)

    label_pad = max(yrange * 0.03, upper * 0.015)

    for bar, val in zip(bars, vals):
        if pd.notna(val):
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + label_pad,
                format_value_for_label(metric, val),
                ha="center",
                va="bottom",
                fontsize=10
            )

# legend_handles = [
#     Patch(facecolor=color_map[g], label=g) for g in group_order
# ]

# fig.legend(
#     handles=legend_handles,
#     loc="upper left",
#     bbox_to_anchor=(0.015, 0.985),
#     ncol=1,
#     frameon=True,
#     fontsize=14,
#     title_fontsize=15,
#     borderaxespad=0.8
# )
```

```
fig.suptitle("Visual 3 — Shelby economic benchmark", fontsize=24, y=0.89)

fig.text(
    0.02,
    0.02,
    "Recovery-period averages (2022–2024). Values are shown directly rather
    fontsize=11
)

plt.tight_layout(rect=[0, 0.05, 1, 0.90])
plt.show()

out_path = OUT_DIR / "shelby_visual_3_economic_benchmark_raw_values_simplifi
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:

```
```

In [ ]:

```
import matplotlib.pyplot as plt
from matplotlib.patches import Patch

legend_handles = [
    Patch(facecolor=color_map[g], label=g) for g in group_order
]

fig_legend, ax_legend = plt.subplots(figsize=(6, 3))
ax_legend.axis("off")

ax_legend.legend(
    handles=legend_handles,
    loc="center",
    ncol=1,
    frameon=True,
    fontsize=14
)

plt.tight_layout()
plt.show()
```

In [ ]:

```
# ============================================================
# VISUAL 3B — Shelby labor force participation benchmark
# Standalone version, no legend
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import PercentFormatter

recovery_years = [2022, 2023, 2024]
lfpr_metric = "lfpr_16p"
lfpr_title = "Labor Force Participation"

group_order = [
```

```
    "Our Shelby Community",
    "Rest of Shelby County",
    "High-Growth Shelby Community",
    "Tennessee",
    "U.S.",
]

color_map = {
    "Our Shelby Community": "#fb8072",
    "Rest of Shelby County": "#bebada",
    "High-Growth Shelby Community": "#8dd3c7",
    "Tennessee": "#2ca02c",
    "U.S.": "#d62728",
}

comparison_label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

def safe_weighted_mean(values: pd.Series, weights: pd.Series):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)

    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())

    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

def normalize_geoids(series):
    return (
        series.astype(str)
        .str.replace(r"\D", "", regex=True)
        .str.zfill(11)
    )

# -----------------------------
# tract -> group lookup
# -----------------------------
group_lookup = (
    three_group_base[["geoid", "final_compare_group"]]
    .drop_duplicates(subset=["geoid"])
    .copy()
)

if "normalize_geoid_series" in globals():
    group_lookup["geoid"] = normalize_geoid_series(group_lookup["geoid"])
else:
    group_lookup["geoid"] = normalize_geoids(group_lookup["geoid"])

group_lookup["group"] = group_lookup["final_compare_group"].map(comparison_l
```

```python
# ----------------------------
# Shelby recovery-period panel
# ----------------------------
shelby_panel = analysis_panel.loc[
    (analysis_panel["display_state"].astype(str) == "Tennessee") &
    (analysis_panel["display_county"].astype(str) == "Shelby County") &
    (analysis_panel["year"].isin(recovery_years))
].copy()

if "normalize_geoid_series" in globals():
    shelby_panel["geoid"] = normalize_geoid_series(shelby_panel["geoid"])
else:
    shelby_panel["geoid"] = normalize_geoids(shelby_panel["geoid"])

shelby_panel = shelby_panel.merge(
    group_lookup[["geoid", "group"]],
    on="geoid",
    how="inner",
    validate="many_to_one"
)

rows = []

# Shelby 3 groups
for grp, df_grp in shelby_panel.groupby("group", dropna=False):
    weights = pd.to_numeric(df_grp["pop_total"], errors="coerce")
    rows.append({
        "group": grp,
        lfpr_metric: safe_weighted_mean(df_grp[lfpr_metric], weights)
    })

# Tennessee
tn_panel = analysis_panel.loc[
    (analysis_panel["display_state"].astype(str) == "Tennessee") &
    (analysis_panel["year"].isin(recovery_years))
].copy()

tn_weights = pd.to_numeric(tn_panel["pop_total"], errors="coerce")
rows.append({
    "group": "Tennessee",
    lfpr_metric: safe_weighted_mean(tn_panel[lfpr_metric], tn_weights)
})

# U.S.
us_panel = analysis_panel.loc[
    analysis_panel["year"].isin(recovery_years)
].copy()

us_weights = pd.to_numeric(us_panel["pop_total"], errors="coerce")
rows.append({
    "group": "U.S.",
    lfpr_metric: safe_weighted_mean(us_panel[lfpr_metric], us_weights)
})

lfpr_benchmark = pd.DataFrame(rows)
```

```
lfpr_benchmark = (
    lfpr_benchmark.set_index("group")
    .reindex(group_order)
    .reset_index()
)

# ----------------------------
# plot
# ----------------------------
vals = pd.to_numeric(lfpr_benchmark[lfpr_metric], errors="coerce").values
x = np.arange(len(group_order))

fig, ax = plt.subplots(figsize=(10, 7.5))

bars = ax.bar(
    x,
    vals,
    color=[color_map[g] for g in lfpr_benchmark["group"]],
    width=0.72
)

ax.set_title(lfpr_title, fontsize=20, pad=18)
ax.set_xticks(x)
ax.set_xticklabels(
    ["Our\nCommunity", "Rest of\nShelby", "High-Growth\nShelby", "Tennessee"
    fontsize=12
)
ax.grid(axis="y", alpha=0.25)
ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))

ymax = np.nanmax(vals) if np.isfinite(np.nanmax(vals)) else 1
ymin = np.nanmin(vals) if np.isfinite(np.nanmin(vals)) else 0
yrange = ymax - ymin

# more space above bars
upper = ymax + max(0.06, yrange * 0.28)
ax.set_ylim(0, upper)

label_pad = max(0.008, yrange * 0.05)

for bar, val in zip(bars, vals):
    if pd.notna(val):
        ax.text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height() + label_pad,
            f"{val:.1%}",
            ha="center",
            va="bottom",
            fontsize=11
        )

fig.text(
    0.02,
    0.02,
    "Recovery-period average (2022–2024).",
    fontsize=11
```

```
)

plt.tight_layout(rect=[0, 0.05, 1, 0.95])
plt.show()

out_path = OUT_DIR / "shelby_visual_3b_lfpr_benchmark.png"
fig.savefig(out_path, bbox_inches="tight", dpi=220)
print("Saved:", out_path)
```

In [ ]:
```
# # =================================================================
# # VISUAL 3C — Shelby ethnicity comparison
# # Our community vs Rest of Shelby County vs High-growth Shelby
# # Recovery-period average composition (2022–2024)
# # =================================================================

# import numpy as np
# import pandas as pd
# import matplotlib.pyplot as plt
# from matplotlib.ticker import PercentFormatter

# # -----------------------------
# # settings
# # -----------------------------
# recovery_years = [2022, 2023, 2024]

# group_order = [
#     "Our Shelby Community",
#     "Rest of Shelby County",
#     "High-Growth Shelby Community",
# ]

# comparison_label_map = {
#     "Selected low-growth community": "Our Shelby Community",
#     "Rest of Shelby County (excluding both selected communities)": "Rest o
#     "Selected high-growth comparator": "High-Growth Shelby Community",
# }

# # Candidate ethnicity/race share columns.
# # The code will use the first one it finds in your panel for each category
# ethnicity_candidates = {
#     "White (non-Hispanic)": [
#         "share_white_nonhisp",
#         "pct_white_nonhisp",
#         "white_nonhisp_share",
#         "white_nonhisp_pct",
#         "white_share",
#         "pct_white",
#     ],
#     "Black (non-Hispanic)": [
#         "share_black_nonhisp",
#         "pct_black_nonhisp",
#         "black_nonhisp_share",
#         "black_nonhisp_pct",
#         "black_share",
#         "pct_black",
#     ],
```

```
#     "Hispanic / Latino": [
#         "share_hispanic",
#         "pct_hispanic",
#         "hispanic_share",
#         "hispanic_pct",
#         "latino_share",
#         "pct_latino",
#     ],
#     "Asian (non-Hispanic)": [
#         "share_asian_nonhisp",
#         "pct_asian_nonhisp",
#         "asian_nonhisp_share",
#         "asian_nonhisp_pct",
#         "asian_share",
#         "pct_asian",
#     ],
#     "Other / Multiracial": [
#         "share_other_nonhisp",
#         "pct_other_nonhisp",
#         "other_nonhisp_share",
#         "other_nonhisp_pct",
#         "other_share",
#         "pct_other",
#         "multiracial_share",
#         "pct_multiracial",
#     ],
# }

# ethnicity_colors = {
#     "White (non-Hispanic)": "#8dd3c7",
#     "Black (non-Hispanic)": "#fb8072",
#     "Hispanic / Latino": "#bebada",
#     "Asian (non-Hispanic)": "#80b1d3",
#     "Other / Multiracial": "#fdb462",
# }

# # ----------------------------
# # helpers
# # ----------------------------
# def safe_weighted_mean(values: pd.Series, weights: pd.Series):
#     values = pd.to_numeric(values, errors="coerce")
#     weights = pd.to_numeric(weights, errors="coerce")
#     mask = values.notna() & weights.notna() & (weights > 0)

#     if mask.sum() == 0:
#         valid = values.notna()
#         if valid.sum() == 0:
#             return np.nan
#         return float(values.loc[valid].mean())

#     return float(np.average(values.loc[mask], weights=weights.loc[mask]))


# def normalize_geoids(series):
#     return (
#         series.astype(str)
```

```
#         .str.replace(r"\D", "", regex=True)
#         .str.zfill(11)
#     )


# def pick_existing_column(df, candidates):
#     for c in candidates:
#         if c in df.columns:
#             return c
#     return None


# def to_share(series):
#     """
#     Converts percent-like values to shares if needed.
#     If values look like 0-100, divide by 100.
#     """
#     s = pd.to_numeric(series, errors="coerce").copy()
#     if s.dropna().max() is not np.nan and s.dropna().max() > 1.5:
#         s = s / 100.0
#     return s


# # ----------------------------
# # tract -> comparison group lookup
# # ----------------------------
# group_lookup = (
#     three_group_base[["geoid", "final_compare_group"]]
#     .drop_duplicates(subset=["geoid"])
#     .copy()
# )

# if "normalize_geoid_series" in globals():
#     group_lookup["geoid"] = normalize_geoid_series(group_lookup["geoid"])
# else:
#     group_lookup["geoid"] = normalize_geoids(group_lookup["geoid"])

# group_lookup["group"] = group_lookup["final_compare_group"].map(comparison

# # ----------------------------
# # Shelby panel for recovery period
# # ----------------------------
# shelby_panel = analysis_panel.loc[
#     (analysis_panel["display_state"].astype(str) == "Tennessee") &
#     (analysis_panel["display_county"].astype(str) == "Shelby County") &
#     (analysis_panel["year"].isin(recovery_years))
# ].copy()

# if "normalize_geoid_series" in globals():
#     shelby_panel["geoid"] = normalize_geoid_series(shelby_panel["geoid"])
# else:
#     shelby_panel["geoid"] = normalize_geoids(shelby_panel["geoid"])

# shelby_panel = shelby_panel.merge(
#     group_lookup[["geoid", "group"]],
#     on="geoid",
```

```
#     how="inner",
#     validate="many_to_one"
# )

# # ----------------------------
# # resolve ethnicity columns
# # ----------------------------
# resolved_ethnicity_cols = {}
# for label, candidates in ethnicity_candidates.items():
#     chosen = pick_existing_column(shelby_panel, candidates)
#     if chosen is not None:
#         resolved_ethnicity_cols[label] = chosen

# if len(resolved_ethnicity_cols) < 3:
#     keyword_cols = [c for c in shelby_panel.columns if any(k in c.lower()
#     print("Possible demographic columns found in analysis_panel:")
#     print(keyword_cols)
#     raise ValueError(
#         "Could not find enough ethnicity/race share columns automatically.
#         "Check the printed column list and update `ethnicity_candidates`."
#     )

# print("Using these ethnicity columns:")
# for k, v in resolved_ethnicity_cols.items():
#     print(f"  {k}: {v}")

# # ----------------------------
# # aggregate weighted composition by group
# # ----------------------------
# rows = []

# for grp, df_grp in shelby_panel.groupby("group", dropna=False):
#     weights = pd.to_numeric(df_grp["pop_total"], errors="coerce")
#     row = {"group": grp}

#     for label, col in resolved_ethnicity_cols.items():
#         row[label] = safe_weighted_mean(to_share(df_grp[col]), weights)

#     rows.append(row)

# ethnicity_compare = pd.DataFrame(rows)
# ethnicity_compare = (
#     ethnicity_compare.set_index("group")
#     .reindex(group_order)
#     .reset_index()
# )

# # ----------------------------
# # if "Other / Multiracial" was not found, infer residual share
# # ----------------------------
# segment_cols = [c for c in ethnicity_compare.columns if c != "group"]

# if "Other / Multiracial" not in segment_cols:
#     ethnicity_compare["Other / Multiracial"] = (
#         1.0 - ethnicity_compare[segment_cols].sum(axis=1)
#     ).clip(lower=0)
```

```
#     segment_cols = [c for c in ethnicity_compare.columns if c != "group"]

# # normalize rows so bars sum to 100%
# row_sums = ethnicity_compare[segment_cols].sum(axis=1).replace(0, np.nan)
# ethnicity_compare[segment_cols] = ethnicity_compare[segment_cols].div(row_

# # keep only columns that actually exist after normalization
# segment_cols = [c for c in ethnicity_colors.keys() if c in ethnicity_compa

# # ----------------------------
# # plot
# # ----------------------------
# fig, ax = plt.subplots(figsize=(12, 6.5))

# y = np.arange(len(group_order))
# left = np.zeros(len(group_order))

# for seg in segment_cols:
#     vals = pd.to_numeric(ethnicity_compare[seg], errors="coerce").values

#     bars = ax.barh(
#         y,
#         vals,
#         left=left,
#         color=ethnicity_colors[seg],
#         edgecolor="white",
#         height=0.62,
#         label=seg
#     )

#     # segment labels if large enough
#     for i, v in enumerate(vals):
#         if pd.notna(v) and v >= 0.07:
#             ax.text(
#                 left[i] + v / 2,
#                 y[i],
#                 f"{v:.0%}",
#                 ha="center",
#                 va="center",
#                 fontsize=10,
#                 color="black"
#             )

#     left += np.nan_to_num(vals, nan=0.0)

# ax.set_yticks(y)
# ax.set_yticklabels(group_order, fontsize=12)
# ax.set_xlim(0, 1)
# ax.xaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
# ax.grid(axis="x", alpha=0.25)
# ax.invert_yaxis()

# ax.set_title("Shelby ethnicity comparison", fontsize=20, pad=15)

# ax.legend(
#     loc="upper center",
```

```
#     bbox_to_anchor=(0.5, 1.12),
#     ncol=min(len(segment_cols), 5),
#     frameon=True,
#     fontsize=11
# )

# fig.text(
#     0.02,
#     0.02,
#     "Recovery-period averages (2022–2024), aggregated from tract-level dem
#     fontsize=11
# )

# plt.tight_layout(rect=[0, 0.05, 1, 0.93])
# plt.show()

# out_path = OUT_DIR / "shelby_visual_3c_ethnicity_comparison.png"
# fig.savefig(out_path, bbox_inches="tight", dpi=220)
# print("Saved:", out_path)

# display(ethnicity_compare)
```

In [ ]:

```
import pandas as pd

print("=== Relevant DataFrames currently in memory ===\n")

keywords = [
    "health", "chr", "care", "medical", "uninsured",
    "heatmap", "county", "cluster", "finalist", "shortlist"
]

found_any = False

for name, obj in list(globals().items()):
    if isinstance(obj, pd.DataFrame):
        lname = name.lower()
        if any(key in lname for key in keywords):
            found_any = True
            print(f"{name}: shape={obj.shape}")
            print("Columns:", list(obj.columns[:30]))
            if len(obj.columns) > 30:
                print("... plus more")
            print("-" * 100)

if not found_any:
    print("No relevant DataFrames found.")
```

In [ ]:

```
# ============================================================
# FINALIST COUNTY HEATMAP
# Uses CURRENT notebook objects:
#   - healthcare_cluster_summary
#   - enrichment_cluster_profiles
# ============================================================

import pandas as pd
```

```python
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.colors import LinearSegmentedColormap

# ---------------------------------------------------
# 1. Validate required tables
# ---------------------------------------------------
required_tables = ["healthcare_cluster_summary", "enrichment_cluster_profile
for name in required_tables:
    if name not in globals():
        raise ValueError(f"Required dataframe `{name}` is not in memory.")

# ---------------------------------------------------
# 2. Merge healthcare-covered finalists with recovery economics
# ---------------------------------------------------
health_df = healthcare_cluster_summary.copy()
econ_df = enrichment_cluster_profiles.copy()

keep_econ_cols = [
    "cluster_id",
    "enrichment_rank",
    "rank",
    "cluster_shortlist_score",
    "igs_economy_recovery",
    "poverty_rate_recovery",
    "unemp_rate_recovery",
    "median_household_income_recovery",
    "lfpr_16p_recovery",
]

base_df = health_df.merge(
    econ_df[keep_econ_cols].drop_duplicates(subset=["cluster_id"]),
    on="cluster_id",
    how="left",
    validate="1:1"
)

# ---------------------------------------------------
# 3. Remove duplicate counties
# Keep the best-ranked finalist cluster in each county
# ---------------------------------------------------
base_df = base_df.sort_values(
    ["enrichment_rank", "cluster_shortlist_score"],
    ascending=[True, False]
).reset_index(drop=True)

county_df = (
    base_df
    .drop_duplicates(subset=["display_state", "display_county"], keep="first
    .reset_index(drop=True)
)

county_df["finalist_county_rank"] = np.arange(1, len(county_df) + 1)
county_df["county_label"] = (
    "#" + county_df["finalist_county_rank"].astype(str)
    + " – "
```

```python
    + county_df["display_county"].astype(str)
    + ", "
    + county_df["display_state"].astype(str)
)

print("Counties retained after removing duplicates:", len(county_df))
display(
    county_df[
        [
            "finalist_county_rank",
            "cluster_id",
            "display_county",
            "display_state",
            "enrichment_rank",
            "cluster_shortlist_score",
        ]
    ]
)

# --------------------------------------------------
# 4. Helper: convert a metric to need intensity
# higher score = worse need / weaker condition
# --------------------------------------------------
def rank_need(series, direction):
    s = pd.to_numeric(series, errors="coerce")

    if s.notna().sum() <= 1:
        return pd.Series(np.nan, index=series.index)

    if direction == "high_is_bad":
        return s.rank(pct=True, method="average")
    elif direction == "low_is_bad":
        return (-s).rank(pct=True, method="average")
    else:
        raise ValueError(f"Unknown direction: {direction}")

# --------------------------------------------------
# 5. Build healthcare composites
# --------------------------------------------------
ranked_parts = pd.DataFrame(index=county_df.index)

# uninsured burden pieces
ranked_parts["uninsured_rate"] = rank_need(county_df["uninsured_rate"], "hig
ranked_parts["uninsured_adults_rate"] = rank_need(county_df["uninsured_adult
ranked_parts["uninsured_children_rate"] = rank_need(county_df["uninsured_chi

# provider access pieces (low supply = worse)
ranked_parts["primary_care_physicians_rate"] = rank_need(county_df["primary_
ranked_parts["dentists_rate"] = rank_need(county_df["dentists_rate"], "low_i
ranked_parts["mental_health_providers_rate"] = rank_need(county_df["mental_h
ranked_parts["other_primary_care_providers_rate"] = rank_need(county_df["oth

# preventable hospital burden
ranked_parts["preventable_hospital_stays_rate"] = rank_need(county_df["preve

# preventive care gap (lower screening / vaccination = worse)
```

```
ranked_parts["mammography_screening_rate"] = rank_need(county_df["mammograph
ranked_parts["flu_vaccinations_rate"] = rank_need(county_df["flu_vaccinatior

# --------------------------------------------------
# 6. Build final heatmap table
# FIX: use positional assignment, then set county labels as index at the enc
# --------------------------------------------------
county_df = county_df.reset_index(drop=True)
ranked_parts = ranked_parts.reset_index(drop=True)

heatmap_df = pd.DataFrame(index=np.arange(len(county_df)))

heatmap_df["Uninsured Burden"] = ranked_parts[
    ["uninsured_rate", "uninsured_adults_rate", "uninsured_children_rate"]
].mean(axis=1, skipna=True).to_numpy()

heatmap_df["Provider Access Gap"] = ranked_parts[
    [
        "primary_care_physicians_rate",
        "dentists_rate",
        "mental_health_providers_rate",
        "other_primary_care_providers_rate",
    ]
].mean(axis=1, skipna=True).to_numpy()

heatmap_df["Preventable Hospital Burden"] = ranked_parts[
    "preventable_hospital_stays_rate"
].to_numpy()

heatmap_df["Preventive Care Gap"] = ranked_parts[
    ["mammography_screening_rate", "flu_vaccinations_rate"]
].mean(axis=1, skipna=True).to_numpy()

# overall healthcare need = mean of the healthcare composites
heatmap_df["Overall Healthcare Need"] = heatmap_df[
    [
        "Uninsured Burden",
        "Provider Access Gap",
        "Preventable Hospital Burden",
        "Preventive Care Gap",
    ]
].mean(axis=1, skipna=True)

# economic indicators from shortlist logic
heatmap_df["IGS Economy"] = rank_need(county_df["igs_economy_recovery"], "lc
heatmap_df["Poverty Rate"] = rank_need(county_df["poverty_rate_recovery"], "
heatmap_df["Unemployment Rate"] = rank_need(county_df["unemp_rate_recovery"]
heatmap_df["Median Income"] = rank_need(county_df["median_household_income_r
heatmap_df["Labor Force Participation"] = rank_need(county_df["lfpr_16p_recc

# final column order
heatmap_df = heatmap_df[
    [
        "Overall Healthcare Need",
        "Uninsured Burden",
        "Provider Access Gap",
```

```
        "Preventable Hospital Burden",
        "Preventive Care Gap",
        "IGS Economy",
        "Poverty Rate",
        "Unemployment Rate",
        "Median Income",
        "Labor Force Participation",
    ]
]

# now set the pretty county labels AFTER all assignments
heatmap_df.index = county_df["county_label"].tolist()

# optional: sort rows by overall healthcare need, highest need at top
heatmap_df = heatmap_df.sort_values("Overall Healthcare Need", ascending=Fal

display(heatmap_df.round(3))
# ----------------------------------------------------
# 7. Plot heatmap
# Green = lower need, Red = higher need
# ----------------------------------------------------
plot_df = heatmap_df.copy()
plot_df.columns = [
    "Overall Need",
    "Uninsured Burden",
    "Provider Gap",
    "Preventable Stays",
    "Prevention Gap",
    "IGS Economy",
    "Poverty",
    "Unemployment",
    "Income",
    "LFP",
]

cmap = LinearSegmentedColormap.from_list(
    "green_to_red_need",
    ["#1a9850", "#ffffbf", "#d73027"]
)

fig, ax = plt.subplots(figsize=(16, max(6, len(plot_df) * 0.7)))

im = ax.imshow(
    plot_df.values,
    aspect="auto",
    cmap=cmap,
    vmin=0,
    vmax=1
)

ax.set_xticks(np.arange(plot_df.shape[1]))
ax.set_xticklabels(plot_df.columns, rotation=0, ha="center")
ax.set_yticks(np.arange(plot_df.shape[0]))
ax.set_yticklabels(plot_df.index)

ax.set_title("Healthcare + Economic Need Comparison Across Finalist Counties
```

```python
ax.set_xlabel("Higher intensity = worse need / weaker economic conditions")
ax.set_ylabel("Finalist county")

cbar = plt.colorbar(im, ax=ax)
cbar.set_label("Relative need intensity")

plt.tight_layout()
plt.show()

# --------------------------------------------------
# 8. Save
# --------------------------------------------------
OUT_DIR.mkdir(parents=True, exist_ok=True)

heatmap_df.to_csv(OUT_DIR / "finalist_county_healthcare_economic_heatmap_tab

fig.savefig(
    OUT_DIR / "finalist_county_healthcare_economic_heatmap.png",
    bbox_inches="tight",
    dpi=220
)

print("\nSaved:")
print("-", OUT_DIR / "finalist_county_healthcare_economic_heatmap_table.csv"
print("-", OUT_DIR / "finalist_county_healthcare_economic_heatmap.png")
```

In [ ]:

```python
# # --------------------------------------------------
# # Split heatmap into two aligned panels
# # Healthcare vs Economic Stress
# # --------------------------------------------------
# import numpy as np
# import matplotlib.pyplot as plt
# from matplotlib.colors import LinearSegmentedColormap
# from matplotlib.patches import Rectangle

# plot_df = heatmap_df.copy()

# # Shorter column labels
# plot_df = plot_df.rename(columns={
#     "Overall Healthcare Need": "Overall Need",
#     "Uninsured Burden": "Uninsured Burden",
#     "Provider Access Gap": "Provider Gap",
#     "Preventable Hospital Burden": "Preventable Stays",
#     "Preventive Care Gap": "Prevention Gap",
#     "IGS Economy": "IGS Economy",
#     "Poverty Rate": "Poverty",
#     "Unemployment Rate": "Unemployment",
#     "Median Income": "Income",
#     "Labor Force Participation": "LFP",
# })

# health_cols = [
#     "Overall Need",
#     "Uninsured Burden",
#     "Provider Gap",
#     "Preventable Stays",
```

```
#     "Prevention Gap",
# ]

# econ_cols = [
#     "IGS Economy",
#     "Poverty",
#     "Unemployment",
#     "Income",
#     "LFP",
# ]

# # Optional: keep the current order, or set your own narrative order here
# # Example narrative order:
# preferred_county_order = [
#     "Apache County, Arizona",
#     "McKinley County, New Mexico",
#     "Wayne County, Michigan",
#     "Shelby County, Tennessee",
#     "Cook County, Illinois",
#     "Cuyahoga County, Ohio",
#     "Milwaukee County, Wisconsin",
#     "Los Angeles County, California",
# ]

# ordered_index = []
# for county_name in preferred_county_order:
#     matches = [idx for idx in plot_df.index if county_name in idx]
#     ordered_index.extend(matches)

# # fallback in case anything was missed
# remaining = [idx for idx in plot_df.index if idx not in ordered_index]
# ordered_index.extend(remaining)

# plot_df = plot_df.loc[ordered_index]

# health_plot = plot_df[health_cols]
# econ_plot = plot_df[econ_cols]

# # cleaner row labels
# row_labels = [
#     idx.split("–", 1)[1].strip() if "–" in idx else idx
#     for idx in plot_df.index
# ]

# # cmap = LinearSegmentedColormap.from_list(
# #     "green_to_red_need",
# #     ["#1a9850", "#ffffbf", "#d73027"]
# # )

# cmap = LinearSegmentedColormap.from_list(
#     "soft_red_to_cream_need",
#     ["#fffaf0", "#fdd9b5", "#f4a582", "#ca0020"]
# )

# fig, (ax1, ax2) = plt.subplots(
#     1, 2,
```

```
#     figsize=(17, max(6, len(plot_df) * 0.7)),
#     gridspec_kw={"width_ratios": [5, 5], "wspace": 0.04},
#     sharey=True
# )

# im1 = ax1.imshow(
#     health_plot.values,
#     aspect="auto",
#     cmap=cmap,
#     vmin=0,
#     vmax=1
# )

# im2 = ax2.imshow(
#     econ_plot.values,
#     aspect="auto",
#     cmap=cmap,
#     vmin=0,
#     vmax=1
# )

# # Titles
# ax1.set_title("Healthcare Need", fontsize=14, pad=10)
# ax2.set_title("Economic Stress", fontsize=14, pad=10)

# # X labels
# ax1.set_xticks(np.arange(health_plot.shape[1]))
# ax1.set_xticklabels(health_plot.columns, rotation=0, ha="center")

# ax2.set_xticks(np.arange(econ_plot.shape[1]))
# ax2.set_xticklabels(econ_plot.columns, rotation=0, ha="center")

# # Y labels
# ax1.set_yticks(np.arange(len(row_labels)))
# ax1.set_yticklabels(row_labels)
# ax2.set_yticks(np.arange(len(row_labels)))
# ax2.tick_params(axis="y", left=False, labelleft=False)

# # Add white gridlines for readability
# for ax, ncols in [(ax1, health_plot.shape[1]), (ax2, econ_plot.shape[1])]:
#     ax.set_xticks(np.arange(-0.5, ncols, 1), minor=True)
#     ax.set_yticks(np.arange(-0.5, len(row_labels), 1), minor=True)
#     ax.grid(which="minor", color="white", linestyle="-", linewidth=1.2)
#     ax.tick_params(which="minor", bottom=False, left=False)

# # Highlight Shelby row
# shelby_idx = next((i for i, label in enumerate(row_labels) if "Shelby Cour
# if shelby_idx is not None:
#     ax1.add_patch(Rectangle(
#         (-0.5, shelby_idx - 0.5),
#         health_plot.shape[1],
#         1,
#         fill=False,
#         edgecolor="black",
#         linewidth=2.5
#     ))
```

```
#     ax2.add_patch(Rectangle(
#         (-0.5, shelby_idx - 0.5),
#         econ_plot.shape[1],
#         1,
#         fill=False,
#         edgecolor="black",
#         linewidth=2.5
#     ))

# fig.suptitle("Finalist county comparison: healthcare need and economic str
# ax1.set_ylabel("Finalist county")

# cbar = fig.colorbar(im2, ax=[ax1, ax2], fraction=0.025, pad=0.02)
# cbar.set_label("Relative need intensity")

# fig.text(
#     0.5, 0.02,
#     "Higher intensity = worse need / weaker economic conditions",
#     ha="center",
#     fontsize=11
# )

# plt.tight_layout(rect=[0, 0.05, 1, 0.95])
# plt.show()

# # Save
# fig.savefig(
#     OUT_DIR / "finalist_county_dual_heatmap_healthcare_economic.png",
#     bbox_inches="tight",
#     dpi=220
# )
# print("Saved:", OUT_DIR / "finalist_county_dual_heatmap_healthcare_economi
```

In [ ]:

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```
# ============================================================
# Cell 10A — load ShelbyCare feasibility / scenario CSV inputs
# Put your 3 CSVs either:
# 1) in data/processed/presentation_ready
# 2) or in data/processed
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd
```

```
SCENARIO_DIR_CANDIDATES = [
    OUT_DIR / "presentation_ready",
    OUT_DIR,
    DATA_DIR,
]

CSV_NAMES = {
    "baseline": "shelbycare_baseline.csv",
    "scenarios": "shelbycare_scenarios.csv",
    "assumptions": "shelbycare_assumptions.csv",
}

def find_csv(filename: str) -> Path:
    for folder in SCENARIO_DIR_CANDIDATES:
        p = folder / filename
        if p.exists():
            return p
    raise FileNotFoundError(
        f"Could not find {filename}. Checked: "
        + ", ".join(str(folder / filename) for folder in SCENARIO_DIR_CANDID
    )

baseline_csv_path = find_csv(CSV_NAMES["baseline"])
scenarios_csv_path = find_csv(CSV_NAMES["scenarios"])
assumptions_csv_path = find_csv(CSV_NAMES["assumptions"])

baseline_df = pd.read_csv(baseline_csv_path)
scenario_inputs = pd.read_csv(scenarios_csv_path)
assumptions_df = pd.read_csv(assumptions_csv_path)

print("Loaded:")
print("-", baseline_csv_path)
print("-", scenarios_csv_path)
print("-", assumptions_csv_path)

display(baseline_df)
display(scenario_inputs.head())
display(assumptions_df.head())
```

In [ ]:

```
# ============================================================
# Cell 10A — load ShelbyCare feasibility / scenario CSV inputs
# Put your 3 CSVs either:
# 1) in data/processed/presentation_ready
# 2) or in data/processed
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd

SCENARIO_DIR_CANDIDATES = [
    OUT_DIR / "presentation_ready",
    OUT_DIR,
    DATA_DIR,
    Path("src/data"),
    Path("src/data/processed"),
```

```python
    Path("src/data/processed/presentation_ready"),
]

CSV_NAMES = {
    "baseline": "shelbycare_baseline.csv",
    "scenarios": "shelbycare_scenarios.csv",
    "assumptions": "shelbycare_assumptions.csv",
}

def find_csv(filename: str) -> Path:
    for folder in SCENARIO_DIR_CANDIDATES:
        p = folder / filename
        if p.exists():
            return p
    raise FileNotFoundError(
        f"Could not find {filename}. Checked: "
        + ", ".join(str(folder / filename) for folder in SCENARIO_DIR_CANDID
    )

baseline_csv_path = find_csv(CSV_NAMES["baseline"])
scenarios_csv_path = find_csv(CSV_NAMES["scenarios"])
assumptions_csv_path = find_csv(CSV_NAMES["assumptions"])

baseline_df = pd.read_csv(baseline_csv_path)
scenario_inputs = pd.read_csv(scenarios_csv_path)
assumptions_df = pd.read_csv(assumptions_csv_path)

print("Loaded:")
print("-", baseline_csv_path)
print("-", scenarios_csv_path)
print("-", assumptions_csv_path)

display(baseline_df)
display(scenario_inputs.head())
display(assumptions_df.head())
```

In [ ]:

```python
# ============================================================
# Cell 10B — patch final baseline values + validate
# ============================================================

# Use the values you just provided
BASELINE_PATCH = {
    "baseline_lfpr_16p": 58.2,
    "baseline_igs_total": 35.8,
    "baseline_igs_economy": 47.8,
    "baseline_poverty_rate": 30.0,
    "baseline_median_household_income": 43421.0,
}

if len(baseline_df) != 1:
    raise ValueError("shelbycare_baseline.csv should contain exactly 1 row."

for col, val in BASELINE_PATCH.items():
    if col not in baseline_df.columns:
        baseline_df[col] = np.nan
    baseline_df.loc[0, col] = val
```

```python
# numeric normalization
numeric_cols = [
    "baseline_population",
    "baseline_target_service_population",
    "labor_force_estimate",
    "baseline_unemp_rate",
    "baseline_lfpr_16p",
    "baseline_igs_total",
    "baseline_igs_economy",
    "baseline_poverty_rate",
    "baseline_median_household_income",
    "baseline_uninsured_adults_pct",
    "baseline_diabetes_pct",
    "baseline_obesity_pct",
    "baseline_dental_visit_gap_pct",
]
for c in numeric_cols:
    if c in baseline_df.columns:
        baseline_df[c] = pd.to_numeric(baseline_df[c], errors="coerce")

scenario_numeric_cols = [
    c for c in scenario_inputs.columns
    if c != "scenario_name"
]
for c in scenario_numeric_cols:
    scenario_inputs[c] = pd.to_numeric(scenario_inputs[c], errors="coerce")

required_baseline_cols = [
    "baseline_target_service_population",
    "labor_force_estimate",
    "baseline_unemp_rate",
    "baseline_lfpr_16p",
    "baseline_igs_total",
    "baseline_igs_economy",
    "baseline_poverty_rate",
    "baseline_median_household_income",
    "baseline_uninsured_adults_pct",
    "baseline_diabetes_pct",
    "baseline_obesity_pct",
    "baseline_dental_visit_gap_pct",
]

missing_baseline = [c for c in required_baseline_cols if c not in baseline_c
if missing_baseline:
    raise ValueError(f"Baseline CSV still has missing required fields: {miss

print("Patched baseline row:")
display(baseline_df)
```

In [ ]:

```python
# ==============================================================
# Cell 10D — compute full ShelbyCare output table
# ==============================================================

baseline = baseline_df.iloc[0].to_dict()
```

```
# compensation assumptions from your feasibility write-up
ROLE_RATES = {
    "family_practitioner_hourly": 120.0,
    "dentist_hourly": 110.0,
    "dietician_hourly": 45.0,
}

# assumption coefficients from shelbycare_assumptions.csv
job_fill_rate = safe_num(get_effect("job_fill_rate", 0.65), 0.65)
dental_effect_per_visit = safe_num(get_effect("projected_dental_visit_gap_pc
obesity_effect_per_bundle = safe_num(get_effect("projected_obesity_pct", -0.
diabetes_effect_per_bundle = safe_num(get_effect("projected_diabetes_pct", -
uninsured_effect_per_medride = safe_num(get_effect("projected_uninsured_adul

rows = []

for _, s in scenario_inputs.iterrows():
    row = {"scenario_name": s["scenario_name"]}

    # ---------------------------------------------------
    # 1) raw scenario inputs
    # ---------------------------------------------------
    n_family_practitioners = safe_num(s["n_family_practitioners"])
    fp_hours_each = safe_num(s["fp_hours_per_week_each"])
    n_dentists = safe_num(s["n_dentists"])
    dentist_hours_each = safe_num(s["dentist_hours_per_week_each"])
    n_dieticians = safe_num(s["n_dieticians"])
    dietician_hours_each = safe_num(s["dietician_hours_per_week_each"])

    n_drivers = safe_num(s["n_drivers"])
    driver_hours_per_week_avg = safe_num(s["driver_hours_per_week_avg"])
    hourly_driver_pay = safe_num(s["hourly_driver_pay"])

    produce_baskets_per_week = safe_num(s["produce_baskets_per_week"])
    seed_cost_per_month = safe_num(s["seed_cost_per_month"])
    greenhouse_op_cost_per_month = safe_num(s["greenhouse_op_cost_per_month"
    mobile_clinic_cost_per_month = safe_num(s["mobile_clinic_cost_per_month"
    admin_cost_per_month = safe_num(s["admin_cost_per_month"])

    subscription_members = safe_num(s["subscription_members"])
    subscription_price_per_month = safe_num(s["subscription_price_per_month"
    avg_medical_visit_minutes = safe_num(s["avg_medical_visit_minutes"], 30.
    avg_rides_per_driver_per_week = safe_num(s["avg_rides_per_driver_per_wee
    medical_ride_share_pct = safe_num(s["medical_ride_share_pct"], 0.10)

    # ---------------------------------------------------
    # 2) staffing / visit capacity
    # ---------------------------------------------------
    fp_hours_total_per_week = n_family_practitioners * fp_hours_each
    dentist_hours_total_per_week = n_dentists * dentist_hours_each
    dietician_hours_total_per_week = n_dieticians * dietician_hours_each
    clinical_hours_total_per_week = (
        fp_hours_total_per_week
        + dentist_hours_total_per_week
        + dietician_hours_total_per_week
    )
```

```
visits_per_hour = 60.0 / avg_medical_visit_minutes if avg_medical_visit_

clinic_visits_per_week = clinical_hours_total_per_week * visits_per_hour
fp_visits_per_week = fp_hours_total_per_week * visits_per_hour
dental_visits_per_week = dentist_hours_total_per_week * visits_per_hour
dietician_visits_per_week = dietician_hours_total_per_week * visits_per_

# --------------------------------------------------
# 3) transportation / jobs
# --------------------------------------------------
driver_hours_total_per_week = n_drivers * driver_hours_per_week_avg
direct_jobs_created = n_drivers
fte_jobs_created = driver_hours_total_per_week / 40.0

rides_per_week = n_drivers * avg_rides_per_driver_per_week
medical_rides_per_week = rides_per_week * medical_ride_share_pct

# --------------------------------------------------
# 4) costs + revenue
# --------------------------------------------------
weekly_driver_wages = driver_hours_total_per_week * hourly_driver_pay
annual_driver_wages = annualize_weekly(weekly_driver_wages)

weekly_fp_cost = fp_hours_total_per_week * ROLE_RATES["family_practition
weekly_dentist_cost = dentist_hours_total_per_week * ROLE_RATES["dentist
weekly_dietician_cost = dietician_hours_total_per_week * ROLE_RATES["die

annual_fp_cost = annualize_weekly(weekly_fp_cost)
annual_dentist_cost = annualize_weekly(weekly_dentist_cost)
annual_dietician_cost = annualize_weekly(weekly_dietician_cost)
annual_clinician_cost = annual_fp_cost + annual_dentist_cost + annual_di

annual_seed_cost = annualize_monthly(seed_cost_per_month)
annual_greenhouse_cost = annualize_monthly(greenhouse_op_cost_per_month)
annual_mobile_clinic_cost = annualize_monthly(mobile_clinic_cost_per_mon
annual_admin_cost = annualize_monthly(admin_cost_per_month)

annual_total_cost = (
    annual_driver_wages
    + annual_clinician_cost
    + annual_seed_cost
    + annual_greenhouse_cost
    + annual_mobile_clinic_cost
    + annual_admin_cost
)

annual_subscription_revenue = subscription_members * subscription_price_
annual_net_operating = annual_subscription_revenue - annual_total_cost

# --------------------------------------------------
# 5) projected community metrics
# These are scenario-based formulas, not causal treatment estimates
# --------------------------------------------------
baseline_unemp_rate = safe_num(baseline["baseline_unemp_rate"])
labor_force_estimate = safe_num(baseline["labor_force_estimate"])
```

```
    baseline_dental_gap = safe_num(baseline["baseline_dental_visit_gap_pct"]
    baseline_obesity = safe_num(baseline["baseline_obesity_pct"])
    baseline_diabetes = safe_num(baseline["baseline_diabetes_pct"])
    baseline_uninsured = safe_num(baseline["baseline_uninsured_adults_pct"])

    projected_unemp_rate = baseline_unemp_rate - (job_fill_rate * direct_job
    projected_unemp_rate = clip_between(projected_unemp_rate, 0.0, 1.0)

    projected_dental_visit_gap_pct = baseline_dental_gap + (dental_effect_pe
    projected_dental_visit_gap_pct = clip_between(projected_dental_visit_gap

    projected_obesity_pct = baseline_obesity + (obesity_effect_per_bundle *
    projected_obesity_pct = clip_between(projected_obesity_pct, 0.0, 100.0)

    projected_diabetes_pct = baseline_diabetes + (diabetes_effect_per_bundle
    projected_diabetes_pct = clip_between(projected_diabetes_pct, 0.0, 100.0

    projected_uninsured_adults_pct = baseline_uninsured + (uninsured_effect_
    projected_uninsured_adults_pct = clip_between(projected_uninsured_adults

    # ----------------------------------------------------
    # 6) relative deltas
    # ----------------------------------------------------
    dental_gap_delta = projected_dental_visit_gap_pct - baseline_dental_gap
    obesity_delta = projected_obesity_pct - baseline_obesity
    diabetes_delta = projected_diabetes_pct - baseline_diabetes
    uninsured_delta = projected_uninsured_adults_pct - baseline_uninsured
    unemp_delta = projected_unemp_rate - baseline_unemp_rate

    # ----------------------------------------------------
    # 7) collect row
    # ----------------------------------------------------
    row.update({
        # baseline anchors
        "baseline_population": baseline["baseline_population"],
        "baseline_target_service_population": baseline["baseline_target_serv
        "labor_force_estimate": labor_force_estimate,
        "baseline_unemp_rate": baseline_unemp_rate,
        "baseline_lfpr_16p": baseline["baseline_lfpr_16p"],
        "baseline_igs_total": baseline["baseline_igs_total"],
        "baseline_igs_economy": baseline["baseline_igs_economy"],
        "baseline_poverty_rate": baseline["baseline_poverty_rate"],
        "baseline_median_household_income": baseline["baseline_median_househ
        "baseline_uninsured_adults_pct": baseline_uninsured,
        "baseline_diabetes_pct": baseline_diabetes,
        "baseline_obesity_pct": baseline_obesity,
        "baseline_dental_visit_gap_pct": baseline_dental_gap,

        # raw scenario inputs
        "n_family_practitioners": n_family_practitioners,
        "fp_hours_per_week_each": fp_hours_each,
        "n_dentists": n_dentists,
        "dentist_hours_per_week_each": dentist_hours_each,
        "n_dieticians": n_dieticians,
        "dietician_hours_per_week_each": dietician_hours_each,
        "n_drivers": n_drivers,
```

```
        "driver_hours_per_week_avg": driver_hours_per_week_avg,
        "hourly_driver_pay": hourly_driver_pay,
        "produce_baskets_per_week": produce_baskets_per_week,
        "seed_cost_per_month": seed_cost_per_month,
        "greenhouse_op_cost_per_month": greenhouse_op_cost_per_month,
        "mobile_clinic_cost_per_month": mobile_clinic_cost_per_month,
        "admin_cost_per_month": admin_cost_per_month,
        "subscription_members": subscription_members,
        "subscription_price_per_month": subscription_price_per_month,
        "avg_medical_visit_minutes": avg_medical_visit_minutes,
        "avg_rides_per_driver_per_week": avg_rides_per_driver_per_week,
        "medical_ride_share_pct": medical_ride_share_pct,

        # derived operations
        "fp_hours_total_per_week": fp_hours_total_per_week,
        "dentist_hours_total_per_week": dentist_hours_total_per_week,
        "dietician_hours_total_per_week": dietician_hours_total_per_week,
        "clinical_hours_total_per_week": clinical_hours_total_per_week,
        "visits_per_hour": visits_per_hour,
        "clinic_visits_per_week": clinic_visits_per_week,
        "fp_visits_per_week": fp_visits_per_week,
        "dental_visits_per_week": dental_visits_per_week,
        "dietician_visits_per_week": dietician_visits_per_week,
        "driver_hours_total_per_week": driver_hours_total_per_week,
        "direct_jobs_created": direct_jobs_created,
        "fte_jobs_created": fte_jobs_created,
        "rides_per_week": rides_per_week,
        "medical_rides_per_week": medical_rides_per_week,

        # finances
        "weekly_driver_wages": weekly_driver_wages,
        "annual_driver_wages": annual_driver_wages,
        "annual_fp_cost": annual_fp_cost,
        "annual_dentist_cost": annual_dentist_cost,
        "annual_dietician_cost": annual_dietician_cost,
        "annual_clinician_cost": annual_clinician_cost,
        "annual_seed_cost": annual_seed_cost,
        "annual_greenhouse_cost": annual_greenhouse_cost,
        "annual_mobile_clinic_cost": annual_mobile_clinic_cost,
        "annual_admin_cost": annual_admin_cost,
        "annual_total_cost": annual_total_cost,
        "annual_subscription_revenue": annual_subscription_revenue,
        "annual_net_operating": annual_net_operating,

        # projected outcomes
        "projected_unemp_rate": projected_unemp_rate,
        "projected_dental_visit_gap_pct": projected_dental_visit_gap_pct,
        "projected_obesity_pct": projected_obesity_pct,
        "projected_diabetes_pct": projected_diabetes_pct,
        "projected_uninsured_adults_pct": projected_uninsured_adults_pct,

        # deltas
        "unemp_rate_delta": unemp_delta,
        "dental_visit_gap_delta": dental_gap_delta,
        "obesity_pct_delta": obesity_delta,
        "diabetes_pct_delta": diabetes_delta,
```

```python
        "uninsured_adults_pct_delta": uninsured_delta,
    })

    rows.append(row)

shelbycare_outputs = pd.DataFrame(rows)

scenario_order = ["baseline", "pilot", "light", "moderate", "full"]
if "scenario_name" in shelbycare_outputs.columns:
    shelbycare_outputs["scenario_name"] = pd.Categorical(
        shelbycare_outputs["scenario_name"],
        categories=scenario_order,
        ordered=True
    )
    shelbycare_outputs = shelbycare_outputs.sort_values("scenario_name").res

display(shelbycare_outputs.round(3))
print("Output shape:", shelbycare_outputs.shape)
```

In [ ]:

```python
# ============================================================
# Cell 10E — save outputs
# ============================================================

PRESENTATION_DIR = OUT_DIR / "presentation_ready"
PRESENTATION_DIR.mkdir(parents=True, exist_ok=True)

full_output_path = PRESENTATION_DIR / "shelbycare_scenario_outputs_full.csv"
summary_output_path = PRESENTATION_DIR / "shelbycare_scenario_outputs_summar

shelbycare_outputs.to_csv(full_output_path, index=False)

summary_cols = [
    "scenario_name",
    "n_drivers",
    "direct_jobs_created",
    "fte_jobs_created",
    "produce_baskets_per_week",
    "clinic_visits_per_week",
    "dental_visits_per_week",
    "medical_rides_per_week",
    "annual_total_cost",
    "annual_subscription_revenue",
    "annual_net_operating",
    "projected_unemp_rate",
    "projected_dental_visit_gap_pct",
    "projected_obesity_pct",
    "projected_diabetes_pct",
    "projected_uninsured_adults_pct",
    "unemp_rate_delta",
    "dental_visit_gap_delta",
    "obesity_pct_delta",
    "diabetes_pct_delta",
    "uninsured_adults_pct_delta",
]
summary_cols = [c for c in summary_cols if c in shelbycare_outputs.columns]
```

```python
shelbycare_outputs[summary_cols].to_csv(summary_output_path, index=False)

print("Saved:")
print("-", full_output_path)
print("-", summary_output_path)

display(shelbycare_outputs[summary_cols].round(3))
```

In [ ]:

```python
# ============================================================
# Cell 10F — quick scenario charts
# ============================================================

import matplotlib.pyplot as plt
import numpy as np

plot_specs = [
    ("projected_dental_visit_gap_pct", "Projected dental visit gap (%)"),
    ("projected_obesity_pct", "Projected obesity (%)"),
    ("projected_diabetes_pct", "Projected diabetes (%)"),
    ("projected_uninsured_adults_pct", "Projected uninsured adults (%)"),
    ("projected_unemp_rate", "Projected unemployment rate"),
    ("annual_net_operating", "Annual net operating ($)"),
]

ncols = 2
nrows = int(np.ceil(len(plot_specs) / ncols))
fig, axes = plt.subplots(nrows, ncols, figsize=(15, 5 * nrows))
axes = np.array(axes).reshape(-1)

x_labels = shelbycare_outputs["scenario_name"].astype(str).tolist()

for ax, (col, title) in zip(axes, plot_specs):
    vals = pd.to_numeric(shelbycare_outputs[col], errors="coerce")
    ax.bar(x_labels, vals)
    ax.set_title(title)
    ax.grid(axis="y", alpha=0.25)

    valid_vals = vals.dropna()
    if len(valid_vals) > 0:
        pad = max(valid_vals.abs().max() * 0.03, 0.02)
        for i, v in enumerate(vals):
            if pd.notna(v):
                ax.text(i, v + pad, f"{v:,.2f}", ha="center", va="bottom", f

for j in range(len(plot_specs), len(axes)):
    axes[j].axis("off")

fig.suptitle("ShelbyCare feasibility + scenario outputs", fontsize=18, y=1.0
fig.text(
    0.02, 0.02,
    "Scenario-based feasibility model using explicit operating assumptions.
    "These are planning projections, not causal estimates.",
    fontsize=10
)
```

```
plt.tight_layout(rect=[0, 0.05, 1, 0.97])
plt.show()
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:
```
# ============================================================
# BOTTOM PATCH A — rebuild Shelby 3-group core objects from files
# Self-contained: does NOT rely on earlier notebook variables
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd

def _norm_geoid(series):
    s = series.astype("string").str.strip()
    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    return s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

def _find_existing(paths):
    for p in paths:
        if p.exists():
            return p
    return None

def _pick_col(df, candidates):
    for c in candidates:
        if c in df.columns:
            return c
    return None

ROOT_CANDIDATES = [
    Path("data/processed/presentation_ready"),
    Path("src/data/processed/presentation_ready"),
    Path("data/processed"),
    Path("src/data/processed"),
]

# --- files we need
three_group_compare_path = _find_existing([
    p / "three_group_selected_vs_comparator_vs_rest.csv" for p in ROOT_CANDI
])

shelby_places_base_path = _find_existing([
```

```
    p / "shelby_places_tract_base.parquet" for p in ROOT_CANDIDATES
])

shelby_full_compare_path = _find_existing([
    p / "shelby_full_community_compare.csv" for p in ROOT_CANDIDATES
])

shelby_full_map_path = _find_existing([
    p / "shelby_full_tract_cluster_map.geojson" for p in ROOT_CANDIDATES
])

missing_paths = {
    "three_group_compare_path": three_group_compare_path,
    "shelby_places_base_path": shelby_places_base_path,
    "shelby_full_compare_path": shelby_full_compare_path,
    "shelby_full_map_path": shelby_full_map_path,
}
missing_paths = {k: v for k, v in missing_paths.items() if v is None}
if missing_paths:
    raise FileNotFoundError(f"Missing required files for reset patch A: {lis

print("Using:")
print(" -", three_group_compare_path)
print(" -", shelby_places_base_path)
print(" -", shelby_full_compare_path)
print(" -", shelby_full_map_path)

# --- load the saved summary table directly
three_group_compare = pd.read_csv(three_group_compare_path)

# --- low-growth selected community tracts from saved Shelby PLACES tract ba
shelby_places_base = pd.read_parquet(shelby_places_base_path).copy()
if "geoid" not in shelby_places_base.columns or "is_selected_community" not
    raise ValueError("shelby_places_tract_base.parquet must contain `geoid`

shelby_places_base["geoid"] = _norm_geoid(shelby_places_base["geoid"])
all_shelby_geoids = set(shelby_places_base["geoid"].dropna().tolist())
low_growth_geoids = set(
    shelby_places_base.loc[
        shelby_places_base["is_selected_community"].fillna(False), "geoid"
    ].dropna().tolist()
)

# --- identify comparator cluster from saved Shelby full-community summary
shelby_full_compare = pd.read_csv(shelby_full_compare_path).copy()

community_id_col = _pick_col(shelby_full_compare, [
    "community_id", "shelby_full_cluster_id", "cluster_id", "community", "co
])
score_col = _pick_col(shelby_full_compare, [
    "igs_total_recovery", "igs_total", "igs_recovery"
])
tract_count_col = _pick_col(shelby_full_compare, [
    "n_tracts", "tract_count"
])
```

```
if community_id_col is None or score_col is None:
    raise ValueError("Could not identify community id / recovery score colum

tmp_compare = shelby_full_compare.copy()
if tract_count_col is not None:
    tmp_compare = tmp_compare.loc[pd.to_numeric(tmp_compare[tract_count_col]

tmp_compare[score_col] = pd.to_numeric(tmp_compare[score_col], errors="coerc
tmp_compare = tmp_compare.dropna(subset=[score_col]).sort_values(score_col,

if tmp_compare.empty:
    raise ValueError("Could not identify a valid high-growth Shelby comparat

BEST_SHELBY_COMPARATOR_ID = tmp_compare.iloc[0][community_id_col]
print("Comparator cluster selected from saved full-community summary:", BEST

# --- load tract map and assign comparator tracts
import geopandas as gpd

shelby_full_map = gpd.read_file(shelby_full_map_path)

map_geoid_col = _pick_col(shelby_full_map, ["geoid", "GEOID"])
map_cluster_col = _pick_col(shelby_full_map, [
    "community_id", "shelby_full_cluster_id", "cluster_id", "community", "co
])

if map_geoid_col is None or map_cluster_col is None:
    raise ValueError("Could not identify geoid/community columns in shelby_f

shelby_full_map["geoid"] = _norm_geoid(shelby_full_map[map_geoid_col])
high_growth_geoids = set(
    shelby_full_map.loc[
        shelby_full_map[map_cluster_col].astype(str) == str(BEST_SHELBY_COMP
        "geoid"
    ].dropna().tolist()
)

# remove overlap just in case
high_growth_geoids = high_growth_geoids - low_growth_geoids
rest_geoids = all_shelby_geoids - low_growth_geoids - high_growth_geoids

print("Low-growth selected community tracts:", len(low_growth_geoids))
print("High-growth selected comparator tracts:", len(high_growth_geoids))
print("Rest of Shelby County tracts:", len(rest_geoids))
print("All Shelby tracts:", len(all_shelby_geoids))

# --- rebuild three_group_base from scratch
three_group_base = pd.DataFrame({"geoid": sorted(all_shelby_geoids)})

def _assign_group(g):
    if g in low_growth_geoids:
        return "Selected low-growth community"
    elif g in high_growth_geoids:
        return "Selected high-growth comparator"
    else:
        return "Rest of Shelby County (excluding both selected communities)"
```

```python
three_group_base["final_compare_group"] = three_group_base["geoid"].map(_ass

group_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

display(three_group_base["final_compare_group"].value_counts())
display(three_group_compare)
```

In [ ]:

```python
# ============================================================
# BOTTOM PATCH B — rebuild three_group_year from eda_panel_clean.parquet
# Self-contained: does NOT rely on earlier notebook variables
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd

def _find_existing(paths):
    for p in paths:
        if p.exists():
            return p
    return None

def _norm_geoid(series):
    s = series.astype("string").str.strip()
    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    return s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

def _safe_weighted_mean(values, weights):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))
```

```
panel_path = _find_existing([
    Path("data/processed/eda_panel_clean.parquet"),
    Path("src/data/processed/eda_panel_clean.parquet"),
    Path("data/processed/igs_x_acs_full.parquet"),
    Path("src/data/processed/igs_x_acs_full.parquet"),
])

if panel_path is None:
    raise FileNotFoundError("Could not find eda_panel_clean.parquet or igs_x

panel_df_reset = pd.read_parquet(panel_path).copy()
panel_df_reset["geoid"] = _norm_geoid(panel_df_reset["geoid"])
panel_df_reset["year"] = pd.to_numeric(panel_df_reset["year"], errors="coerc
panel_df_reset = panel_df_reset.loc[panel_df_reset["year"].isin([2017,2018,2
panel_df_reset["year"] = panel_df_reset["year"].astype(int)

# filter Shelby County from geoid prefix if county_fips not present
if "county_fips" in panel_df_reset.columns:
    shelby_panel_reset = panel_df_reset.loc[
        panel_df_reset["county_fips"].astype(str).str.zfill(5) == "47157"
    ].copy()
else:
    shelby_panel_reset = panel_df_reset.loc[
        panel_df_reset["geoid"].astype(str).str[:5] == "47157"
    ].copy()

if shelby_panel_reset.empty:
    raise ValueError("No Shelby County rows found in panel file.")

group_lookup = three_group_base[["geoid", "final_compare_group"]].drop_dupli
group_lookup["geoid"] = _norm_geoid(group_lookup["geoid"])
group_lookup = group_lookup.rename(columns={"final_compare_group": "group"})

shelby_panel_reset = shelby_panel_reset.merge(
    group_lookup,
    on="geoid",
    how="inner",
    validate="many_to_one"
)

required_cols = ["year", "igs_total", "poverty_rate", "median_household_inco
missing = [c for c in required_cols if c not in shelby_panel_reset.columns]
if missing:
    raise ValueError(f"Panel file is missing required columns for three_grou

unemp_col = "unemp_rate" if "unemp_rate" in shelby_panel_reset.columns else
if unemp_col is None:
    raise ValueError("Could not find unemployment column in panel file.")

rows = []
for (grp, yr), df_grp in shelby_panel_reset.groupby(["group", "year"], dropr
    weights = pd.to_numeric(df_grp["pop_total"], errors="coerce")
    rows.append({
        "group": grp,
        "year": int(yr),
```

```
        "igs_total": _safe_weighted_mean(df_grp["igs_total"], weights),
        "poverty_rate": _safe_weighted_mean(df_grp["poverty_rate"], weights)
        "median_household_income": _safe_weighted_mean(df_grp["median_househ
        "lfpr_16p": _safe_weighted_mean(df_grp["lfpr_16p"], weights),
        "unemp_rate": _safe_weighted_mean(df_grp[unemp_col], weights),
        "population_sum": float(weights.sum()),
        "n_tracts": int(df_grp["geoid"].nunique()),
    })

three_group_year = pd.DataFrame(rows).sort_values(["group", "year"]).reset_i

display(three_group_year.head(12))
```

In [ ]:

```
# ============================================================
# BOTTOM PATCH C — load education_compare + demographic_compare from saved C
# Self-contained: does NOT rely on earlier notebook variables
# ============================================================

from pathlib import Path
import pandas as pd

def _find_existing(paths):
    for p in paths:
        if p.exists():
            return p
    return None

education_path = _find_existing([
    Path("data/processed/presentation_ready/shelby_education_compare.csv"),
    Path("src/data/processed/presentation_ready/shelby_education_compare.csv
    Path("data/processed/shelby_education_compare.csv"),
    Path("src/data/processed/shelby_education_compare.csv"),
])

demographic_path = _find_existing([
    Path("data/processed/presentation_ready/shelby_demographic_compare.csv")
    Path("src/data/processed/presentation_ready/shelby_demographic_compare.c
    Path("data/processed/shelby_demographic_compare.csv"),
    Path("src/data/processed/shelby_demographic_compare.csv"),
])

if education_path is None:
    raise FileNotFoundError("Could not find shelby_education_compare.csv")

education_compare = pd.read_csv(education_path)

if demographic_path is None:
    raise FileNotFoundError(
        "Could not find shelby_demographic_compare.csv. "
        "If that file is not already in your repo, skip the demographic char
    )

demographic_compare = pd.read_csv(demographic_path)

DEMO_OUT_DIR = Path("src/data/processed/presentation_ready")
DEMO_OUT_DIR.mkdir(parents=True, exist_ok=True)
```

```
print("Loaded education_compare from:", education_path)
print("Loaded demographic_compare from:", demographic_path)

display(education_compare)
display(demographic_compare)
```

In [ ]:
```
# ============================================================
# Cell 11A — load ShelbyCare CSVs + configure output folder
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# ------------------------------
# Search paths for the 3 completed CSVs
# ------------------------------
CSV_SEARCH_DIRS = [
    Path("src/data"),
    Path("src/data/processed"),
    Path("src/data/processed/presentation_ready"),
    Path("data"),
    Path("data/processed"),
    Path("data/processed/presentation_ready"),
]

CSV_FILES = {
    "baseline": "shelbycare_baseline.csv",
    "scenarios": "shelbycare_scenarios.csv",
    "assumptions": "shelbycare_assumptions.csv",
}

def find_csv(filename: str) -> Path:
    for folder in CSV_SEARCH_DIRS:
        p = folder / filename
        if p.exists():
            return p
    raise FileNotFoundError(f"Could not find {filename} in: {CSV_SEARCH_DIRS

baseline_csv_path = find_csv(CSV_FILES["baseline"])
scenarios_csv_path = find_csv(CSV_FILES["scenarios"])
assumptions_csv_path = find_csv(CSV_FILES["assumptions"])

baseline_df = pd.read_csv(baseline_csv_path)
scenario_inputs = pd.read_csv(scenarios_csv_path)
assumptions_df = pd.read_csv(assumptions_csv_path)

# ------------------------------
# Output folder
# ------------------------------
if "OUT_DIR" in globals():
    OUT_BASE = Path(OUT_DIR)
elif "DATA_DIR" in globals():
    OUT_BASE = Path(DATA_DIR) / "processed"
```

```python
else:
    OUT_BASE = Path("src/data/processed")

PRESENTATION_DIR = OUT_BASE if OUT_BASE.name == "presentation_ready" else (O
PRESENTATION_DIR.mkdir(parents=True, exist_ok=True)

print("Loaded:")
print(" -", baseline_csv_path)
print(" -", scenarios_csv_path)
print(" -", assumptions_csv_path)
print("Output dir:", PRESENTATION_DIR)

display(baseline_df)
display(scenario_inputs.head())
display(assumptions_df.head())
```

In [ ]:

```python
# ============================================================
# Cell 11B — helpers + explicit configuration assumptions
# ============================================================

def safe_num(x, default=0.0):
    x = pd.to_numeric(pd.Series([x]), errors="coerce").iloc[0]
    return default if pd.isna(x) else float(x)

def annualize_weekly(x):
    return safe_num(x) * 52.0

def annualize_monthly(x):
    return safe_num(x) * 12.0

def clip_between(value, min_value=None, max_value=None):
    out = value
    if min_value is not None and pd.notna(min_value):
        out = max(out, min_value)
    if max_value is not None and pd.notna(max_value):
        out = min(out, max_value)
    return out

def get_assumption_row(metric_name: str):
    match = assumptions_df.loc[assumptions_df["metric_name"] == metric_name]
    if match.empty:
        return None
    return match.iloc[0].to_dict()

def get_assumption_effect(metric_name: str, default=np.nan):
    row = get_assumption_row(metric_name)
    if row is None:
        return default
    return safe_num(row.get("effect_per_unit"), default)

def pct_or_share_to_pct(x):
    """
    Convert:
    - 0.118 -> 11.8
    - 11.8 -> 11.8
    """
```

```python
    x = safe_num(x, np.nan)
    if pd.isna(x):
        return np.nan
    return x * 100.0 if x <= 1.0 else x

def pct_or_share_to_share(x):
    """
    Convert:
    - 0.118 -> 0.118
    - 11.8 -> 0.118
    """
    x = safe_num(x, np.nan)
    if pd.isna(x):
        return np.nan
    return x / 100.0 if x > 1.0 else x

# ------------------------------------------------------------
# Feasibility-model compensation assumptions
# ------------------------------------------------------------
ROLE_RATES = {
    "family_practitioner_hourly": 120.0,
    "dentist_hourly": 110.0,
    "dietician_hourly": 45.0,
}

# ------------------------------------------------------------
# Driver-pay sensitivity assumptions
# This is what lets higher driver pay affect community metrics.
# If you want pay to affect ONLY costs, set USE_DRIVER_PAY_FILL_EFFECT = Fal
# ------------------------------------------------------------
USE_DRIVER_PAY_FILL_EFFECT = True
REFERENCE_DRIVER_PAY = 20.0         # reference pay level in your scenarios
BASE_DRIVER_FILL_RATE = 0.80        # active-driver fill rate at reference
DRIVER_FILL_RATE_PER_DOLLAR = 0.03  # +3 percentage points per $1 above ref
MIN_DRIVER_FILL_RATE = 0.50
MAX_DRIVER_FILL_RATE = 0.98

# ------------------------------------------------------------
# Read coefficients from assumptions CSV
# ------------------------------------------------------------
JOB_FILL_RATE = get_assumption_effect("job_fill_rate", 0.65)
DENTAL_EFFECT_PER_VISIT = get_assumption_effect("projected_dental_visit_gap_
OBESITY_EFFECT_PER_BUNDLE = get_assumption_effect("projected_obesity_pct", -
DIABETES_EFFECT_PER_BUNDLE = get_assumption_effect("projected_diabetes_pct",
UNINSURED_EFFECT_PER_MEDRIDE = get_assumption_effect("projected_uninsured_ac

print("Assumption coefficients:")
print("JOB_FILL_RATE =", JOB_FILL_RATE)
print("DENTAL_EFFECT_PER_VISIT =", DENTAL_EFFECT_PER_VISIT)
print("OBESITY_EFFECT_PER_BUNDLE =", OBESITY_EFFECT_PER_BUNDLE)
print("DIABETES_EFFECT_PER_BUNDLE =", DIABETES_EFFECT_PER_BUNDLE)
print("UNINSURED_EFFECT_PER_MEDRIDE =", UNINSURED_EFFECT_PER_MEDRIDE)
```

In [ ]:

```python
# ============================================================
# Cell 11C — scenario engine: compute one scenario row
# ============================================================
```

```
if len(baseline_df) != 1:
    raise ValueError("shelbycare_baseline.csv should contain exactly 1 row."

baseline = baseline_df.iloc[0].copy()

# normalize numeric baseline fields
baseline_numeric_cols = [
    "baseline_population",
    "baseline_target_service_population",
    "labor_force_estimate",
    "baseline_unemp_rate",
    "baseline_lfpr_16p",
    "baseline_igs_total",
    "baseline_igs_economy",
    "baseline_poverty_rate",
    "baseline_median_household_income",
    "baseline_uninsured_adults_pct",
    "baseline_diabetes_pct",
    "baseline_obesity_pct",
    "baseline_dental_visit_gap_pct",
]
for c in baseline_numeric_cols:
    if c in baseline.index:
        baseline[c] = safe_num(baseline[c], np.nan)

for c in scenario_inputs.columns:
    if c != "scenario_name":
        scenario_inputs[c] = pd.to_numeric(scenario_inputs[c], errors="coerc

def compute_driver_fill_rate(hourly_driver_pay: float) -> float:
    if not USE_DRIVER_PAY_FILL_EFFECT:
        return BASE_DRIVER_FILL_RATE
    fill_rate = BASE_DRIVER_FILL_RATE + DRIVER_FILL_RATE_PER_DOLLAR * (hourl
    return clip_between(fill_rate, MIN_DRIVER_FILL_RATE, MAX_DRIVER_FILL_RAT

def compute_scenario_row(s: pd.Series, baseline: pd.Series) -> dict:
    row = {"scenario_name": s["scenario_name"]}

    # ----------------------------
    # Baseline anchors
    # ----------------------------
    baseline_unemp_rate_share = pct_or_share_to_share(baseline["baseline_une
    baseline_unemp_rate_pct = pct_or_share_to_pct(baseline["baseline_unemp_r

    baseline_uninsured_pct = safe_num(baseline["baseline_uninsured_adults_pc
    baseline_diabetes_pct = safe_num(baseline["baseline_diabetes_pct"])
    baseline_obesity_pct = safe_num(baseline["baseline_obesity_pct"])
    baseline_dental_gap_pct = safe_num(baseline["baseline_dental_visit_gap_p
    labor_force_estimate = safe_num(baseline["labor_force_estimate"])
    target_service_pop = safe_num(baseline["baseline_target_service_populati

    # ----------------------------
    # Scenario inputs
    # ----------------------------
    n_family_practitioners = safe_num(s["n_family_practitioners"])
```

```
    fp_hours_per_week_each = safe_num(s["fp_hours_per_week_each"])
    n_dentists = safe_num(s["n_dentists"])
    dentist_hours_per_week_each = safe_num(s["dentist_hours_per_week_each"])
    n_dieticians = safe_num(s["n_dieticians"])
    dietician_hours_per_week_each = safe_num(s["dietician_hours_per_week_eac

    n_drivers = safe_num(s["n_drivers"])
    driver_hours_per_week_avg = safe_num(s["driver_hours_per_week_avg"])
    hourly_driver_pay = safe_num(s["hourly_driver_pay"])

    produce_baskets_per_week = safe_num(s["produce_baskets_per_week"])
    seed_cost_per_month = safe_num(s["seed_cost_per_month"])
    greenhouse_op_cost_per_month = safe_num(s["greenhouse_op_cost_per_month"
    mobile_clinic_cost_per_month = safe_num(s["mobile_clinic_cost_per_month"
    admin_cost_per_month = safe_num(s["admin_cost_per_month"])

    subscription_members = safe_num(s["subscription_members"])
    subscription_price_per_month = safe_num(s["subscription_price_per_month"
    avg_medical_visit_minutes = safe_num(s["avg_medical_visit_minutes"], 30.
    avg_rides_per_driver_per_week = safe_num(s["avg_rides_per_driver_per_wee
    medical_ride_share_pct = safe_num(s["medical_ride_share_pct"], 0.10)

    # ------------------------------
    # Driver pay -> active drivers logic
    # ------------------------------
    driver_fill_rate = compute_driver_fill_rate(hourly_driver_pay)
    active_drivers = n_drivers * driver_fill_rate
    direct_jobs_created = active_drivers
    driver_hours_total_per_week = active_drivers * driver_hours_per_week_avg
    fte_jobs_created = driver_hours_total_per_week / 40.0

    # ------------------------------
    # Transportation capacity
    # ------------------------------
    rides_per_week = active_drivers * avg_rides_per_driver_per_week
    medical_rides_per_week = rides_per_week * medical_ride_share_pct

    # ------------------------------
    # Clinic capacity
    # ------------------------------
    fp_hours_total_per_week = n_family_practitioners * fp_hours_per_week_eac
    dentist_hours_total_per_week = n_dentists * dentist_hours_per_week_each
    dietician_hours_total_per_week = n_dieticians * dietician_hours_per_week

    clinical_hours_total_per_week = (
        fp_hours_total_per_week +
        dentist_hours_total_per_week +
        dietician_hours_total_per_week
    )

    visits_per_hour = 60.0 / avg_medical_visit_minutes if avg_medical_visit_

    clinic_visits_per_week = clinical_hours_total_per_week * visits_per_hour
    fp_visits_per_week = fp_hours_total_per_week * visits_per_hour
    dental_visits_per_week = dentist_hours_total_per_week * visits_per_hour
    dietician_visits_per_week = dietician_hours_total_per_week * visits_per_
```

```
# -----------------------------
# Costs
# -----------------------------
weekly_driver_wages = driver_hours_total_per_week * hourly_driver_pay
annual_driver_wages = annualize_weekly(weekly_driver_wages)

annual_fp_cost = annualize_weekly(fp_hours_total_per_week * ROLE_RATES["
annual_dentist_cost = annualize_weekly(dentist_hours_total_per_week * RO
annual_dietician_cost = annualize_weekly(dietician_hours_total_per_week
annual_clinician_cost = annual_fp_cost + annual_dentist_cost + annual_di

annual_seed_cost = annualize_monthly(seed_cost_per_month)
annual_greenhouse_cost = annualize_monthly(greenhouse_op_cost_per_month)
annual_mobile_clinic_cost = annualize_monthly(mobile_clinic_cost_per_mon
annual_admin_cost = annualize_monthly(admin_cost_per_month)

annual_total_cost = (
    annual_driver_wages +
    annual_clinician_cost +
    annual_seed_cost +
    annual_greenhouse_cost +
    annual_mobile_clinic_cost +
    annual_admin_cost
)

# -----------------------------
# Revenue
# -----------------------------
annual_subscription_revenue = subscription_members * subscription_price_
annual_net_operating = annual_subscription_revenue - annual_total_cost

# -----------------------------
# Projected community outcomes
# These are scenario projections, not causal treatment estimates.
# -----------------------------
projected_unemp_rate_share = baseline_unemp_rate_share - (JOB_FILL_RATE
projected_unemp_rate_share = clip_between(projected_unemp_rate_share, 0.
projected_unemp_rate_pct = projected_unemp_rate_share * 100.0

projected_dental_gap_pct = baseline_dental_gap_pct + (DENTAL_EFFECT_PER_
projected_dental_gap_pct = clip_between(projected_dental_gap_pct, 0.0, 1

projected_obesity_pct = baseline_obesity_pct + (OBESITY_EFFECT_PER_BUNDL
projected_obesity_pct = clip_between(projected_obesity_pct, 0.0, 100.0)

projected_diabetes_pct = baseline_diabetes_pct + (DIABETES_EFFECT_PER_BU
projected_diabetes_pct = clip_between(projected_diabetes_pct, 0.0, 100.0

projected_uninsured_pct = baseline_uninsured_pct + (UNINSURED_EFFECT_PER
projected_uninsured_pct = clip_between(projected_uninsured_pct, 0.0, 100

# -----------------------------
# Deltas
# -----------------------------
unemp_rate_delta_pct_points = projected_unemp_rate_pct - baseline_unemp_
```

```
    dental_gap_delta_pct_points = projected_dental_gap_pct - baseline_dental
    obesity_delta_pct_points = projected_obesity_pct - baseline_obesity_pct
    diabetes_delta_pct_points = projected_diabetes_pct - baseline_diabetes_p
    uninsured_delta_pct_points = projected_uninsured_pct - baseline_uninsure

    row.update({
        # baseline context
        "baseline_population": safe_num(baseline["baseline_population"]),
        "baseline_target_service_population": target_service_pop,
        "labor_force_estimate": labor_force_estimate,
        "baseline_unemp_rate_pct": baseline_unemp_rate_pct,
        "baseline_lfpr_16p": safe_num(baseline["baseline_lfpr_16p"]),
        "baseline_igs_total": safe_num(baseline["baseline_igs_total"]),
        "baseline_igs_economy": safe_num(baseline["baseline_igs_economy"]),
        "baseline_poverty_rate": safe_num(baseline["baseline_poverty_rate"])
        "baseline_median_household_income": safe_num(baseline["baseline_medi
        "baseline_uninsured_adults_pct": baseline_uninsured_pct,
        "baseline_diabetes_pct": baseline_diabetes_pct,
        "baseline_obesity_pct": baseline_obesity_pct,
        "baseline_dental_visit_gap_pct": baseline_dental_gap_pct,

        # scenario inputs
        "n_family_practitioners": n_family_practitioners,
        "fp_hours_per_week_each": fp_hours_per_week_each,
        "n_dentists": n_dentists,
        "dentist_hours_per_week_each": dentist_hours_per_week_each,
        "n_dieticians": n_dieticians,
        "dietician_hours_per_week_each": dietician_hours_per_week_each,
        "n_drivers_budgeted": n_drivers,
        "driver_hours_per_week_avg": driver_hours_per_week_avg,
        "hourly_driver_pay": hourly_driver_pay,
        "produce_baskets_per_week": produce_baskets_per_week,
        "seed_cost_per_month": seed_cost_per_month,
        "greenhouse_op_cost_per_month": greenhouse_op_cost_per_month,
        "mobile_clinic_cost_per_month": mobile_clinic_cost_per_month,
        "admin_cost_per_month": admin_cost_per_month,
        "subscription_members": subscription_members,
        "subscription_price_per_month": subscription_price_per_month,
        "avg_medical_visit_minutes": avg_medical_visit_minutes,
        "avg_rides_per_driver_per_week": avg_rides_per_driver_per_week,
        "medical_ride_share_pct": medical_ride_share_pct,

        # pay / staffing mechanics
        "driver_fill_rate": driver_fill_rate,
        "active_drivers": active_drivers,
        "direct_jobs_created": direct_jobs_created,
        "fte_jobs_created": fte_jobs_created,

        # operations
        "driver_hours_total_per_week": driver_hours_total_per_week,
        "rides_per_week": rides_per_week,
        "medical_rides_per_week": medical_rides_per_week,
        "fp_hours_total_per_week": fp_hours_total_per_week,
        "dentist_hours_total_per_week": dentist_hours_total_per_week,
        "dietician_hours_total_per_week": dietician_hours_total_per_week,
        "clinical_hours_total_per_week": clinical_hours_total_per_week,
```

```
        "visits_per_hour": visits_per_hour,
        "clinic_visits_per_week": clinic_visits_per_week,
        "fp_visits_per_week": fp_visits_per_week,
        "dental_visits_per_week": dental_visits_per_week,
        "dietician_visits_per_week": dietician_visits_per_week,

        # finance
        "weekly_driver_wages": weekly_driver_wages,
        "annual_driver_wages": annual_driver_wages,
        "annual_fp_cost": annual_fp_cost,
        "annual_dentist_cost": annual_dentist_cost,
        "annual_dietician_cost": annual_dietician_cost,
        "annual_clinician_cost": annual_clinician_cost,
        "annual_seed_cost": annual_seed_cost,
        "annual_greenhouse_cost": annual_greenhouse_cost,
        "annual_mobile_clinic_cost": annual_mobile_clinic_cost,
        "annual_admin_cost": annual_admin_cost,
        "annual_total_cost": annual_total_cost,
        "annual_subscription_revenue": annual_subscription_revenue,
        "annual_net_operating": annual_net_operating,

        # projected outcomes
        "projected_unemp_rate_pct": projected_unemp_rate_pct,
        "projected_dental_visit_gap_pct": projected_dental_gap_pct,
        "projected_obesity_pct": projected_obesity_pct,
        "projected_diabetes_pct": projected_diabetes_pct,
        "projected_uninsured_adults_pct": projected_uninsured_pct,

        # deltas vs baseline
        "unemp_rate_delta_pct_points": unemp_rate_delta_pct_points,
        "dental_visit_gap_delta_pct_points": dental_gap_delta_pct_points,
        "obesity_delta_pct_points": obesity_delta_pct_points,
        "diabetes_delta_pct_points": diabetes_delta_pct_points,
        "uninsured_adults_delta_pct_points": uninsured_delta_pct_points,
    })

    return row
```

In [ ]:

```
# ============================================================
# Cell 11D — full output table
# ============================================================

scenario_rows = [compute_scenario_row(s, baseline) for _, s in scenario_inpu
shelbycare_outputs = pd.DataFrame(scenario_rows)

scenario_order = ["baseline", "pilot", "light", "moderate", "full"]
if "scenario_name" in shelbycare_outputs.columns:
    shelbycare_outputs["scenario_name"] = pd.Categorical(
        shelbycare_outputs["scenario_name"],
        categories=scenario_order,
        ordered=True
    )
    shelbycare_outputs = shelbycare_outputs.sort_values("scenario_name").res

full_output_path = PRESENTATION_DIR / "shelbycare_scenario_outputs_full.csv"
summary_output_path = PRESENTATION_DIR / "shelbycare_scenario_outputs_summar
```

```python
summary_cols = [
    "scenario_name",
    "hourly_driver_pay",
    "active_drivers",
    "direct_jobs_created",
    "fte_jobs_created",
    "clinic_visits_per_week",
    "dental_visits_per_week",
    "medical_rides_per_week",
    "produce_baskets_per_week",
    "annual_total_cost",
    "annual_subscription_revenue",
    "annual_net_operating",
    "projected_unemp_rate_pct",
    "projected_dental_visit_gap_pct",
    "projected_uninsured_adults_pct",
    "projected_obesity_pct",
    "projected_diabetes_pct",
    "unemp_rate_delta_pct_points",
    "dental_visit_gap_delta_pct_points",
    "uninsured_adults_delta_pct_points",
    "obesity_delta_pct_points",
    "diabetes_delta_pct_points",
]

shelbycare_outputs.to_csv(full_output_path, index=False)
shelbycare_outputs[summary_cols].to_csv(summary_output_path, index=False)

print("Saved:")
print(" -", full_output_path)
print(" -", summary_output_path)

display(shelbycare_outputs[summary_cols].round(3))
```

In [ ]:

```python
# ============================================================
# Cell 11E — sensitivity table
# Tests what happens when you change one lever at a time
# ============================================================

SENSITIVITY_BASE_SCENARIO = "full"  # change to "moderate" if you prefer

base_match = scenario_inputs.loc[scenario_inputs["scenario_name"].astype(str
if base_match.empty:
    raise ValueError(f"Could not find base sensitivity scenario: {SENSITIVIT

base_scenario = base_match.iloc[0].copy()
base_output = compute_scenario_row(base_scenario, baseline)

# ------------------------------------------------------------
# Sensitivity cases
# You can add/remove cases here
# ------------------------------------------------------------
sensitivity_cases = [
    {"test_name": "Driver pay +10%", "field": "hourly_driver_pay", "mode": "
    {"test_name": "Driver pay +20%", "field": "hourly_driver_pay", "mode": "
```

```
    {"test_name": "Driver pay -10%", "field": "hourly_driver_pay", "mode": "

    {"test_name": "Drivers +10%", "field": "n_drivers", "mode": "pct", "valu
    {"test_name": "Drivers +20%", "field": "n_drivers", "mode": "pct", "valu
    {"test_name": "Drivers -10%", "field": "n_drivers", "mode": "pct", "valu

    {"test_name": "Seed cost +25%", "field": "seed_cost_per_month", "mode":
    {"test_name": "Seed cost +50%", "field": "seed_cost_per_month", "mode":

    {"test_name": "Mobile clinic cost +25%", "field": "mobile_clinic_cost_pe
    {"test_name": "Mobile clinic cost +50%", "field": "mobile_clinic_cost_pe

    {"test_name": "Produce baskets +10%", "field": "produce_baskets_per_week
    {"test_name": "Produce baskets +20%", "field": "produce_baskets_per_week

    {"test_name": "Subscription members +10%", "field": "subscription_member
    {"test_name": "Subscription members +20%", "field": "subscription_member
]

def apply_sensitivity(base_scenario: pd.Series, field: str, mode: str, value
    s = base_scenario.copy()
    current = safe_num(s[field])

    if mode == "pct":
        s[field] = current * (1.0 + value)
    elif mode == "add":
        s[field] = current + value
    else:
        raise ValueError(f"Unknown sensitivity mode: {mode}")

    # keep counts reasonable
    if field in ["n_drivers", "n_family_practitioners", "n_dentists", "n_die
        s[field] = max(0, round(s[field]))

    return s

sensitivity_rows = []

for case in sensitivity_cases:
    scenario_variant = apply_sensitivity(
        base_scenario=base_scenario,
        field=case["field"],
        mode=case["mode"],
        value=case["value"]
    )
    out = compute_scenario_row(scenario_variant, baseline)

    sensitivity_rows.append({
        "base_scenario": SENSITIVITY_BASE_SCENARIO,
        "test_name": case["test_name"],
        "changed_field": case["field"],
        "change_mode": case["mode"],
        "change_value": case["value"],

        "base_field_value": safe_num(base_scenario[case["field"]]),
        "test_field_value": safe_num(scenario_variant[case["field"]]),
```

```python
        "annual_total_cost": out["annual_total_cost"],
        "annual_subscription_revenue": out["annual_subscription_revenue"],
        "annual_net_operating": out["annual_net_operating"],
        "direct_jobs_created": out["direct_jobs_created"],
        "clinic_visits_per_week": out["clinic_visits_per_week"],
        "medical_rides_per_week": out["medical_rides_per_week"],

        "projected_unemp_rate_pct": out["projected_unemp_rate_pct"],
        "projected_dental_visit_gap_pct": out["projected_dental_visit_gap_pc
        "projected_uninsured_adults_pct": out["projected_uninsured_adults_pc
        "projected_obesity_pct": out["projected_obesity_pct"],
        "projected_diabetes_pct": out["projected_diabetes_pct"],

        "delta_annual_total_cost": out["annual_total_cost"] - base_output["a
        "delta_annual_subscription_revenue": out["annual_subscription_revenu
        "delta_annual_net_operating": out["annual_net_operating"] - base_out
        "delta_direct_jobs_created": out["direct_jobs_created"] - base_outpu
        "delta_clinic_visits_per_week": out["clinic_visits_per_week"] - base
        "delta_medical_rides_per_week": out["medical_rides_per_week"] - base
        "delta_projected_unemp_rate_pct_points": out["projected_unemp_rate_p
        "delta_projected_dental_gap_pct_points": out["projected_dental_visit
        "delta_projected_uninsured_pct_points": out["projected_uninsured_adu
        "delta_projected_obesity_pct_points": out["projected_obesity_pct"] -
        "delta_projected_diabetes_pct_points": out["projected_diabetes_pct"]
    })

sensitivity_table = pd.DataFrame(sensitivity_rows)

sensitivity_output_path = PRESENTATION_DIR / "shelbycare_sensitivity_table.c
sensitivity_table.to_csv(sensitivity_output_path, index=False)

print("Saved:", sensitivity_output_path)
display(sensitivity_table.round(3))
```

In [ ]:

```python
# ============================================================
# Cell 11F — presentation-ready charts
# ============================================================

plt.rcParams["figure.figsize"] = (10, 6)

# ------------------------------
# Chart 1 — operations
# ------------------------------
fig, ax = plt.subplots()
ops_cols = ["direct_jobs_created", "clinic_visits_per_week", "medical_rides_
ops_labels = ["Direct jobs", "Clinic visits / week", "Medical rides / week",

plot_df = shelbycare_outputs[["scenario_name"] + ops_cols].copy()
plot_df = plot_df.set_index("scenario_name")
plot_df.columns = ops_labels
plot_df.plot(kind="bar", ax=ax)
ax.set_title("ShelbyCare operating capacity by scenario")
ax.set_ylabel("Count")
ax.grid(axis="y", alpha=0.25)
plt.xticks(rotation=0)
```

```python
plt.tight_layout()
ops_chart_path = PRESENTATION_DIR / "shelbycare_chart_operations.png"
fig.savefig(ops_chart_path, bbox_inches="tight", dpi=220)
plt.show()
print("Saved:", ops_chart_path)

# ------------------------------
# Chart 2 — finance
# ------------------------------
fig, ax = plt.subplots()
finance_cols = ["annual_total_cost", "annual_subscription_revenue", "annual_
finance_labels = ["Annual cost", "Annual revenue", "Annual net operating"]

plot_df = shelbycare_outputs[["scenario_name"] + finance_cols].copy()
plot_df = plot_df.set_index("scenario_name")
plot_df.columns = finance_labels
plot_df.plot(kind="bar", ax=ax)
ax.set_title("ShelbyCare financial profile by scenario")
ax.set_ylabel("Dollars")
ax.grid(axis="y", alpha=0.25)
plt.xticks(rotation=0)
plt.tight_layout()
finance_chart_path = PRESENTATION_DIR / "shelbycare_chart_finance.png"
fig.savefig(finance_chart_path, bbox_inches="tight", dpi=220)
plt.show()
print("Saved:", finance_chart_path)

# ------------------------------
# Chart 3 — community health/access metrics
# ------------------------------
fig, ax = plt.subplots()
community_cols = [
    "projected_dental_visit_gap_pct",
    "projected_uninsured_adults_pct",
    "projected_obesity_pct",
    "projected_diabetes_pct",
]
community_labels = [
    "Dental visit gap",
    "Uninsured adults",
    "Obesity",
    "Diabetes",
]

plot_df = shelbycare_outputs[["scenario_name"] + community_cols].copy()
plot_df = plot_df.set_index("scenario_name")
plot_df.columns = community_labels
plot_df.plot(kind="bar", ax=ax)
ax.set_title("Projected community health/access metrics by scenario")
ax.set_ylabel("Percent")
ax.grid(axis="y", alpha=0.25)
plt.xticks(rotation=0)
plt.tight_layout()
community_chart_path = PRESENTATION_DIR / "shelbycare_chart_community_metric
fig.savefig(community_chart_path, bbox_inches="tight", dpi=220)
plt.show()
```

```
print("Saved:", community_chart_path)

# ------------------------------
# Chart 4 — projected unemployment
# ------------------------------
fig, ax = plt.subplots()
plot_df = shelbycare_outputs[["scenario_name", "projected_unemp_rate_pct"]].
ax.bar(plot_df["scenario_name"].astype(str), plot_df["projected_unemp_rate_p
ax.set_title("Projected unemployment rate by scenario")
ax.set_ylabel("Percent")
ax.grid(axis="y", alpha=0.25)

for i, v in enumerate(plot_df["projected_unemp_rate_pct"]):
    ax.text(i, v + 0.05, f"{v:.2f}%", ha="center", va="bottom", fontsize=9)

plt.xticks(rotation=0)
plt.tight_layout()
unemp_chart_path = PRESENTATION_DIR / "shelbycare_chart_unemployment.png"
fig.savefig(unemp_chart_path, bbox_inches="tight", dpi=220)
plt.show()
print("Saved:", unemp_chart_path)

# ------------------------------
# Chart 5 — sensitivity on annual net operating
# ------------------------------
fig, ax = plt.subplots(figsize=(11, 7))
sens_plot = sensitivity_table.sort_values("delta_annual_net_operating")
ax.barh(sens_plot["test_name"], sens_plot["delta_annual_net_operating"])
ax.set_title(f"Sensitivity: impact on annual net operating ({SENSITIVITY_BAS
ax.set_xlabel("Change in annual net operating ($)")
ax.grid(axis="x", alpha=0.25)
plt.tight_layout()
sens_finance_chart_path = PRESENTATION_DIR / "shelbycare_chart_sensitivity_r
fig.savefig(sens_finance_chart_path, bbox_inches="tight", dpi=220)
plt.show()
print("Saved:", sens_finance_chart_path)

# ------------------------------
# Chart 6 — sensitivity on unemployment + dental gap
# ------------------------------
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

sens_plot = sensitivity_table.sort_values("delta_projected_unemp_rate_pct_pc
axes[0].barh(sens_plot["test_name"], sens_plot["delta_projected_unemp_rate_p
axes[0].set_title(f"Sensitivity: unemployment ({SENSITIVITY_BASE_SCENARIO})"
axes[0].set_xlabel("Δ unemployment rate (pct points)")
axes[0].grid(axis="x", alpha=0.25)

sens_plot = sensitivity_table.sort_values("delta_projected_dental_gap_pct_pc
axes[1].barh(sens_plot["test_name"], sens_plot["delta_projected_dental_gap_p
axes[1].set_title(f"Sensitivity: dental visit gap ({SENSITIVITY_BASE_SCENARI
axes[1].set_xlabel("Δ dental visit gap (pct points)")
axes[1].grid(axis="x", alpha=0.25)

plt.tight_layout()
sens_metrics_chart_path = PRESENTATION_DIR / "shelbycare_chart_sensitivity_m
```

```python
fig.savefig(sens_metrics_chart_path, bbox_inches="tight", dpi=220)
plt.show()
print("Saved:", sens_metrics_chart_path)
```

In [ ]:

```python
```

In [ ]:

```python
# ============================================================
# Cell 8Q — selected low-growth Shelby community tract geometry
# Goal:
# build a geometry table for the 96 tracts in our selected community
# ============================================================

import geopandas as gpd
import pandas as pd
import numpy as np

# ---------------------------------------------------
# 1) Recover selected tract GEOIDs
# ---------------------------------------------------
selected_geoids = set()

if "selected_benchmark_tracts" in globals():
    selected_geoids = set(
        normalize_geoid_series(selected_benchmark_tracts["geoid"]).dropna().
    )

if not selected_geoids and "TARGET_GEOIDS" in globals():
    selected_geoids = set(
        normalize_geoid_series(pd.Series(TARGET_GEOIDS)).dropna().tolist()
    )

if not selected_geoids and "tract_cluster_map" in globals():
    SELECTED_CLUSTER_ID = "Tennessee | Shelby County | cluster_77"
    tmp = tract_cluster_map.copy()
    if "geoid" in tmp.columns and "cluster_id" in tmp.columns:
        tmp["geoid"] = normalize_geoid_series(tmp["geoid"])
        selected_geoids = set(
            tmp.loc[tmp["cluster_id"].astype(str) == SELECTED_CLUSTER_ID, "g
            .dropna()
            .tolist()
        )

if not selected_geoids:
    raise ValueError("Could not recover the selected Shelby community tract

print("Selected low-growth community tract count:", len(selected_geoids))

# ---------------------------------------------------
# 2) Build selected tract geometry table
# ---------------------------------------------------
if "shelby_geo" not in globals():
    raise ValueError("Expected shelby_geo to exist. Run the Shelby geometry

selected_tracts_gdf = shelby_geo.copy()
selected_tracts_gdf["geoid"] = normalize_geoid_series(selected_tracts_gdf["g
```

```python
selected_tracts_gdf = selected_tracts_gdf.loc[
    selected_tracts_gdf["geoid"].isin(selected_geoids)
].copy()

if selected_tracts_gdf.empty:
    raise ValueError("Selected tract geometry table is empty.")

print("Selected tract geometry rows:", selected_tracts_gdf.shape)
display(selected_tracts_gdf[["geoid"]].head())

# Save a reusable version
selected_tracts_gdf.to_file(
    OUT_DIR / "selected_low_growth_96_tracts.geojson",
    driver="GeoJSON"
)

print("Saved:", OUT_DIR / "selected_low_growth_96_tracts.geojson")
```

In [ ]:

```python
# ============================================================
# Cell 8R — load Census ZCTA polygons
# Goal:
# get ZIP/ZCTA geometry for overlap analysis
# ============================================================

from pathlib import Path
import zipfile
import requests
import geopandas as gpd

ZCTA_ROOT_DIR = RAW_DIR / "zcta"
ZCTA_ROOT_DIR.mkdir(parents=True, exist_ok=True)

# 2024 nationwide ZCTA shapefile
zcta_zip_name = "tl_2024_us_zcta520.zip"
zcta_shp_name = "tl_2024_us_zcta520.shp"
zcta_zip_path = ZCTA_ROOT_DIR / zcta_zip_name
zcta_extract_dir = ZCTA_ROOT_DIR / "tl_2024_us_zcta520"
zcta_shp_path = zcta_extract_dir / zcta_shp_name

if not zcta_shp_path.exists():
    zcta_extract_dir.mkdir(parents=True, exist_ok=True)

    if not zcta_zip_path.exists():
        zcta_url = f"https://www2.census.gov/geo/tiger/TIGER2024/ZCTA520/{zc
        print("Downloading:", zcta_url)
        r = requests.get(zcta_url, timeout=120)
        r.raise_for_status()
        with open(zcta_zip_path, "wb") as f:
            f.write(r.content)

    print("Extracting ZCTA shapefile...")
    with zipfile.ZipFile(zcta_zip_path, "r") as zf:
        zf.extractall(zcta_extract_dir)

if not zcta_shp_path.exists():
    raise FileNotFoundError(f"Could not find extracted ZCTA shapefile: {zcta
```

```
zcta_gdf = gpd.read_file(zcta_shp_path).copy()

# Standardize ZIP/ZCTA code column
zcta_code_col = None
for c in ["ZCTA5CE20", "GEOID20", "ZCTA5CE10", "GEOID10", "GEOID"]:
    if c in zcta_gdf.columns:
        zcta_code_col = c
        break

if zcta_code_col is None:
    raise ValueError(f"Could not find a ZCTA code column. Columns: {list(zct

zcta_gdf["zcta"] = zcta_gdf[zcta_code_col].astype(str).str.extract(r"(\d{5})

zcta_gdf = zcta_gdf.dropna(subset=["zcta", "geometry"]).copy()
zcta_gdf = zcta_gdf[zcta_gdf.geometry.notna()].copy()
zcta_gdf = zcta_gdf[~zcta_gdf.geometry.is_empty].copy()

print("ZCTA rows loaded:", zcta_gdf.shape)
display(zcta_gdf[["zcta"]].head())
```

In [ ]:

```
# ============================================================
# Cell 8S — compute ZIP/ZCTA overlaps with the selected 96 tracts
# Outputs:
# - overlapping ZIP/ZCTA list
# - area overlap table
# - community-level overlap shares
# ============================================================

import geopandas as gpd
import pandas as pd
import numpy as np

if "selected_tracts_gdf" not in globals():
    raise ValueError("Run Cell 8Q first.")
if "zcta_gdf" not in globals():
    raise ValueError("Run Cell 8R first.")

tracts_work = selected_tracts_gdf.copy()
zcta_work = zcta_gdf.copy()

# Reproject to equal-area CRS for area calculations
equal_area_crs = 5070  # NAD83 / CONUS Albers
tracts_work = tracts_work.to_crs(equal_area_crs)
zcta_work = zcta_work.to_crs(equal_area_crs)

# Dissolve all 96 tracts into one community shape
selected_community_union = tracts_work.dissolve()
selected_community_union = selected_community_union.reset_index(drop=True)
selected_community_union["community_id"] = "Selected low-growth community"

# Intersect tract-by-ZCTA for detailed overlap
tract_zcta_overlap = gpd.overlay(
    tracts_work[["geoid", "geometry"]],
    zcta_work[["zcta", "geometry"]],
```

```
    how="intersection"
)

if tract_zcta_overlap.empty:
    raise ValueError("No tract/ZCTA overlaps were found.")

tract_zcta_overlap["overlap_sq_m"] = tract_zcta_overlap.geometry.area
tract_zcta_overlap["overlap_sq_miles"] = tract_zcta_overlap["overlap_sq_m"]

# tract area for share-of-tract covered by each ZCTA
tract_area = tracts_work[["geoid", "geometry"]].copy()
tract_area["tract_sq_m"] = tract_area.geometry.area
tract_area = tract_area[["geoid", "tract_sq_m"]]

tract_zcta_overlap = tract_zcta_overlap.merge(tract_area, on="geoid", how="l
tract_zcta_overlap["share_of_tract_area"] = tract_zcta_overlap["overlap_sq_m

# community-level overlap by ZCTA
community_zcta_overlap = gpd.overlay(
    selected_community_union[["community_id", "geometry"]],
    zcta_work[["zcta", "geometry"]],
    how="intersection"
)

community_zcta_overlap["overlap_sq_m"] = community_zcta_overlap.geometry.are
community_zcta_overlap["overlap_sq_miles"] = community_zcta_overlap["overlap

community_total_sq_m = selected_community_union.geometry.area.iloc[0]
community_zcta_overlap["share_of_community_area"] = community_zcta_overlap["

community_zcta_summary = (
    community_zcta_overlap[["zcta", "overlap_sq_miles", "share_of_community_
    .sort_values(["share_of_community_area", "overlap_sq_miles"], ascending=
    .reset_index(drop=True)
)

# Distinct overlapping ZIP/ZCTA list
overlapping_zctas = community_zcta_summary["zcta"].dropna().astype(str).toli

print("Overlapping ZIP/ZCTAs:")
print(overlapping_zctas)
print()
print("Count of overlapping ZIP/ZCTAs:", len(overlapping_zctas))

display(community_zcta_summary.head(25))

# Save outputs
community_zcta_summary.to_csv(OUT_DIR / "selected_low_growth_community_overl
tract_zcta_overlap.drop(columns="geometry").to_csv(
    OUT_DIR / "selected_low_growth_tract_zcta_overlap_detail.csv",
    index=False
)
community_zcta_overlap.to_file(
    OUT_DIR / "selected_low_growth_community_zcta_overlap.geojson",
    driver="GeoJSON"
)
```

```
print("Saved:")
print("-", OUT_DIR / "selected_low_growth_community_overlapping_zctas.csv")
print("-", OUT_DIR / "selected_low_growth_tract_zcta_overlap_detail.csv")
print("-", OUT_DIR / "selected_low_growth_community_zcta_overlap.geojson")
```

In [ ]:
```
# ============================================================
# Cell 8T – final ZIP/ZCTA list only
# ============================================================

if "community_zcta_summary" not in globals():
    raise ValueError("Run Cell 8S first.")

zip_list_df = community_zcta_summary[["zcta", "share_of_community_area", "ov
zip_list_df["share_of_community_area_pct"] = 100 * zip_list_df["share_of_com

display(
    zip_list_df[["zcta", "share_of_community_area_pct", "overlap_sq_miles"]]
    .round({"share_of_community_area_pct": 2, "overlap_sq_miles": 2})
)

print("ZIP/ZCTAs overlapping the selected 96-tract community:")
print(zip_list_df["zcta"].tolist())
```

In [ ]:
```
# ============================================================
# BOTTOM PATCH – Shelby insurance coverage over time
# Truthful label:
# "Estimated health insurance coverage rate (ACS-based tract-year panel)"
#
# What this does:
# 1) Reuses existing three_group_base if it exists
# 2) Otherwise rebuilds the 3 Shelby groups from the saved presentation file
# 3) Reloads the FULL tract-year panel (not analysis_panel) so insurance col
# 4) Finds the best insurance variable available in the panel
# 5) Builds a 2017-2024 3-group time series
# 6) Saves a CSV + PNG
# ============================================================

from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.ticker import PercentFormatter

# ------------------------------------------------------------
# helpers
# ------------------------------------------------------------
def _find_existing(paths):
    for p in paths:
        p = Path(p)
        if p.exists():
            return p
    return None

def _norm_geoid(series):
```

```python
    s = series.astype("string").str.strip()
    s = s.str.replace(r"\.0$", "", regex=True)
    s = s.str.replace(r"\D", "", regex=True)
    s = s.str.zfill(11)
    return s.where(s.str.fullmatch(r"\d{11}"), pd.NA)

def _safe_weighted_mean(values, weights):
    values = pd.to_numeric(values, errors="coerce")
    weights = pd.to_numeric(weights, errors="coerce")
    mask = values.notna() & weights.notna() & (weights > 0)
    if mask.sum() == 0:
        valid = values.notna()
        if valid.sum() == 0:
            return np.nan
        return float(values.loc[valid].mean())
    return float(np.average(values.loc[mask], weights=weights.loc[mask]))

def _pick_col(df, preferred_names):
    lower_map = {str(c).lower(): c for c in df.columns}
    for name in preferred_names:
        if str(name).lower() in lower_map:
            return lower_map[str(name).lower()]
    return None

def _find_fuzzy_cols(df, include_tokens):
    cols = []
    for c in df.columns:
        lc = str(c).lower()
        if any(tok in lc for tok in include_tokens):
            cols.append(c)
    return cols

def _to_rate(series):
    s = pd.to_numeric(series, errors="coerce")
    valid = s.dropna()
    if valid.empty:
        return s
    # if values look like 0-100 percentages, convert to 0-1
    if valid.quantile(0.95) > 1.5 and valid.quantile(0.95) <= 100:
        return s / 100.0
    return s

def _derive_rate_from_counts(numerator, denominator):
    num = pd.to_numeric(numerator, errors="coerce")
    den = pd.to_numeric(denominator, errors="coerce")
    out = pd.Series(np.nan, index=num.index, dtype=float)
    mask = num.notna() & den.notna() & (den > 0)
    out.loc[mask] = num.loc[mask] / den.loc[mask]
    return out

# ------------------------------------------------------------
# paths / settings
# ------------------------------------------------------------
DATA_DIR = Path("data")
RAW_DIR = DATA_DIR / "raw"
OUT_DIR = DATA_DIR / "processed"
```

```
PRESENTATION_DIR = OUT_DIR / "presentation_ready"
PRESENTATION_DIR.mkdir(parents=True, exist_ok=True)

ANALYSIS_YEARS = list(range(2017, 2025))
SHELBY_COUNTY_FIPS = "47157"

group_order = [
    "Selected low-growth community",
    "Rest of Shelby County (excluding both selected communities)",
    "Selected high-growth comparator",
]

label_map = {
    "Selected low-growth community": "Our Shelby Community",
    "Rest of Shelby County (excluding both selected communities)": "Rest of
    "Selected high-growth comparator": "High-Growth Shelby Community",
}

color_map = {
    "Selected low-growth community": "#fb8072",
    "Rest of Shelby County (excluding both selected communities)": "#bebada"
    "Selected high-growth comparator": "#8dd3c7",
}

# ------------------------------------------------------------
# A. recover / rebuild the 3-group Shelby tract lookup
# ------------------------------------------------------------
need_rebuild_three_group = (
    ("three_group_base" not in globals()) or
    (three_group_base is None) or
    (not isinstance(three_group_base, pd.DataFrame)) or
    (len(three_group_base) == 0)
)

if not need_rebuild_three_group:
    tgb = three_group_base.copy()
    tgb_cols = {str(c).lower(): c for c in tgb.columns}
    geoid_col = tgb_cols.get("geoid")
    group_col = tgb_cols.get("final_compare_group", tgb_cols.get("group"))
    if geoid_col is None or group_col is None:
        need_rebuild_three_group = True

if need_rebuild_three_group:
    three_group_compare_path = _find_existing([
        "data/processed/presentation_ready/three_group_selected_vs_comparato
        "src/data/processed/presentation_ready/three_group_selected_vs_compa
        "data/processed/three_group_selected_vs_comparator_vs_rest.csv",
        "src/data/processed/three_group_selected_vs_comparator_vs_rest.csv",
    ])

    shelby_places_base_path = _find_existing([
        "data/processed/presentation_ready/shelby_places_tract_base.parquet"
        "src/data/processed/presentation_ready/shelby_places_tract_base.parc
        "data/processed/shelby_places_tract_base.parquet",
        "src/data/processed/shelby_places_tract_base.parquet",
    ])
```

```
shelby_full_compare_path = _find_existing([
    "data/processed/presentation_ready/shelby_full_community_compare.csv
    "src/data/processed/presentation_ready/shelby_full_community_compare
    "data/processed/shelby_full_community_compare.csv",
    "src/data/processed/shelby_full_community_compare.csv",
])

shelby_full_map_path = _find_existing([
    "data/processed/presentation_ready/shelby_full_tract_cluster_map.geo
    "src/data/processed/presentation_ready/shelby_full_tract_cluster_map
    "data/processed/shelby_full_tract_cluster_map.geojson",
    "src/data/processed/shelby_full_tract_cluster_map.geojson",
])

missing_paths = []
if three_group_compare_path is None:
    missing_paths.append("three_group_selected_vs_comparator_vs_rest.csv
if shelby_places_base_path is None:
    missing_paths.append("shelby_places_tract_base.parquet")
if shelby_full_compare_path is None:
    missing_paths.append("shelby_full_community_compare.csv")
if shelby_full_map_path is None:
    missing_paths.append("shelby_full_tract_cluster_map.geojson")

if missing_paths:
    raise FileNotFoundError(
        "Could not rebuild three_group_base. Missing required saved file
        + ", ".join(missing_paths)
    )

# low-growth selected community from saved Shelby PLACES tract base
shelby_places_base = pd.read_parquet(shelby_places_base_path).copy()
if "geoid" not in shelby_places_base.columns or "is_selected_community"
    raise ValueError(
        "shelby_places_tract_base.parquet must contain `geoid` and `is_s
    )

shelby_places_base["geoid"] = _norm_geoid(shelby_places_base["geoid"])
all_shelby_geoids = set(shelby_places_base["geoid"].dropna().tolist())
low_growth_geoids = set(
    shelby_places_base.loc[
        shelby_places_base["is_selected_community"].fillna(False),
        "geoid"
    ].dropna().tolist()
)

# identify best high-growth comparator from saved full-community summary
shelby_full_compare = pd.read_csv(shelby_full_compare_path).copy()
community_id_col = _pick_col(shelby_full_compare, [
    "community_id", "shelby_full_cluster_id", "cluster_id", "community"
])
score_col = _pick_col(shelby_full_compare, [
    "igs_total_recovery", "igs_total", "igs_recovery"
])
tract_count_col = _pick_col(shelby_full_compare, [
```

```
        "n_tracts", "tract_count"
    ])

    if community_id_col is None or score_col is None:
        raise ValueError(
            "Could not identify community id / recovery score columns in she
        )

    tmp_compare = shelby_full_compare.copy()
    if tract_count_col is not None:
        tmp_compare = tmp_compare.loc[
            pd.to_numeric(tmp_compare[tract_count_col], errors="coerce").fil
        ].copy()

    tmp_compare[score_col] = pd.to_numeric(tmp_compare[score_col], errors="c
    tmp_compare = (
        tmp_compare
        .dropna(subset=[score_col])
        .sort_values(score_col, ascending=False)
        .reset_index(drop=True)
    )

    if tmp_compare.empty:
        raise ValueError("Could not identify a valid high-growth Shelby comp

    BEST_SHELBY_COMPARATOR_ID = tmp_compare.iloc[0][community_id_col]

    # load tract-level full-community map to recover comparator GEOIDs
    import geopandas as gpd

    shelby_full_map = gpd.read_file(shelby_full_map_path)
    map_geoid_col = _pick_col(shelby_full_map, ["geoid", "GEOID"])
    map_cluster_col = _pick_col(shelby_full_map, [
        "community_id", "shelby_full_cluster_id", "cluster_id", "community"
    ])

    if map_geoid_col is None or map_cluster_col is None:
        raise ValueError(
            "Could not identify geoid/community columns in shelby_full_tract
        )

    shelby_full_map["geoid"] = _norm_geoid(shelby_full_map[map_geoid_col])

    high_growth_geoids = set(
        shelby_full_map.loc[
            shelby_full_map[map_cluster_col].astype(str) == str(BEST_SHELBY_
            "geoid"
        ].dropna().tolist()
    )

    # remove overlap just in case
    high_growth_geoids = high_growth_geoids - low_growth_geoids
    rest_geoids = all_shelby_geoids - low_growth_geoids - high_growth_geoids

    three_group_base = pd.DataFrame({"geoid": sorted(all_shelby_geoids)})
```

```
    def _assign_group(g):
        if g in low_growth_geoids:
            return "Selected low-growth community"
        elif g in high_growth_geoids:
            return "Selected high-growth comparator"
        else:
            return "Rest of Shelby County (excluding both selected communiti

    three_group_base["final_compare_group"] = three_group_base["geoid"].map(

else:
    three_group_base = three_group_base.copy()
    tgb_cols = {str(c).lower(): c for c in three_group_base.columns}
    geoid_col = tgb_cols.get("geoid")
    group_col = tgb_cols.get("final_compare_group", tgb_cols.get("group"))
    three_group_base = three_group_base.rename(
        columns={geoid_col: "geoid", group_col: "final_compare_group"}
    )
    three_group_base["geoid"] = _norm_geoid(three_group_base["geoid"])

three_group_base = (
    three_group_base[["geoid", "final_compare_group"]]
    .dropna(subset=["geoid", "final_compare_group"])
    .drop_duplicates(subset=["geoid"])
    .copy()
)

three_group_base["final_compare_group"] = pd.Categorical(
    three_group_base["final_compare_group"],
    categories=group_order,
    ordered=True
)

print("Three-group tract counts:")
display(
    three_group_base["final_compare_group"]
    .value_counts(dropna=False)
    .rename_axis("group")
    .reset_index(name="tract_count")
)

# ------------------------------------------------------------
# B. load FULL tract-year panel (not analysis_panel)
# ------------------------------------------------------------
panel_path = _find_existing([
    "data/processed/eda_panel_clean.parquet",
    "src/data/processed/eda_panel_clean.parquet",
    "data/processed/igs_x_acs_full.parquet",
    "src/data/processed/igs_x_acs_full.parquet",
])

if panel_path is None:
    raise FileNotFoundError(
        "Could not find eda_panel_clean.parquet or igs_x_acs_full.parquet."
    )
```

```python
panel_df_full = pd.read_parquet(panel_path).copy()
panel_df_full["geoid"] = _norm_geoid(panel_df_full["geoid"])
panel_df_full["year"] = pd.to_numeric(panel_df_full["year"], errors="coerce"
panel_df_full = panel_df_full.loc[panel_df_full["year"].isin(ANALYSIS_YEARS)
panel_df_full["year"] = panel_df_full["year"].astype(int)

# Shelby County filter
if "county_fips" in panel_df_full.columns:
    shelby_panel = panel_df_full.loc[
        panel_df_full["county_fips"].astype(str).str.zfill(5) == SHELBY_COUN
    ].copy()
elif ("display_state" in panel_df_full.columns) and ("display_county" in par
    shelby_panel = panel_df_full.loc[
        (panel_df_full["display_state"].astype(str) == "Tennessee") &
        (panel_df_full["display_county"].astype(str) == "Shelby County")
    ].copy()
else:
    shelby_panel = panel_df_full.loc[
        panel_df_full["geoid"].astype(str).str[:5] == SHELBY_COUNTY_FIPS
    ].copy()

if shelby_panel.empty:
    raise ValueError("No Shelby County rows found in the full panel file.")

# join in 3-group labels
group_lookup = (
    three_group_base[["geoid", "final_compare_group"]]
    .drop_duplicates(subset=["geoid"])
    .copy()
)

shelby_panel = shelby_panel.merge(
    group_lookup,
    on="geoid",
    how="inner",
    validate="many_to_one"
)

if shelby_panel.empty:
    raise ValueError("Shelby panel has no overlap with three_group_base.")

# ------------------------------------------------------------
# C. find the best insurance variable available
# ------------------------------------------------------------
direct_coverage_candidates = [
    # direct insured / coverage rate names
    "insured_rate",
    "insurance_coverage_rate",
    "health_insurance_coverage_rate",
    "insured_pct",
    "insured_share",
    "pct_insured",
    "share_insured",
    "health_insured_rate",
    "insured_population_rate",
    "health_coverage_rate",
```

```python
    "has_health_insurance_rate",
]

uninsured_rate_candidates = [
    # direct uninsured rate names
    "uninsured_rate",
    "uninsured_pct",
    "uninsured_share",
    "pct_uninsured",
    "share_uninsured",
    "no_health_insurance_rate",
    "no_health_insurance_pct",
    "without_health_insurance_rate",
    "without_health_insurance_pct",
    "uninsured_population_rate",
]

insured_count_candidates = [
    "insured_count",
    "insured_total",
    "health_insured_count",
    "population_with_health_insurance",
    "with_health_insurance_count",
]

uninsured_count_candidates = [
    "uninsured_count",
    "uninsured_total",
    "health_uninsured_count",
    "population_without_health_insurance",
    "without_health_insurance_count",
    "no_health_insurance_count",
]

denominator_candidates = [
    "insurance_universe",
    "health_insurance_universe",
    "civilian_noninstitutionalized_pop",
    "civilian_noninstitutionalized_population",
    "civilian_population_for_insurance",
    "insured_uninsured_total",
    "population_total_for_insurance",
    "pop_total",
]

selected_metric_source = None
selected_metric_note = None
coverage_rate = None

# 1) direct insured / coverage rate
direct_col = _pick_col(shelby_panel, direct_coverage_candidates)
if direct_col is not None:
    coverage_rate = _to_rate(shelby_panel[direct_col])
    selected_metric_source = direct_col
    selected_metric_note = "used direct coverage / insured rate column"
```

```python
# 2) direct uninsured rate -> convert to coverage
if coverage_rate is None:
    unins_col = _pick_col(shelby_panel, uninsured_rate_candidates)
    if unins_col is not None:
        uninsured_rate = _to_rate(shelby_panel[unins_col])
        coverage_rate = 1 - uninsured_rate
        selected_metric_source = unins_col
        selected_metric_note = "derived coverage as 1 - uninsured rate"

# 3) insured counts / denominator
if coverage_rate is None:
    insured_num_col = _pick_col(shelby_panel, insured_count_candidates)
    denom_col = _pick_col(shelby_panel, denominator_candidates)
    if insured_num_col is not None and denom_col is not None:
        coverage_rate = _derive_rate_from_counts(
            shelby_panel[insured_num_col],
            shelby_panel[denom_col]
        )
        selected_metric_source = f"{insured_num_col} / {denom_col}"
        selected_metric_note = "derived coverage from insured count / denomi

# 4) uninsured counts / denominator -> convert to coverage
if coverage_rate is None:
    uninsured_num_col = _pick_col(shelby_panel, uninsured_count_candidates)
    denom_col = _pick_col(shelby_panel, denominator_candidates)
    if uninsured_num_col is not None and denom_col is not None:
        uninsured_rate = _derive_rate_from_counts(
            shelby_panel[uninsured_num_col],
            shelby_panel[denom_col]
        )
        coverage_rate = 1 - uninsured_rate
        selected_metric_source = f"{uninsured_num_col} / {denom_col}"
        selected_metric_note = "derived coverage as 1 - (uninsured count / c

# 5) fuzzy fallback
if coverage_rate is None:
    fuzzy_insurance_cols = _find_fuzzy_cols(shelby_panel, ["insur", "coverag
    fuzzy_numeric_cols = []
    for c in fuzzy_insurance_cols:
        s = pd.to_numeric(shelby_panel[c], errors="coerce")
        if s.notna().sum() > 0:
            fuzzy_numeric_cols.append(c)

    picked = None
    picked_mode = None

    # prefer fuzzy uninsured-like names first, then direct coverage-like nam
    for c in fuzzy_numeric_cols:
        lc = str(c).lower()
        if any(tok in lc for tok in ["uninsured", "no_health", "without_heal
            picked = c
            picked_mode = "uninsured"
            break

    if picked is None:
        for c in fuzzy_numeric_cols:
```

```
            lc = str(c).lower()
            if any(tok in lc for tok in ["insured", "coverage", "covered"]):
                picked = c
                picked_mode = "coverage"
                break

    if picked is not None:
        series_rate = _to_rate(shelby_panel[picked])
        if picked_mode == "uninsured":
            coverage_rate = 1 - series_rate
            selected_metric_source = picked
            selected_metric_note = "derived coverage as 1 - fuzzy uninsured-
        else:
            coverage_rate = series_rate
            selected_metric_source = picked
            selected_metric_note = "used fuzzy coverage-style column"

# hard fail with useful debug info
if coverage_rate is None:
    possible_matches = [c for c in shelby_panel.columns if ("insur" in str(c
    raise ValueError(
        "Could not locate an insurance coverage variable in the full panel.\
        f"Possible insurance-related columns found: {possible_matches}"
    )

shelby_panel["insurance_coverage_rate_est"] = coverage_rate.clip(lower=0, up
shelby_panel["uninsured_rate_est"] = 1 - shelby_panel["insurance_coverage_ra

print("Insurance source chosen:", selected_metric_source)
print("Insurance source note:", selected_metric_note)

# -------------------------------------------------------------
# D. aggregate to group-year
# -------------------------------------------------------------
rows = []
for (grp, yr), df_grp in shelby_panel.groupby(["final_compare_group", "year"
    weights = pd.to_numeric(df_grp.get("pop_total"), errors="coerce")
    if weights is None or weights.notna().sum() == 0:
        weights = pd.Series(1.0, index=df_grp.index)

    row = {
        "group": grp,
        "year": int(yr),
        "n_tracts": int(df_grp["geoid"].nunique()),
        "population_sum": float(pd.to_numeric(df_grp.get("pop_total"), error
                          if "pop_total" in df_grp.columns else np.nan,
        "insurance_coverage_rate_est": _safe_weighted_mean(
            df_grp["insurance_coverage_rate_est"], weights
        ),
        "uninsured_rate_est": _safe_weighted_mean(
            df_grp["uninsured_rate_est"], weights
        ),
        "source_metric": selected_metric_source,
        "source_note": selected_metric_note,
    }
    rows.append(row)
```

```
three_group_insurance_year = (
    pd.DataFrame(rows)
    .sort_values(["group", "year"])
    .reset_index(drop=True)
)

three_group_insurance_year["group"] = pd.Categorical(
    three_group_insurance_year["group"],
    categories=group_order,
    ordered=True
)

display(three_group_insurance_year.round(4))

# ------------------------------------------------------------
# E. plot
# ------------------------------------------------------------
fig, ax = plt.subplots(figsize=(14, 8))

for grp in group_order:
    line_df = (
        three_group_insurance_year.loc[three_group_insurance_year["group"] =
        .sort_values("year")
        .copy()
    )
    if line_df.empty:
        continue

    ax.plot(
        line_df["year"],
        line_df["insurance_coverage_rate_est"],
        marker="o",
        linewidth=2.7,
        color=color_map[grp],
        label=label_map[grp],
    )

# period shading
ax.axvspan(2017 - 0.5, 2019 + 0.5, alpha=0.08, color="gray")
ax.axvspan(2020 - 0.5, 2021 + 0.5, alpha=0.12, color="gray")
ax.axvspan(2022 - 0.5, 2024 + 0.5, alpha=0.08, color="gray")

ax.set_title("Estimated health insurance coverage rate over time", fontsize=
ax.set_ylabel("Estimated insured share")
ax.set_xticks(sorted(three_group_insurance_year["year"].dropna().unique().to
ax.yaxis.set_major_formatter(PercentFormatter(xmax=1.0, decimals=0))
ax.grid(alpha=0.25)
ax.legend(loc="upper center", ncol=3, frameon=True, bbox_to_anchor=(0.5, 1.1

footer = (
    "ACS-based tract-year insurance measure.\n"
    f"Source used here: {selected_metric_source} ({selected_metric_note}).\n
    "Population-weighted tract averages using tract population from your par
    "This is insurance coverage, not a direct healthcare-access measure."
)
```

```python
fig.text(0.02, 0.01, footer, fontsize=10)

plt.tight_layout(rect=[0, 0.08, 1, 0.93])
plt.show()

# ------------------------------------------------------------
# F. save
# ------------------------------------------------------------
csv_path = PRESENTATION_DIR / "shelby_insurance_coverage_over_time.csv"
png_path = PRESENTATION_DIR / "shelby_insurance_coverage_over_time.png"

three_group_insurance_year.to_csv(csv_path, index=False)
fig.savefig(png_path, bbox_inches="tight", dpi=220)

print("Saved:")
print("-", csv_path)
print("-", png_path)
```

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